Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**DESIGN CALCULATION** **3310 ROUGHER AND SCAVENGER FLOTATION** **DESIGN OF FOUNDATION**
Document Number:	**MD-216-3310-EG-CV-CAL-1213**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to prepare design of **Rougher and Scavenger Flotation (3310)** Structure Foundation. In this document design & analysis of foundation marked F3, F4, F5, F6, F6A, F7, F7A & F11 has been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

3. SCOPE

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Analysis of foundation F4 has been carried out by Staad pro & Excel. Load has been taken from super structure STAAD File 'FLOTATION STRUCTURE 29.01.2020' &. For all other foundation & member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Sliding & overturning check has been done in excel sheet. Base Pressure check has been done both in excel sheet & from STAAD Pro (for F4) output. For F4 The STAAD modelling has been done with plate elements. Plate mat with subgrade modulus of 40500 kN/m2/m has been considered (as per soil report).

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa C01

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

1 DEAD LOAD (DL)
2 NORMAL OPERATING FLUID LOAD (FL1)
3 MAX. FLUID LOAD (FL2)
4 LIVE LOAD (LL)
5 SEISMIC LOAD + X (SLX1)
6 SEISMIC LOAD - X (SLX2)

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7 SEISMIC LOAD +Z (SLZ1)
8 SEISMIC LOAD -Z (SLZ2)
9 WIND LOAD + X (WLX1)
10 WIND LOAD - X (WLX2)
11 WIND LOAD + Z (WLZ1)
12 WIND LOAD - Z (WLZ2)
13 TEMPERATURE LOAD (TL)
14 LATERAL PIPE LOAD (PL)

4.3 Load combination

A. The following load combination are considered for base pressure, sliding & overturning check.

201 1.0DL+1.0FL1
202 1.0DL+1.0FL2
203 1.0DL+1.0FL1+1.0LL+1.0TL
204 1.0DL+1.0FL2+1.0LL+1.0TL
205 1.0DL+1.0FL1+0.75LL+0.75TL
206 1.0DL+1.0FL2+0.75LL+0.75TL
207 1.0DL+1.0FL1+1.0PL
208 1.0DL+1.0FL2+1.0PL
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
213 1.0DL+1.0FL1+1.0WLX1
214 1.0DL+1.0FL1+1.0WLX2
215 1.0DL+1.0FL1+1.0WLZ1
216 1.0DL+1.0FL1+1.0WLZ2
217 1.0DL+1.0FL2+1.0WLX1
218 1.0DL+1.0FL2+1.0WLX2
219 1.0DL+1.0FL2+1.0WLZ1
220 1.0DL+1.0FL2+1.0WLZ2
221 1.0DL+1.0FL1+0.7SLX1
222 1.0DL+1.0FL1+0.7SLX2
223 1.0DL+1.0FL1+0.7SLZ1
224 1.0DL+1.0FL1+0.7SLZ2
225 1.0DL+1.0FL2+0.7SLX1
226 1.0DL+1.0FL2+0.7SLX2

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227 1.0DL+1.0FL2+0.7SLZ1

228 1.0DL+1.0FL2+0.7SLZ2

229 1.0DL+1.0FL1+0.75WLX1+0.75LL

230 1.0DL+1.0FL1+0.75WLX2+0.75LL

231 1.0DL+1.0FL1+0.75WLZ1+0.75LL

232 1.0DL+1.0FL1+0.75WLZ2+0.75LL

233 1.0DL+1.0FL2+0.75WLX1+0.75LL

234 1.0DL+1.0FL2+0.75WLX2+0.75LL

235 1.0DL+1.0FL2+0.75WLZ1+0.75LL

236 1.0DL+1.0FL2+0.75WLZ2+0.75LL

237 1.0DL+1.0FL1+0.525SLX1+0.75LL

238 1.0DL+1.0FL1+0.525SLX2+0.75LL

239 1.0DL+1.0FL1+0.525SLZ1+0.75LL

240 1.0DL+1.0FL1+0.525SLZ2+0.75LL

241 1.0DL+1.0FL2+0.525SLX1+0.75LL

242 1.0DL+1.0FL2+0.525SLX2+0.75LL

243 1.0DL+1.0FL2+0.525SLZ1+0.75LL

244 1.0DL+1.0FL2+0.525SLZ2+0.75LL

245 0.6DL+1.0WLX1

246 0.6DL+1.0WLX2

247 0.6DL+1.0WLZ1

248 0.6DL+1.0WLZ2

249 0.6DL+0.6FL1+0.7SLX1

250 0.6DL+0.6FL1+0.7SLX2

251 0.6DL+0.6FL1+0.7SLZ1

252 0.6DL+0.6FL1+0.7SLZ2

B) The following load combination are considered for base pressure, sliding & overturning check.

101 1.4DL+1.4FL1

102 1.4DL+1.4FL2

103 1.2DL+1.2FL1+1.6LL

104 1.2DL+1.2FL2+1.6LL

105 1.2DL+1.2FL1+1.2TL+1.6LL

106 1.2DL+1.2FL2+1.2TL+1.6LL

107 1.4DL+1.4FL1+1.4PL

108 1.4DL+1.4FL2+1.4PL

109 1.2DL+1.2FL1+1.6LL+1.2PL

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110 1.2DL+1.2FL2+1.6LL+1.2PL

111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL

112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

113 1.2DL+0.8WLX1

114 1.2DL+0.8WLX2

115 1.2DL+0.8WLZ1

116 1.2DL+0.8WLZ2

117 1.2DL+1.6WLX1+0.5LL

118 1.2DL+1.6WLX2+0.5LL

119 1.2DL+1.6WLZ1+0.5LL

120 1.2DL+1.6WLZ2+0.5LL

121 1.2DL+1.0SLX1+0.5LL+0.5FL1

122 1.2DL+1.0SLX2+0.5LL+0.5FL1

123 1.2DL+1.0SLZ1+0.5LL+0.5FL1

124 1.2DL+1.0SLZ2+0.5LL+0.5FL1

125 0.9DL+1.6WLX1

126 0.9DL+1.6WLX2

127 0.9DL+1.6WLZ1

128 0.9DL+1.6WLZ2

129 0.9DL+1.0SLX1

130 0.9DL+1.0SLX2

131 0.9DL+1.0SLZ1

132 0.9DL+1.0SLZ2

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum Reinforcing Steel (Rebar) Size

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

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4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.5m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-10 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.5m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-10 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

5. DESIGN OF MISCELLANEOUS COMPONENTS

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APPENDIX – A

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F3

STAAD Node - 7851 7072 7091 7076 7429
Calculation has been done for node no 7091

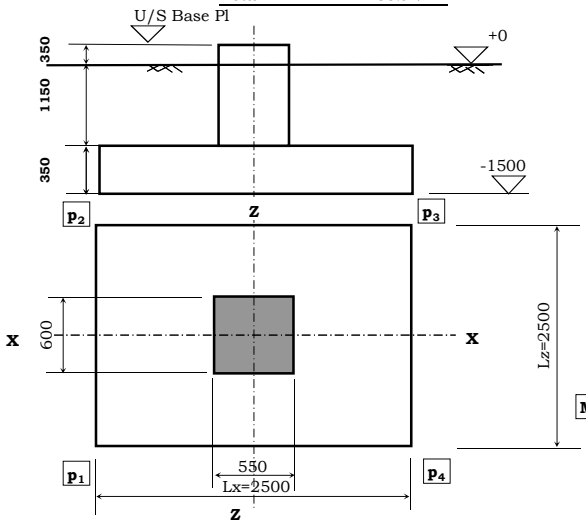
	Along-X	Along-Z
Size of column :	0.55m	0.60m
Size of fdn. raft :	2.50m	2.50m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

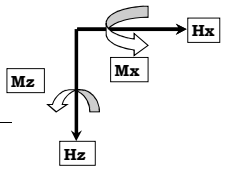
Weight of raft & Col.= (2.5*2.5*0.35*24)+(0.55*0.6*1.5*24) = 64.38KN
Weight of soil = ((2.5*2.5)-(0.55*0.6))*1.15*18 = 122.54KN
Total = 186.92KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 294mm = 0.294m
Effective depth of slab along Z direction = 282mm = 0.282m
Clear projection along Z direction = 0.95m
Clear projection along X direction = 0.975m



H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

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O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F3

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	78.8	-0.5	0.0	0.1	0.0	265.71	-0.49	-0.90	0.08	0.15
202	1.0DL+1.0FL2	80.3	-0.4	0.0	0.1	0.0	267.25	-0.44	-0.82	0.07	0.14
203	1.0DL+1.0FL1+1.0LL+1.0TL	101.8	4.5	0.0	-0.4	0.0	288.72	4.52	8.37	-0.40	-0.73
204	1.0DL+1.0FL2+1.0LL+1.0TL	103.3	4.6	0.0	-0.4	0.0	290.26	4.57	8.45	-0.41	-0.75
205	1.0DL+1.0FL1+0.75LL+0.75TL	96.0	3.3	0.0	-0.3	0.0	282.96	3.27	6.05	-0.28	-0.51
206	1.0DL+1.0FL2+0.75LL+0.75TL	97.6	3.3	0.0	-0.3	0.0	284.51	3.31	6.13	-0.29	-0.53
207	1.0DL+1.0FL1+1.0PL	291.4	5.9	0.0	6.1	0.0	478.30	5.85	10.82	6.07	11.24
208	1.0DL+1.0FL2+1.0PL	292.9	5.9	0.0	6.1	0.0	479.85	5.89	10.90	6.06	11.22
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	314.4	10.9	0.0	5.6	0.0	501.31	10.86	20.09	5.59	10.35
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	315.9	10.9	0.0	5.6	0.0	502.86	10.90	20.17	5.58	10.33
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	308.6	9.6	0.0	5.7	0.0	495.56	9.61	17.77	5.71	10.57
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	310.2	9.6	0.0	5.7	0.0	497.11	9.65	17.85	5.70	10.55
213	1.0DL+1.0FL1+1.0WLX1	-33.3	-6.6	0.0	0.1	0.0	153.64	-6.55	-12.12	0.14	0.25
214	1.0DL+1.0FL1+1.0WLX2	191.3	5.6	0.0	0.0	0.0	378.23	5.60	10.35	0.03	0.06
215	1.0DL+1.0FL1+1.0WLZ1	88.2	-0.3	0.0	4.5	0.0	275.09	-0.29	-0.53	4.49	8.30
216	1.0DL+1.0FL1+1.0WLZ2	69.4	-0.7	0.0	-4.3	0.0	256.32	-0.69	-1.27	-4.32	-8.00
217	1.0DL+1.0FL2+1.0WLX1	-31.7	-6.5	0.0	0.1	0.0	155.18	-6.51	-12.04	0.13	0.23
218	1.0DL+1.0FL2+1.0WLX2	192.9	5.6	0.0	0.0	0.0	379.77	5.64	10.43	0.02	0.04
219	1.0DL+1.0FL2+1.0WLZ1	89.7	-0.2	0.0	4.5	0.0	276.64	-0.24	-0.45	4.48	8.28
220	1.0DL+1.0FL2+1.0WLZ2	70.9	-0.6	0.0	-4.3	0.0	257.87	-0.64	-1.19	-4.33	-8.01
221	1.0DL+1.0FL1+0.7SLX1	22.5	-6.0	0.0	0.1	0.0	209.44	-6.04	-11.18	0.14	0.25
222	1.0DL+1.0FL1+0.7SLX2	135.1	5.1	0.0	0.0	0.0	321.97	5.07	9.39	0.03	0.05
223	1.0DL+1.0FL1+0.7SLZ1	73.6	-0.7	0.0	1.0	0.0	260.53	-0.74	-1.37	1.00	1.85
224	1.0DL+1.0FL1+0.7SLZ2	84.0	-0.2	0.0	-0.8	0.0	270.88	-0.23	-0.42	-0.84	-1.55
225	1.0DL+1.0FL2+0.7SLX1	24.1	-6.0	0.0	0.1	0.0	210.99	-6.00	-11.10	0.13	0.23
226	1.0DL+1.0FL2+0.7SLX2	136.6	5.1	0.0	0.0	0.0	323.52	5.12	9.47	0.02	0.04
227	1.0DL+1.0FL2+0.7SLZ1	75.2	-0.7	0.0	1.0	0.0	262.08	-0.70	-1.29	0.99	1.84
228	1.0DL+1.0FL2+0.7SLZ2	85.5	-0.2	0.0	-0.8	0.0	272.43	-0.19	-0.35	-0.85	-1.57
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	16.8	-5.1	0.0	0.1	0.0	203.71	-5.10	-9.43	0.14	0.25
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	185.2	4.0	0.0	0.1	0.0	372.15	4.02	7.43	0.06	0.10
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	107.9	-0.4	0.0	3.4	0.0	294.80	-0.40	-0.73	3.40	6.29
232	1.0DL+1.0FL1+0.75WLZ2+0.75LL	93.8	-0.7	0.0	-3.2	0.0	280.72	-0.70	-1.29	-3.21	-5.93

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MD-216-3310-EG-CV-CAL-1213				Mansourah Massarah Gold Project							
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION					Date: 10.11.2023		
O19045_450_C_K_ 24091 _001_R2				AS BUILT					Rev: D01		
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	18.3	-5.1	0.0	0.1	0.0	205.25	-5.05	-9.35	0.13	0.23	
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	186.8	4.1	0.0	0.0	0.0	373.70	4.06	7.51	0.05	0.09	
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	109.4	-0.4	0.0	3.4	0.0	296.35	-0.35	-0.65	3.39	6.27	
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	95.3	-0.7	0.0	-3.2	0.0	282.27	-0.65	-1.21	-3.22	-5.95	
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	58.6	-4.7	0.0	0.1	0.0	245.56	-4.72	-8.72	0.14	0.25	
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	143.0	3.6	0.0	0.1	0.0	329.96	3.62	6.70	0.06	0.10	
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	97.0	-0.7	0.0	0.8	0.0	283.88	-0.74	-1.37	0.79	1.45	
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	104.7	-0.4	0.0	-0.6	0.0	291.64	-0.36	-0.66	-0.59	-1.10	
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	60.2	-4.7	0.0	0.1	0.0	247.11	-4.67	-8.65	0.13	0.23	
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	144.6	3.7	0.0	0.0	0.0	331.51	3.67	6.78	0.05	0.09	
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	98.5	-0.7	0.0	0.8	0.0	285.43	-0.70	-1.29	0.78	1.44	
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	106.3	-0.3	0.0	-0.6	0.0	293.19	-0.31	-0.58	-0.60	-1.12	
245 0.6DL+1.0WLX1	-67.5	-6.5	0.0	0.1	0.0	119.43	-6.47	-11.96	0.13	0.23	
246 0.6DL+1.0WLX2	157.1	5.7	0.0	0.0	0.0	344.02	5.68	10.51	0.02	0.04	
247 0.6DL+1.0WLZ1	54.0	-0.2	0.0	4.5	0.0	240.89	-0.20	-0.37	4.48	8.29	
248 0.6DL+1.0WLZ2	35.2	-0.6	0.0	-4.3	0.0	222.12	-0.60	-1.11	-4.33	-8.01	
249 0.6DL+0.6FL1+0.7SLX1	-9.0	-5.9	0.0	0.1	0.0	177.93	-5.85	-10.82	0.10	0.19	
250 0.6DL+0.6FL1+0.7SLX2	103.5	5.3	0.0	0.0	0.0	290.46	5.27	9.75	0.00	-0.01	
251 0.6DL+0.6FL1+0.7SLZ1	42.1	-0.5	0.0	1.0	0.0	229.02	-0.55	-1.01	0.97	1.79	
252 0.6DL+0.6FL1+0.7SLZ2	52.4	0.0	0.0	-0.9	0.0	239.37	-0.04	-0.06	-0.87	-1.61	
Ultimate Conditions											
101 1.4DL+1.4FL1	110.3	-0.7	0.0	0.1	0.0	297.22	-0.68	-1.26	0.12	0.21	
102 1.4DL+1.4FL2	112.5	-0.6	0.0	0.1	0.0	299.39	-0.62	-1.15	0.10	0.19	
103 1.2DL+1.2FL1+1.6LL	141.6	-0.7	0.0	0.1	0.0	328.51	-0.71	-1.32	0.13	0.23	
104 1.2DL+1.2FL2+1.6LL	143.4	-0.7	0.0	0.1	0.0	330.37	-0.66	-1.22	0.12	0.21	
105 1.2DL+1.2FL1+1.2TL+1.6LL	133.9	5.4	0.0	-0.5	0.0	320.84	5.39	9.98	-0.47	-0.87	
106 1.2DL+1.2FL2+1.2TL+1.6LL	135.8	5.4	0.0	-0.5	0.0	322.69	5.45	10.07	-0.48	-0.89	
107 1.4DL+1.4FL1+1.4PL	407.9	8.2	0.0	8.5	0.0	594.86	8.19	15.15	8.50	15.73	
108 1.4DL+1.4FL2+1.4PL	410.1	8.3	0.0	8.5	0.0	597.02	8.25	15.26	8.49	15.71	
109 1.2DL+1.2FL1+1.6LL+1.2PL	396.7	6.9	0.0	7.3	0.0	583.63	6.89	12.75	7.32	13.53	
110 1.2DL+1.2FL2+1.6LL+1.2PL	398.6	6.9	0.0	7.3	0.0	585.48	6.94	12.84	7.30	13.51	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	389.0	13.0	0.0	6.7	0.0	575.95	13.00	24.04	6.72	12.43	

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	390.9	13.0	0.0	6.7	0.0	577.81	13.05	24.14	6.71	12.41
113 1.2DL+0.8WLX1	-0.5	-5.7	0.0	0.2	0.0	186.43	-5.65	-10.46	0.19	0.35
114 1.2DL+0.8WLX2	179.2	4.1	0.0	0.1	0.0	366.10	4.06	7.52	0.11	0.20
115 1.2DL+0.8WLZ1	96.7	-0.6	0.0	3.7	0.0	283.60	-0.64	-1.19	3.67	6.80
116 1.2DL+0.8WLZ2	81.7	-1.0	0.0	-3.4	0.0	268.58	-0.96	-1.78	-3.38	-6.24
117 1.2DL+1.6WLX1+0.5LL	-75.4	-10.5	0.0	0.2	0.0	111.48	-10.55	-19.51	0.24	0.45
118 1.2DL+1.6WLX2+0.5LL	283.9	8.9	0.0	0.1	0.0	470.82	8.89	16.44	0.07	0.13
119 1.2DL+1.6WLZ1+0.5LL	118.9	-0.5	0.0	7.2	0.0	305.81	-0.52	-0.97	7.21	13.33
120 1.2DL+1.6WLZ2+0.5LL	88.9	-1.2	0.0	-6.9	0.0	275.78	-1.16	-2.15	-6.89	-12.75
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	25.7	-8.7	0.0	0.2	0.0	212.65	-8.69	-16.08	0.21	0.39
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	186.5	7.2	0.0	0.1	0.0	373.41	7.19	13.30	0.06	0.11
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	98.7	-1.1	0.0	1.5	0.0	285.64	-1.12	-2.06	1.45	2.68
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	113.5	-0.4	0.0	-1.2	0.0	300.43	-0.39	-0.71	-1.18	-2.18
125 0.9DL+1.6WLX1	-112.4	-10.3	0.0	0.2	0.0	74.48	-10.31	-19.07	0.20	0.36
126 0.9DL+1.6WLX2	246.9	9.1	0.0	0.0	0.0	433.83	9.13	16.89	0.03	0.05
127 0.9DL+1.6WLZ1	81.9	-0.3	0.0	7.2	0.0	268.81	-0.28	-0.52	7.16	13.24
128 0.9DL+1.6WLZ2	51.9	-0.9	0.0	-6.9	0.0	238.78	-0.92	-1.70	-6.94	-12.83
129 0.9DL+1.0SLX1	-13.5	-8.5	0.0	0.2	0.0	173.42	-8.54	-15.80	0.19	0.35
130 0.9DL+1.0SLX2	147.3	7.3	0.0	0.0	0.0	334.18	7.34	13.58	0.04	0.06
131 0.9DL+1.0SLZ1	59.5	-1.0	0.0	1.4	0.0	246.40	-0.97	-1.79	1.43	2.64
132 0.9DL+1.0SLZ2	74.3	-0.2	0.0	-1.2	0.0	261.19	-0.24	-0.44	-1.20	-2.22

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0FL1	265.7	-0.5	-0.9	0.1	0.2	370.2	2163.0	219.1	1280.5
202	1.0DL+1.0FL2	267.3	-0.4	-0.8	0.1	0.1	408.5	2473.7	241.9	1464.4
203	1.0DL+1.0FL1+1.0LL+1.0TL	288.7	4.5	8.4	-0.4	-0.7	43.1	491.4	25.5	290.9
204	1.0DL+1.0FL2+1.0LL+1.0TL	290.3	4.6	8.4	-0.4	-0.8	43.0	481.9	25.4	285.3
205	1.0DL+1.0FL1+0.75LL+0.75TL	283.0	3.3	6.0	-0.3	-0.5	58.5	690.2	34.6	408.6
206	1.0DL+1.0FL2+0.75LL+0.75TL	284.5	3.3	6.1	-0.3	-0.5	58.0	669.8	34.4	396.5
207	1.0DL+1.0FL1+1.0PL	478.3	5.9	10.8	6.1	11.2	55.2	53.2	32.7	31.5
208	1.0DL+1.0FL2+1.0PL	479.9	5.9	10.9	6.1	11.2	55.0	53.5	32.6	31.7
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	501.3	10.9	20.1	5.6	10.3	31.2	60.6	18.5	35.8
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	502.9	10.9	20.2	5.6	10.3	31.2	60.8	18.5	36.0
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	495.6	9.6	17.8	5.7	10.6	34.9	58.6	20.6	34.7

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
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212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	497.1	9.6	17.9	5.7	10.6	34.8	58.9	20.6	34.9
213 1.0DL+1.0FL1+1.0WLX1	153.6	-6.6	-12.1	0.1	0.2	15.8	768.9	9.4	455.2
214 1.0DL+1.0FL1+1.0WLX2	378.2	5.6	10.4	0.0	0.1	45.7	8518.6	27.0	5043.0
215 1.0DL+1.0FL1+1.0WLZ1	275.1	-0.3	-0.5	4.5	8.3	652.2	41.4	386.1	24.5
216 1.0DL+1.0FL1+1.0WLZ2	256.3	-0.7	-1.3	-4.3	-8.0	252.8	40.1	149.7	23.7
217 1.0DL+1.0FL2+1.0WLX1	155.2	-6.5	-12.0	0.1	0.2	16.1	832.2	9.5	492.6
218 1.0DL+1.0FL2+1.0WLX2	379.8	5.6	10.4	0.0	0.0	45.5	12830.1	26.9	7595.4
219 1.0DL+1.0FL2+1.0WLZ1	276.6	-0.2	-0.4	4.5	8.3	772.4	41.7	457.3	24.7
220 1.0DL+1.0FL2+1.0WLZ2	257.9	-0.6	-1.2	-4.3	-8.0	271.4	40.2	160.7	23.8
221 1.0DL+1.0FL1+0.7SLX1	209.4	-6.0	-11.2	0.1	0.3	23.4	1040.6	13.9	616.0
222 1.0DL+1.0FL1+0.7SLX2	322.0	5.1	9.4	0.0	0.1	42.9	7501.7	25.4	4441.0
223 1.0DL+1.0FL1+0.7SLZ1	260.5	-0.7	-1.4	1.0	1.9	237.9	175.7	140.8	104.0
224 1.0DL+1.0FL1+0.7SLZ2	270.9	-0.2	-0.4	-0.8	-1.5	799.3	218.7	473.2	129.5
225 1.0DL+1.0FL2+0.7SLX1	211.0	-6.0	-11.1	0.1	0.2	23.8	1122.5	14.1	664.5
226 1.0DL+1.0FL2+0.7SLX2	323.5	5.1	9.5	0.0	0.0	42.7	10929.7	25.3	6470.4
227 1.0DL+1.0FL2+0.7SLZ1	262.1	-0.7	-1.3	1.0	1.8	254.1	178.3	150.4	105.6
228 1.0DL+1.0FL2+0.7SLZ2	272.4	-0.2	-0.3	-0.8	-1.6	984.4	217.6	582.7	128.8
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	203.7	-5.1	-9.4	0.1	0.2	27.0	1019.5	16.0	603.6
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	372.1	4.0	7.4	0.1	0.1	62.6	4571.8	37.1	2706.5
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	294.8	-0.4	-0.7	3.4	6.3	503.0	58.6	297.8	34.7
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	280.7	-0.7	-1.3	-3.2	-5.9	272.5	59.1	161.3	35.0
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	205.3	-5.1	-9.3	0.1	0.2	27.4	1109.5	16.2	656.8
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	373.7	4.1	7.5	0.0	0.1	62.2	5489.1	36.8	3249.5
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	296.3	-0.4	-0.7	3.4	6.3	567.2	59.1	335.8	35.0
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	282.3	-0.7	-1.2	-3.2	-6.0	292.1	59.3	172.9	35.1
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	245.6	-4.7	-8.7	0.1	0.3	35.2	1220.0	20.8	722.2
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	330.0	3.6	6.7	0.1	0.1	61.5	4053.5	36.4	2399.7
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	283.9	-0.7	-1.4	0.8	1.5	259.9	244.3	153.9	144.7
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	291.6	-0.4	-0.7	-0.6	-1.1	555.1	331.7	328.6	196.4
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	247.1	-4.7	-8.6	0.1	0.2	35.7	1325.1	21.2	784.5
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	331.5	3.7	6.8	0.0	0.1	61.1	4869.3	36.2	2882.7
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	285.4	-0.7	-1.3	0.8	1.4	277.5	248.5	164.3	147.1
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	293.2	-0.3	-0.6	-0.6	-1.1	634.9	328.0	375.9	194.2
245 0.6DL+1.0WLX1	119.4	-6.5	-12.0	0.1	0.2	12.5	635.4	7.4	376.2
246 0.6DL+1.0WLX2	344.0	5.7	10.5	0.0	0.0	40.9	11069.0	24.2	6552.8
247 0.6DL+1.0WLZ1	240.9	-0.2	-0.4	4.5	8.3	809.8	36.3	479.4	21.5
248 0.6DL+1.0WLZ2	222.1	-0.6	-1.1	-4.3	-8.0	249.7	34.7	147.8	20.5
249 0.6DL+0.6FL1+0.7SLX1	177.9	-5.9	-10.8	0.1	0.2	20.6	1167.2	12.2	691.0
250 0.6DL+0.6FL1+0.7SLX2	290.5	5.3	9.7	0.0	0.0	37.3	49063.7	22.1	29045.7
251 0.6DL+0.6FL1+0.7SLZ1	229.0	-0.5	-1.0	1.0	1.8	283.4	159.7	167.8	94.5
252 0.6DL+0.6FL1+0.7SLZ2	239.4	0.0	-0.1	-0.9	-1.6	4621.0	185.9	2735.6	110.1
					Min.=	12.5	34.7	7.4	20.5
						OK	OK	OK	OK

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DESIGN OF FOUNDATION MKD. F3

Load Combination & Calculation of Base pressure

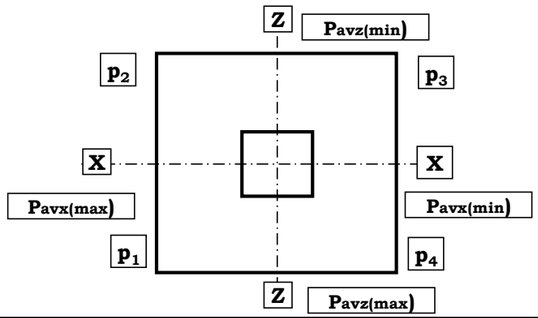
Max. Base pressure = p_{max} = $(V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = p_{min} = $(V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	Base Pressure (kN/m²) p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
201	1.0DL+1.0FL1	265.71	-0.90	0.15	42.51	0.34	0.06	42.92	42.80	42.23	42.11	15.92	42.86	42.17	42.58	42.46	O.K
202	1.0DL+1.0FL2	267.25	-0.82	0.14	42.76	0.31	0.05	43.13	43.02	42.50	42.39	16.13	43.08	42.45	42.82	42.71	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	288.72	8.37	-0.73	46.19	3.21	0.28	49.69	49.12	43.26	42.70	22.69	49.41	42.98	46.48	45.91	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	290.26	8.45	-0.75	46.44	3.24	0.29	49.97	49.40	43.49	42.91	22.97	49.69	43.20	46.73	46.16	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	282.96	6.05	-0.51	45.27	2.32	0.20	47.79	47.40	43.15	42.75	20.79	47.60	42.95	45.47	45.08	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	284.51	6.13	-0.53	45.52	2.35	0.20	48.08	47.67	43.37	42.96	21.08	47.88	43.17	45.73	45.32	O.K
207	1.0DL+1.0FL1+1.0PL	478.30	10.82	11.24	76.53	4.16	4.31	85.00	76.37	76.69	68.06	58.00	80.69	72.38	80.85	72.22	O.K
208	1.0DL+1.0FL2+1.0PL	479.85	10.90	11.22	76.78	4.19	4.31	85.27	76.66	76.90	68.28	58.27	80.97	72.59	81.09	72.47	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	501.31	20.09	10.35	80.21	7.71	3.97	91.90	83.95	76.47	68.52	64.90	87.93	72.50	84.19	76.24	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	502.86	20.17	10.33	80.46	7.74	3.97	92.17	84.23	76.68	68.75	65.17	88.20	72.72	84.43	76.49	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+	495.56	17.77	10.57	79.29	6.82	4.06	90.17	82.05	76.52	68.41	63.17	86.11	72.47	83.35	75.23	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+	497.11	17.85	10.55	79.54	6.85	4.05	90.44	82.34	76.73	68.63	63.44	86.39	72.68	83.59	75.49	O.K
213	1.0DL+1.0FL1+1.0WLX1	153.64	-12.12	0.25	24.58	4.65	0.10	29.33	29.14	20.02	19.83	2.33	29.24	19.93	24.68	24.49	O.K
214	1.0DL+1.0FL1+1.0WLX2	378.23	10.35	0.06	60.52	3.98	0.02	64.51	64.47	56.56	56.52	37.51	64.49	56.54	60.54	60.50	O.K
215	1.0DL+1.0FL1+1.0WLZ1	275.09	-0.53	8.30	44.01	0.20	3.19	47.41	41.03	47.00	40.62	20.41	44.22	43.81	47.21	40.83	O.K
216	1.0DL+1.0FL1+1.0WLZ2	256.32	-1.27	-8.00	41.01	0.49	3.07	44.57	38.43	43.60	37.45	17.57	41.50	40.53	44.09	37.94	O.K
217	1.0DL+1.0FL2+1.0WLX1	155.18	-12.04	0.23	24.83	4.62	0.09	29.54	29.36	20.30	20.12	2.54	29.45	20.21	24.92	24.74	O.K
218	1.0DL+1.0FL2+1.0WLX2	379.77	10.43	0.04	60.76	4.01	0.01	64.78	64.76	56.77	56.74	37.78	64.77	56.76	60.78	60.75	O.K
219	1.0DL+1.0FL2+1.0WLZ1	276.64	-0.45	8.28	44.26	0.17	3.18	47.62	41.25	47.27	40.91	20.62	44.44	44.09	47.45	41.08	O.K
220	1.0DL+1.0FL2+1.0WLZ2	257.87	-1.19	-8.01	41.26	0.46	3.08	44.79	38.64	43.88	37.73	17.79	41.72	40.81	44.34	38.19	O.K

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

221 1.0DL+1.0FL1+0.7SLX1	209.44	-11.18	0.25	33.51	4.29	0.10	37.90	37.71	29.31	29.12	10.90	37.81	29.22	33.61	33.42	O.K
222 1.0DL+1.0FL1+0.7SLX2	321.97	9.39	0.05	51.52	3.60	0.02	55.14	55.10	47.93	47.89	28.14	55.12	47.91	51.54	51.50	O.K
223 1.0DL+1.0FL1+0.7SLZ1	260.53	-1.37	1.85	41.69	0.53	0.71	42.92	41.50	41.87	40.45	15.92	42.21	41.16	42.40	40.98	O.K
224 1.0DL+1.0FL1+0.7SLZ2	270.88	-0.42	-1.55	43.34	0.16	0.59	44.10	42.91	43.77	42.58	17.10	43.51	43.18	43.94	42.75	O.K
225 1.0DL+1.0FL2+0.7SLX1	210.99	-11.10	0.23	33.76	4.26	0.09	38.11	37.93	29.59	29.40	11.11	38.02	29.50	33.85	33.67	O.K
226 1.0DL+1.0FL2+0.7SLX2	323.52	9.47	0.04	51.76	3.64	0.01	55.41	55.38	48.14	48.11	28.41	55.40	48.13	51.78	51.75	O.K
227 1.0DL+1.0FL2+0.7SLZ1	262.08	-1.29	1.84	41.93	0.50	0.71	43.13	41.72	42.14	40.73	16.13	42.43	41.44	42.64	41.23	O.K
228 1.0DL+1.0FL2+0.7SLZ2	272.43	-0.35	-1.57	43.59	0.13	0.60	44.32	43.12	44.06	42.85	17.32	43.72	43.46	44.19	42.99	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75I	203.71	-9.43	0.25	32.59	3.62	0.10	36.31	36.12	29.07	28.88	9.31	36.22	28.98	32.69	32.50	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75I	372.15	7.43	0.10	59.54	2.85	0.04	62.44	62.36	56.73	56.65	35.44	62.40	56.69	59.59	59.51	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75I	294.80	-0.73	6.29	47.17	0.28	2.41	49.86	45.03	49.30	44.47	22.86	47.45	46.89	49.58	44.75	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75I	280.72	-1.29	-5.93	44.92	0.49	2.28	47.69	43.13	46.70	42.14	20.69	45.41	44.42	47.20	42.64	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75I	205.25	-9.35	0.23	32.84	3.59	0.09	36.52	36.34	29.34	29.16	9.52	36.43	29.25	32.93	32.75	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75I	373.70	7.51	0.09	59.79	2.88	0.03	62.71	62.64	56.94	56.88	35.71	62.68	56.91	59.83	59.76	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75I	296.35	-0.65	6.27	47.42	0.25	2.41	50.07	45.26	49.57	44.76	23.07	47.67	47.17	49.82	45.01	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75I	282.27	-1.21	-5.95	45.16	0.46	2.29	47.91	43.34	46.99	42.41	20.91	45.63	44.70	47.45	42.88	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	245.56	-8.72	0.25	39.29	3.35	0.10	42.74	42.54	36.04	35.84	15.74	42.64	35.94	39.39	39.19	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	329.96	6.70	0.10	52.79	2.57	0.04	55.41	55.33	50.26	50.18	28.41	55.37	50.22	52.84	52.76	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	283.88	-1.37	1.45	45.42	0.52	0.56	46.50	45.39	45.45	44.34	19.50	45.95	44.90	45.98	44.87	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	291.64	-0.66	-1.10	46.66	0.25	0.42	47.34	46.49	46.83	45.99	20.34	46.92	46.41	47.09	46.24	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	247.11	-8.65	0.23	39.54	3.32	0.09	42.95	42.77	36.31	36.13	15.95	42.86	36.22	39.63	39.45	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	331.51	6.78	0.09	53.04	2.60	0.03	55.68	55.61	50.47	50.40	28.68	55.65	50.44	53.08	53.01	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	285.43	-1.29	1.44	45.67	0.49	0.55	46.71	45.61	45.73	44.62	19.71	46.16	45.18	46.22	45.12	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	293.19	-0.58	-1.12	46.91	0.22	0.43	47.56	46.70	47.12	46.26	20.56	47.13	46.69	47.34	46.48	O.K
245 0.6DL+1.0WLX1	119.43	-11.96	0.23	19.11	4.59	0.09	23.79	23.61	14.61	14.42	-3.21	23.70	14.52	19.20	19.02	O.K
246 0.6DL+1.0WLX2	344.02	10.51	0.04	55.04	4.04	0.01	59.09	59.06	51.02	50.99	32.09	59.08	51.01	55.06	55.03	O.K
247 0.6DL+1.0WLZ1	240.89	-0.37	8.29	38.54	0.14	3.18	41.87	35.50	41.58	35.22	14.87	38.69	38.40	41.73	35.36	O.K
248 0.6DL+1.0WLZ2	222.12	-1.11	-8.01	35.54	0.43	3.08	39.04	32.89	38.19	32.04	12.04	35.97	35.12	38.62	32.47	O.K
249 0.6DL+0.6FL1+0.7SLX1	177.93	-10.82	0.19	28.47	4.16	0.07	32.70	32.55	24.39	24.24	5.70	32.63	24.32	28.55	28.40	O.K
250 0.6DL+0.6FL1+0.7SLX2	290.46	9.75	-0.01	46.47	3.74	0.00	50.22	50.21	42.73	42.73	23.22	50.22	42.73	46.48	46.47	O.K
251 0.6DL+0.6FL1+0.7SLZ1	229.02	-1.01	1.79	36.64	0.39	0.69	37.72	36.34	36.94	35.57	10.72	37.03	36.26	37.33	35.96	O.K
252 0.6DL+0.6FL1+0.7SLZ2	239.37	-0.06	-1.61	38.30	0.02	0.62	38.94	37.71	38.89	37.66	11.94	38.33	38.28	38.92	37.69	O.K

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Ultimate Conditions

101 1.4DL+1.4FL1	297.22	-1.26	0.21	47.56	0.48	0.08	48.12	47.96	47.16	46.99	21.12	48.04	47.08	47.64	47.48	
102 1.4DL+1.4FL2	299.39	-1.15	0.19	47.90	0.44	0.07	48.41	48.27	47.54	47.39	21.41	48.34	47.47	47.98	47.83	
103 1.2DL+1.2FL1+1.6LL	328.51	-1.32	0.23	52.56	0.51	0.09	53.16	52.98	52.14	51.97	26.16	53.07	52.06	52.65	52.48	
104 1.2DL+1.2FL2+1.6LL	330.37	-1.22	0.21	52.86	0.47	0.08	53.41	53.25	52.47	52.31	26.41	53.33	52.39	52.94	52.78	
105 1.2DL+1.2FL1+1.2TL+1.6LL	320.84	9.98	-0.87	51.33	3.83	0.33	55.50	54.83	47.84	47.17	28.50	55.17	47.51	51.67	51.00	
106 1.2DL+1.2FL2+1.2TL+1.6LL	322.69	10.07	-0.89	51.63	3.87	0.34	55.84	55.16	48.11	47.42	28.84	55.50	47.77	51.98	51.29	
107 1.4DL+1.4FL1+1.4PL	594.86	15.15	15.73	95.18	5.82	6.04	107.04	94.96	95.40	83.32	80.04	101.00	89.36	101.22	89.14	
108 1.4DL+1.4FL2+1.4PL	597.02	15.26	15.71	95.52	5.86	6.03	107.42	95.35	95.69	83.63	80.42	101.39	89.66	101.56	89.49	
109 1.2DL+1.2FL1+1.6LL+1.2PL	583.63	12.75	13.53	93.38	4.89	5.20	103.47	93.08	93.68	83.29	76.47	98.28	88.49	98.58	88.19	
110 1.2DL+1.2FL2+1.6LL+1.2PL	585.48	12.84	13.51	93.68	4.93	5.19	103.80	93.42	93.93	83.56	76.80	98.61	88.75	98.87	88.49	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	575.95	24.04	12.43	92.15	9.23	4.77	106.16	96.61	87.69	78.15	79.16	101.39	82.92	96.93	87.38	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	577.81	24.14	12.41	92.45	9.27	4.77	106.48	96.95	87.95	78.41	79.48	101.72	83.18	97.22	87.68	
113 1.2DL+0.8WLX1	186.43	-10.46	0.35	29.83	4.02	0.14	33.98	33.71	25.95	25.68	6.98	33.85	25.82	29.97	29.70	
114 1.2DL+0.8WLX2	366.10	7.52	0.20	58.58	2.89	0.08	61.54	61.39	55.76	55.61	34.54	61.47	55.69	58.65	58.50	
115 1.2DL+0.8WLZ1	283.60	-1.19	6.80	45.38	0.46	2.61	48.44	43.22	47.53	42.31	21.44	45.83	44.92	47.99	42.77	
116 1.2DL+0.8WLZ2	268.58	-1.78	-6.24	42.97	0.68	2.40	46.05	41.26	44.69	39.89	19.05	43.66	42.29	45.37	40.58	
117 1.2DL+1.6WLX1+0.5LL	111.48	-19.51	0.45	17.84	7.49	0.17	25.50	25.16	10.51	10.17	-1.50	25.33	10.34	18.01	17.67	
118 1.2DL+1.6WLX2+0.5LL	470.82	16.44	0.13	75.33	6.31	0.05	81.70	81.59	69.07	68.97	54.70	81.65	69.02	75.39	75.28	
119 1.2DL+1.6WLZ1+0.5LL	305.81	-0.97	13.33	48.93	0.37	5.12	54.42	44.18	53.68	43.44	27.42	49.30	48.56	54.05	43.81	
120 1.2DL+1.6WLZ2+0.5LL	275.78	-2.15	-12.75	44.12	0.83	4.90	49.85	40.05	48.19	38.40	22.85	44.95	43.30	49.02	39.23	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	212.65	-16.08	0.39	34.02	6.17	0.15	40.35	40.05	28.00	27.70	13.35	40.20	27.85	34.18	33.88	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	373.41	13.30	0.11	59.75	5.11	0.04	64.90	64.81	54.68	54.59	37.90	64.86	54.64	59.79	59.70	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	285.64	-2.06	2.68	45.70	0.79	1.03	47.52	45.46	45.94	43.88	20.52	46.49	44.91	46.73	44.67	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	300.43	-0.71	-2.18	48.07	0.27	0.84	49.18	47.51	48.63	46.96	22.18	48.35	47.80	48.91	47.24	
125 0.9DL+1.6WLX1	74.48	-19.07	0.36	11.92	7.32	0.14	19.38	19.10	4.73	4.46	-7.62	19.24	4.60	12.06	11.78	
126 0.9DL+1.6WLX2	433.83	16.89	0.05	69.41	6.49	0.02	75.92	75.88	62.94	62.91	48.92	75.90	62.93	69.43	69.40	
127 0.9DL+1.6WLZ1	268.81	-0.52	13.24	43.01	0.20	5.09	48.30	38.12	47.90	37.72	21.30	43.21	42.81	48.10	37.92	
128 0.9DL+1.6WLZ2	238.78	-1.70	-12.83	38.21	0.65	4.93	43.79	33.93	42.48	32.62	16.79	38.86	37.55	43.14	33.28	
129 0.9DL+1.0SLX1	173.42	-15.80	0.35	27.75	6.07	0.13	33.95	33.68	21.81	21.55	6.95	33.82	21.68	27.88	27.62	
130 0.9DL+1.0SLX2	334.18	13.58	0.06	53.47	5.21	0.02	58.71	58.66	48.28	48.23	31.71	58.69	48.26	53.50	53.45	
131 0.9DL+1.0SLZ1	246.40	-1.79	2.64	39.42	0.69	1.01	41.12	39.10	39.75	37.73	14.12	40.11	38.74	40.44	38.42	
132 0.9DL+1.0SLZ2	261.19	-0.44	-2.22	41.79	0.17	0.85	42.81	41.10	42.48	40.77	15.81	41.96	41.63	42.65	40.94	

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Max :	597	24	16	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**15.92 kN/m^2**
Min :	74	-20	-13	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**25.92 kN/m^2**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**65.17 kN/m^2**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**75.17 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) =	42.76+0.31+0.05=	43.13 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(43.13+43.02)/2= 43.07 kN/m^2
p_2=	42.76+0.31-0.05=	43.02 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(42.39+42.50)/2= 42.45 kN/m^2
p_3=	42.76-(0.31+0.05)=	42.39 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(43.13+42.50)/2= 42.81 kN/m^2
p_4=	42.76-0.31+0.05=	42.50 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(42.39+43.02)/2= 42.71 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-γh)=	43.13-1.5*18=	16.13 kN/m^2		

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

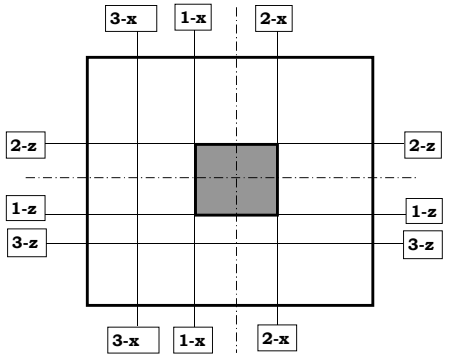
DESIGN OF FOUNDATION MKD. F3

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.55+2*0.5*0.282)+(0.6+2*0.5*0.282))*2
= 3.428 m
Punching Area= ((0.55+2*0.5*0.282)*(0.6+2*0.5*0.282))
= 0.73 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)
= 29.10 kN/m^2
Area of footing, A_f= 6.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	47.67	47.45	47.78	8.94	8.61	12.81	101.87
102	1.4DL+1.4FL2	48.00	47.81	48.10	9.09	8.79	13.02	103.79
103	1.2DL+1.2FL1+1.6LL	52.68	52.45	52.79	11.33	10.98	16.23	129.51
104	1.2DL+1.2FL2+1.6LL	52.96	52.76	53.07	11.46	11.13	16.41	131.15
105	1.2DL+1.2FL1+1.2TL+1.6LL	52.18	50.50	53.08	11.92	9.22	17.04	122.73
106	1.2DL+1.2FL2+1.2TL+1.6LL	52.49	50.78	53.39	12.07	9.35	17.26	124.37
107	1.4DL+1.4FL1+1.4PL	96.46	93.90	97.83	33.46	29.36	47.88	364.75
108	1.4DL+1.4FL2+1.4PL	96.82	94.23	98.19	33.64	29.51	48.14	366.66
109	1.2DL+1.2FL1+1.6LL+1.2PL	94.46	92.31	95.61	32.28	28.83	46.20	354.83
110	1.2DL+1.2FL2+1.6LL+1.2PL	94.76	92.60	95.92	32.43	28.96	46.42	356.47
111	1.2DL+1.2FL1+1.2TL+1.6LL+	94.19	90.12	96.36	33.22	26.72	47.52	348.05
112	1.2DL+1.2FL2+1.2TL+1.6LL+	94.49	90.41	96.67	33.37	26.85	47.73	349.69

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MD-216-3310-EG-CV-CAL-1213			Mansourah Massarah Gold Project				
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2			DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION				Date: 10.11.2023
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113 1.2DL+0.8WLX1	30.72	28.95	31.66	1.76	-1.06	2.49	4.02
114 1.2DL+0.8WLX2	59.22	57.94	59.90	15.03	13.00	21.51	162.71
115 1.2DL+0.8WLZ1	45.48	45.27	45.58	7.90	7.58	11.31	89.84
116 1.2DL+0.8WLZ2	43.13	42.82	43.29	6.84	6.35	9.79	76.58
117 1.2DL+1.6WLX1+0.5LL	19.48	16.19	21.25	-2.72	-7.99	-3.96	-62.18
118 1.2DL+1.6WLX2+0.5LL	76.72	73.95	78.21	24.20	19.75	34.62	255.20
119 1.2DL+1.6WLZ1+0.5LL	49.01	48.85	49.10	9.56	9.30	13.69	109.46
120 1.2DL+1.6WLZ2+0.5LL	44.31	43.94	44.50	7.43	6.85	10.64	82.93
121 1.2DL+1.0SLX1+0.5LL+0.5FL	35.38	32.67	36.84	4.51	0.17	6.41	27.18
122 1.2DL+1.0SLX2+0.5LL+0.5FL	60.87	58.63	62.08	16.37	12.77	23.40	169.16
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	45.87	45.53	46.06	8.17	7.61	11.70	91.64
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	48.14	48.01	48.20	9.12	8.92	13.06	104.70
125 0.9DL+1.6WLX1	13.53	10.31	15.25	-5.59	-10.74	-8.07	-94.85
126 0.9DL+1.6WLX2	70.84	67.99	72.37	21.44	16.88	30.67	222.52
127 0.9DL+1.6WLZ1	43.05	42.97	43.10	6.68	6.54	9.57	76.78
128 0.9DL+1.6WLZ2	38.35	38.06	38.50	4.56	4.10	6.53	50.26
129 0.9DL+1.0SLX1	29.09	26.41	30.51	1.49	-2.78	2.09	-7.47
130 0.9DL+1.0SLX2	54.62	52.33	55.85	13.42	9.75	19.18	134.51
131 0.9DL+1.0SLZ1	39.58	39.27	39.74	5.15	4.67	7.37	56.99
132 0.9DL+1.0SLZ2	41.83	41.76	41.87	6.09	5.98	8.73	70.05

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} = 48.04-(48.04-47.08)*0.975/2.5 = 47.67 kN/m^2 Overhang = 0.975 m

p_{2x} = 47.08+(48.04-47.08)*0.975/2.5 = 47.45 kN/m^2 Effective depth = 0.294 m

p_{3x} = 48.04-(48.04-47.08)*(0.975-0.294)/2.5 = 47.78 kN/m^2

BM_{1x} = (47.67-29.1)*0.975*0.975/2+0.5*(48.04-47.67)*0.975*0.975*0.67 = 8.94 kN-m/m

BM_{2x} = (47.45-29.1)*0.975*0.975/2-0.5*(-47.08+47.45)*0.975*0.975*0.67 = 8.61 kN-m/m

SF_{3x} = (47.78-29.1)*(0.975-0.294)+0.5*(48.04-47.78)*(0.975-0.294) = 12.81 kN/m

$S.F_{(punching)}$ = (6.25-0.73)*(47.56-29.1) = 101.87 kN

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	33.64	0.39	0.52	0.001	366.42	529.20	12ϕT	213.71	300	150	753.98
At Top	-10.74	-0.12	0.17	0.000	116.30	264.60	12ϕT	427.43	300	300	376.99

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 48.14 kN/m

Shear stress , τ_v= (48.14*1000)/(1000*294)

= 0.16 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 366.66 kN

Strength of Concrete in shear Vc1 = 3161.89 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1846.01 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1987.47 kN **Table 22.6.5.2_ (c)**

Design Vc= 1846.01 kN

Allowable shear stress, (ϕ Vc)/bd = 1.37 N/mm^2

Design punching shear stress, τ = (366.66*1000)/(3428*294)

= 0.36 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	47.58	47.54	47.60	8.36	8.30	12.37
102	1.4DL+1.4FL2	47.92	60.38	47.94	8.51	17.69	12.60
103	1.2DL+1.2FL1+1.6LL	52.59	63.35	52.60	10.62	18.21	15.72
104	1.2DL+1.2FL2+1.6LL	52.88	64.72	52.90	10.75	19.56	15.91
105	1.2DL+1.2FL1+1.2TL+1.6LL	51.42	63.70	51.49	10.15	19.48	15.02
106	1.2DL+1.2FL2+1.2TL+1.6LL	51.72	63.47	51.80	10.29	19.07	15.22
107	1.4DL+1.4FL1+1.4PL	96.63	87.03	97.99	31.86	22.65	47.10
108	1.4DL+1.4FL2+1.4PL	96.97	64.86	98.33	32.01	3.93	47.33
109	1.2DL+1.2FL1+1.6LL+1.2PL	94.63	77.68	95.80	30.76	16.73	45.49
110	1.2DL+1.2FL2+1.6LL+1.2PL	94.93	72.80	96.10	30.90	10.05	45.68
111	1.2DL+1.2FL1+1.2TL+1.6LL+	93.30	70.93	94.38	30.07	8.90	44.46
112	1.2DL+1.2FL2+1.2TL+1.6LL+	93.59	63.83	94.67	30.20	3.85	44.65

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113 1.2DL+0.8WLX1	29.87	41.51	29.90	0.38	11.42	0.56
114 1.2DL+0.8WLX2	58.59	54.30	58.61	13.33	7.37	19.73
115 1.2DL+0.8WLZ1	46.01	43.37	46.60	8.23	4.87	12.15
116 1.2DL+0.8WLZ2	43.55	37.93	44.09	7.07	2.62	10.44
117 1.2DL+1.6WLX1+0.5LL	17.88	30.54	17.92	-5.02	6.99	-7.44
118 1.2DL+1.6WLX2+0.5LL	75.35	62.79	75.36	20.88	8.61	30.91
119 1.2DL+1.6WLZ1+0.5LL	50.16	43.86	51.31	10.68	6.32	15.75
120 1.2DL+1.6WLZ2+0.5LL	45.30	37.19	46.40	8.43	2.59	12.43
121 1.2DL+1.0SLX1+0.5LL+0.5FL	34.07	40.68	34.10	2.28	8.57	3.37
122 1.2DL+1.0SLX2+0.5LL+0.5FL	59.76	53.22	59.77	13.84	7.26	20.49
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	45.95	44.49	46.18	7.84	6.49	11.59
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	48.28	41.71	48.46	8.84	2.91	13.08
125 0.9DL+1.6WLX1	11.95	29.95	11.99	-7.71	9.32	-11.41
126 0.9DL+1.6WLX2	69.42	61.87	69.42	18.20	9.61	26.94
127 0.9DL+1.6WLZ1	44.23	41.45	45.38	8.00	5.93	11.78
128 0.9DL+1.6WLZ2	39.39	33.15	40.51	5.78	1.71	8.50
129 0.9DL+1.0SLX1	27.78	39.86	27.81	-0.57	10.87	-0.84
130 0.9DL+1.0SLX2	53.48	52.07	53.49	11.01	8.23	16.29
131 0.9DL+1.0SLZ1	39.67	41.85	39.90	5.00	6.07	7.39
132 0.9DL+1.0SLZ2	42.00	40.35	42.19	6.02	4.73	8.90

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 47.64-(47.64-47.48)*0.95/2.5= 47.58 kN/m^2

p_{2z} = 47.48+(47.64-47.48)*0.95/2.5= 47.54 kN/m^2 Overhang = 0.95 m

p_{3z} = 47.64-(47.64-47.48)*(0.95-0.282)/2.5= 47.60 kN/m^2 Effective depth = 0.282 m

BM_{1z} = (47.58-29.1)*0.95*0.95/2+0.5*(47.64-47.58)*0.95*0.95*0.67 = 8.36 kN-m/m

BM_{2z} = (47.54-29.1)*0.95*0.95/2-0.5*(-47.48+47.54)*0.95*0.95*0.67 = 8.30 kN-m/m

SF_{3z} = (47.60-29.1)*(0.95-0.282)+0.5*(47.64-47.60)*(0.95-0.282) = 12.37 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	32.01	0.403	0.537	0.0013	379.19	507.60	12ϕT	222.81	300	150	754.0
Top	-7.71	-0.097	-0.129	-0.0003	-0.87	253.80	12ϕT	445.62	300	300	377.0

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 47.33 kN/m

Shear stress , τ_v= (47.33*1000)/(1000*282)

= 0.17 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **300** c/c

Along Z Direction :

At Bottom **12ϕT** **150** c/c

At Top **12ϕT** **300** c/c

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DESIGN OF FOUNDATION MKD. F4 (STAAD Node -7852 to 7855)

Size of Columns		Along-X	Along-Z
Node	7854	0.55m	0.60m
Node	7855	0.55m	0.60m
Node	7852	0.80m	0.60m
Node	7853	0.55m	0.65m
Size of fdn. raft:		7.20m	7.20m
Thickness of raft:		0.60 m	
Original Gr. Level:		0.00 m	
Finished Gr. Level:		0.00 m	
U/S of B/Pl Above GL		0.35 m	
U/S level of fdn:		-1.50 m	
Unit weight of conc:		25 kN/m^3	
Unit weight of soil:		18 kN/m^3	

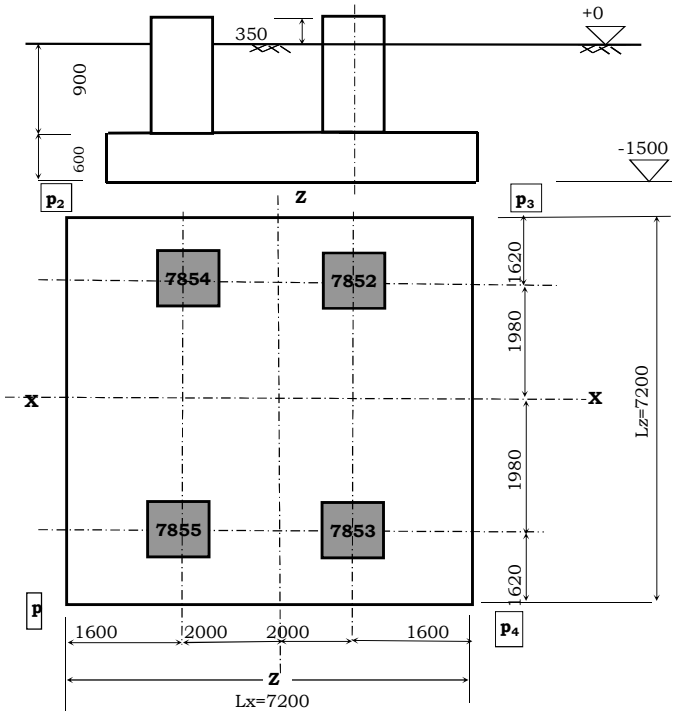
Calculation of Weight of Foundation :

Weight of raft = (7.2*7.2*0.6*25) = 777.60KN

Weight of soil = ((7.2*7.2)-3*((0.55*0.6)+(0.8*0.6))*0.9* 815.99KN

Total = 1593.59KN

Submerged wt. = 1593.59-7.2*7.2*1.5*10 = 815.99KN

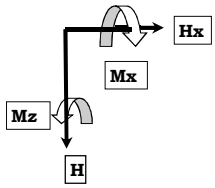


Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 534mm = 0.534m

Effective depth of slab along Z direction = 542mm = 0.542m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding:

Node	7854					7852				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	122.798	43.614	0	1.258	0	858.599	-22.412	0	6.175	0
202 1.0DL+1.0FL2	136.143	50.695	0	1.43	0	1005.427	-25.502	0	6.456	0
203 1.0DL+1.0FL1+1.0LL+1.0TL	390.014	140.657	0	3.583	0	708.334	-21.688	0	23.511	0
204 1.0DL+1.0FL2+1.0LL+1.0TL	403.36	147.738	0	3.754	0	855.162	-24.778	0	23.793	0
205 1.0DL+1.0FL1+0.75LL+0.75TL	323.21	116.397	0	3.002	0	745.9	-21.869	0	19.177	0
206 1.0DL+1.0FL2+0.75LL+0.75TL	336.555	123.477	0	3.173	0	892.728	-24.959	0	19.459	0
207 1.0DL+1.0FL1+1.0PL	168.228	59.602	0	-16.167	0	646.608	-5.482	0	-23.946	0
208 1.0DL+1.0FL2+1.0PL	181.573	66.682	0	-15.995	0	793.436	-8.572	0	-23.665	0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1	435.444	156.645	0	-13.843	0	496.343	-4.759	0	-6.61	0
210 1.0DL+1.0FL2+1.0LL+1.0TL+1	448.789	163.726	0	-13.671	0	643.171	-7.848	0	-6.328	0
211 1.0DL+1.0FL1+0.75LL+0.75TL	368.64	132.384	0	-14.424	0	533.909	-4.94	0	-10.944	0
212 1.0DL+1.0FL2+0.75LL+0.75TL	381.985	139.465	0	-14.252	0	680.737	-8.029	0	-10.663	0
213 1.0DL+1.0FL1+1.0WLX1	-20.974	-0.987	0	1.525	0	1000.977	-47.278	0	7.829	0
214 1.0DL+1.0FL1+1.0WLX2	267.064	88.328	0	0.99	0	715.744	2.507	0	4.517	0
215 1.0DL+1.0FL1+1.0WLZ1	-53.164	12.15	0	-16.897	0	730.555	-17.792	0	-37.68	0
216 1.0DL+1.0FL1+1.0WLZ2	298.76	75.078	0	19.414	0	986.643	-27.032	0	50.03	0
217 1.0DL+1.0FL2+1.0WLX1	-7.629	6.094	0	1.697	0	1147.805	-50.367	0	8.11	0
218 1.0DL+1.0FL2+1.0WLX2	280.409	95.409	0	1.162	0	862.572	-0.583	0	4.798	0
219 1.0DL+1.0FL2+1.0WLZ1	-39.819	19.231	0	-16.726	0	877.383	-20.882	0	-37.399	0
220 1.0DL+1.0FL2+1.0WLZ2	312.105	82.158	0	19.586	0	1133.471	-30.122	0	50.312	0
221 1.0DL+1.0FL1+0.7SLX1	9.028	-12.583	0	1.556	0	975.366	-57.559	0	8.034	0
222 1.0DL+1.0FL1+0.7SLX2	236.567	99.81	0	0.96	0	741.832	12.734	0	4.316	0
223 1.0DL+1.0FL1+0.7SLZ1	62.196	25.641	0	-3.874	0	765.927	-24.4	0	-39.236	0
224 1.0DL+1.0FL1+0.7SLZ2	183.399	61.587	0	6.39	0	951.272	-20.424	0	51.586	0
225 1.0DL+1.0FL2+0.7SLX1	22.374	-5.502	0	1.728	0	1122.194	-60.648	0	8.315	0
226 1.0DL+1.0FL2+0.7SLX2	249.912	106.891	0	1.132	0	888.66	9.645	0	4.598	0
227 1.0DL+1.0FL2+0.7SLZ1	75.542	32.722	0	-3.702	0	912.755	-27.49	0	-38.955	0
228 1.0DL+1.0FL2+0.7SLZ2	196.744	68.667	0	6.562	0	1098.099	-23.514	0	51.867	0
229 1.0DL+1.0FL1+0.75WLX1+0.7	32.476	13.483	0	2.193	0	1021.423	-43.02	0	12.824	0
230 1.0DL+1.0FL1+0.75WLX2+0.7	248.505	80.469	0	1.792	0	807.498	-5.682	0	10.34	0
231 1.0DL+1.0FL1+0.75WLZ1+0.7	8.334	23.336	0	-11.624	0	818.606	-20.906	0	-21.308	0
232 1.0DL+1.0FL1+0.75WLZ2+0.7	272.276	70.531	0	15.61	0	1010.672	-27.836	0	44.475	0
233 1.0DL+1.0FL2+0.75WLX1+0.7	45.821	20.564	0	2.365	0	1168.25	-46.11	0	13.105	0
234 1.0DL+1.0FL2+0.75WLX2+0.7	261.85	87.55	0	1.964	0	954.326	-8.771	0	10.621	0
235 1.0DL+1.0FL2+0.75WLZ1+0.7	21.679	30.417	0	-11.452	0	965.434	-23.995	0	-21.027	0
236 1.0DL+1.0FL2+0.75WLZ2+0.7	285.622	77.612	0	15.782	0	1157.5	-30.926	0	44.756	0
237 1.0DL+1.0FL1+0.525SLX1+0.1	54.978	4.786	0	2.216	0	1002.214	-50.731	0	12.977	0
238 1.0DL+1.0FL1+0.525SLX2+0.1	225.632	89.081	0	1.77	0	827.064	1.989	0	10.19	0
239 1.0DL+1.0FL1+0.525SLZ1+0.7	94.854	33.454	0	-1.856	0	845.135	-25.862	0	-22.475	0
240 1.0DL+1.0FL1+0.525SLZ2+0.7	185.756	60.413	0	5.842	0	984.144	-22.88	0	45.642	0
241 1.0DL+1.0FL2+0.525SLX1+0.1	68.324	11.867	0	2.388	0	1149.042	-53.82	0	13.259	0
242 1.0DL+1.0FL2+0.525SLX2+0.1	238.977	96.162	0	1.941	0	973.892	-1.101	0	10.471	0
243 1.0DL+1.0FL2+0.525SLZ1+0.7	108.2	40.535	0	-1.684	0	991.963	-28.952	0	-22.193	0
244 1.0DL+1.0FL2+0.525SLZ2+0.7	199.101	67.494	0	6.014	0	1130.971	-25.97	0	45.923	0
245 0.6DL+1.0WLX1	-104.618	-36.562	0	0.578	0	284.803	-30.512	0	4.638	0
246 0.6DL+1.0WLX2	183.42	52.753	0	0.043	0	-0.43	19.272	0	1.327	0
247 0.6DL+1.0WLZ1	-136.807	-23.425	0	-17.845	0	14.381	-1.027	0	-40.871	0
248 0.6DL+1.0WLZ2	215.116	39.502	0	18.467	0	270.469	-10.267	0	46.84	0
249 0.6DL+0.6FL1+0.7SLX1	-40.091	-30.028	0	1.053	0	631.927	-48.594	0	5.564	0
250 0.6DL+0.6FL1+0.7SLX2	187.448	82.365	0	0.457	0	398.393	21.699	0	1.846	0
251 0.6DL+0.6FL1+0.7SLZ1	13.077	8.196	0	-4.377	0	422.487	-15.435	0	-41.706	0
252 0.6DL+0.6FL1+0.7SLZ2	134.28	44.141	0	5.887	0	607.832	-11.459	0	49.116	0

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Node	7855					7853				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	46.194	12.737	0	0.349	0	104.106	-10.966	0	-5.418	0
202 1.0DL+1.0FL2	41.438	11.934	0	0.522	0	106.654	-11.186	0	-5.58	0
203 1.0DL+1.0FL1+1.0LL+1.0TL	315.342	132.172	0	-0.452	0	-116.986	26.027	0	-22.092	0
204 1.0DL+1.0FL2+1.0LL+1.0TL	310.585	131.369	0	-0.279	0	-114.437	25.807	0	-22.254	0
205 1.0DL+1.0FL1+0.75LL+0.75TL	248.055	102.314	0	-0.252	0	-61.713	16.779	0	-17.923	0
206 1.0DL+1.0FL2+0.75LL+0.75TL	243.298	101.51	0	-0.079	0	-59.165	16.559	0	-18.085	0
207 1.0DL+1.0FL1+1.0PL	447.729	82.263	0	-17.005	0	86.594	-1.188	0	-32.848	0
208 1.0DL+1.0FL2+1.0PL	442.972	81.459	0	-16.832	0	89.142	-1.408	0	-33.01	0
209 1.0DL+1.0FL1+1.0LL+1.0TL+1	716.876	201.698	0	-17.805	0	-134.497	35.806	0	-49.522	0
210 1.0DL+1.0FL2+1.0LL+1.0TL+1	712.119	200.895	0	-17.632	0	-131.949	35.585	0	-49.684	0
211 1.0DL+1.0FL1+0.75LL+0.75TL	649.589	171.839	0	-17.605	0	-79.224	26.557	0	-45.354	0
212 1.0DL+1.0FL2+0.75LL+0.75TL	644.832	171.036	0	-17.432	0	-76.676	26.337	0	-45.516	0
213 1.0DL+1.0FL1+1.0WLX1	-71.074	-15.378	0	0.588	0	208.791	-34.459	0	-5.518	0
214 1.0DL+1.0FL1+1.0WLX2	163.86	40.923	0	0.109	0	-0.932	12.587	0	-5.316	0
215 1.0DL+1.0FL1+1.0WLZ1	227.324	44.18	0	-17.804	0	227.103	-21.554	0	-49.413	0
216 1.0DL+1.0FL1+1.0WLZ2	-134.936	-18.706	0	18.501	0	-18.891	-0.378	0	38.577	0
217 1.0DL+1.0FL2+1.0WLX1	-75.831	-16.182	0	0.761	0	211.34	-34.679	0	-5.68	0
218 1.0DL+1.0FL2+1.0WLX2	159.103	40.12	0	0.282	0	1.616	12.366	0	-5.478	0
219 1.0DL+1.0FL2+1.0WLZ1	222.568	43.377	0	-17.631	0	229.651	-21.774	0	-49.575	0
220 1.0DL+1.0FL2+1.0WLZ2	-139.693	-19.509	0	18.674	0	-16.343	-0.599	0	38.415	0
221 1.0DL+1.0FL1+0.7SLX1	-13.488	-1.609	0	0.637	0	153.488	-21.679	0	-4.481	0
222 1.0DL+1.0FL1+0.7SLX2	105.877	27.083	0	0.06	0	54.723	-0.253	0	-6.355	0
223 1.0DL+1.0FL1+0.7SLZ1	103.097	25.445	0	-4.78	0	200.51	-15.381	0	-52.631	0
224 1.0DL+1.0FL1+0.7SLZ2	-10.708	0.029	0	5.477	0	7.701	-6.551	0	41.795	0
225 1.0DL+1.0FL2+0.7SLX1	-18.245	-2.413	0	0.81	0	156.036	-21.899	0	-4.643	0
226 1.0DL+1.0FL2+0.7SLX2	101.12	26.28	0	0.233	0	57.272	-0.474	0	-6.517	0
227 1.0DL+1.0FL2+0.7SLZ1	98.34	24.642	0	-4.607	0	203.058	-15.601	0	-52.793	0
228 1.0DL+1.0FL2+0.7SLZ2	-15.465	-0.775	0	5.651	0	10.249	-6.771	0	41.633	0
229 1.0DL+1.0FL1+0.75WLX1+0.7	-29.922	-5.476	0	-0.116	0	199.82	-29.867	0	-11.34	0
230 1.0DL+1.0FL1+0.75WLX2+0.7	146.279	36.75	0	-0.475	0	42.527	5.417	0	-11.189	0
231 1.0DL+1.0FL1+0.75WLZ1+0.7	193.877	39.193	0	-13.91	0	213.554	-20.188	0	-44.261	0
232 1.0DL+1.0FL1+0.75WLZ2+0.7	-77.818	-7.971	0	13.319	0	29.058	-4.306	0	21.731	0
233 1.0DL+1.0FL2+0.75WLX1+0.7	-34.678	-6.279	0	0.057	0	202.368	-30.087	0	-11.502	0
234 1.0DL+1.0FL2+0.75WLX2+0.7	141.522	35.947	0	-0.302	0	45.076	5.197	0	-11.351	0
235 1.0DL+1.0FL2+0.75WLZ1+0.7	189.12	38.39	0	-13.737	0	216.102	-20.408	0	-44.424	0
236 1.0DL+1.0FL2+0.75WLZ2+0.7	-82.575	-8.775	0	13.492	0	31.606	-4.527	0	21.569	0
237 1.0DL+1.0FL1+0.525SLX1+0.1	13.268	4.851	0	-0.079	0	158.343	-20.282	0	-10.562	0
238 1.0DL+1.0FL1+0.525SLX2+0.1	102.791	26.371	0	-0.512	0	84.269	-4.213	0	-11.968	0
239 1.0DL+1.0FL1+0.525SLZ1+0.7	100.706	25.142	0	-4.142	0	193.609	-15.558	0	-46.675	0
240 1.0DL+1.0FL1+0.525SLZ2+0.7	15.353	6.08	0	3.551	0	49.002	-8.936	0	24.144	0
241 1.0DL+1.0FL2+0.525SLX1+0.1	8.511	4.048	0	0.094	0	160.891	-20.502	0	-10.724	0
242 1.0DL+1.0FL2+0.525SLX2+0.1	98.035	25.567	0	-0.339	0	86.817	-4.433	0	-12.13	0
243 1.0DL+1.0FL2+0.525SLZ1+0.7	95.95	24.339	0	-3.969	0	196.157	-15.779	0	-46.837	0
244 1.0DL+1.0FL2+0.525SLZ2+0.7	10.596	5.276	0	3.724	0	51.551	-9.156	0	23.982	0
245 0.6DL+1.0WLX1	-77.883	-18.514	0	0	0	161.162	-29.598	0	-2.951	0
246 0.6DL+1.0WLX2	157.051	37.787	0	-0.479	0	-48.561	17.448	0	-2.75	0
247 0.6DL+1.0WLZ1	220.516	41.044	0	-18.391	0	179.474	-16.693	0	-46.846	0
248 0.6DL+1.0WLZ2	-141.745	-21.842	0	17.914	0	-66.521	4.483	0	41.143	0
249 0.6DL+0.6FL1+0.7SLX1	-31.966	-6.704	0	0.498	0	111.846	-17.292	0	-2.313	0
250 0.6DL+0.6FL1+0.7SLX2	87.399	21.989	0	-0.079	0	13.081	4.133	0	-4.188	0
251 0.6DL+0.6FL1+0.7SLZ1	84.619	20.351	0	-4.92	0	158.868	-10.994	0	-50.463	0
252 0.6DL+0.6FL1+0.7SLZ2	-29.186	-5.066	0	5.338	0	-33.941	-2.165	0	43.962	0

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MA'ADEN Saudi Arabian Mining Company معادن شركة التعدين العربية السعودية

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

	Load at top of foundation					Load at bottom of foundation				
L/C	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0FL1	1131.7	23.0	1587.4	2.4	1645.6	2725.3	23.0	1629.9	2.4	1649.9
202 1.0DL+1.0FL2	1289.7	25.9	1869.0	2.8	1967.1	2883.3	25.9	1917.0	2.8	1972.3
203 1.0DL+1.0FL1+1.0LL+1.0TL	1296.7	277.2	-228.0	4.6	1782.0	2890.3	277.2	284.7	4.6	1790.4
204 1.0DL+1.0FL2+1.0LL+1.0TL	1454.7	280.1	53.6	5.0	2103.5	3048.3	280.1	571.8	5.0	2112.8
205 1.0DL+1.0FL1+0.75LL+0.75TL	1255.5	213.6	225.8	4.0	1747.9	2849.0	213.6	621.0	4.0	1755.3
206 1.0DL+1.0FL2+0.75LL+0.75TL	1413.4	216.6	507.4	4.5	2069.4	3007.0	216.6	908.1	4.5	2077.7
207 1.0DL+1.0FL1+1.0PL	1349.2	135.2	234.5	-90.0	555.4	2942.7	135.2	484.6	-90.0	389.0
208 1.0DL+1.0FL2+1.0PL	1507.1	138.2	516.1	-89.5	876.9	3100.7	138.2	771.7	-89.5	711.4
209 1.0DL+1.0FL1+1.0LL+1.0TL+1	1514.2	389.4	-1580.9	-87.8	691.8	3107.8	389.4	-860.6	-87.8	529.4
210 1.0DL+1.0FL2+1.0LL+1.0TL+1	1672.1	392.4	-1299.4	-87.3	1013.3	3265.7	392.4	-573.5	-87.3	851.8
211 1.0DL+1.0FL1+0.75LL+0.75TL	1472.9	325.8	-1127.1	-88.3	657.7	3066.5	325.8	-524.3	-88.3	494.3
212 1.0DL+1.0FL2+0.75LL+0.75TL	1630.9	328.8	-845.5	-87.9	979.2	3224.5	328.8	-237.2	-87.9	816.7
213 1.0DL+1.0FL1+1.0WLX1	1117.7	-98.1	2603.6	4.4	1667.7	2711.3	-98.1	2422.1	4.4	1675.9
214 1.0DL+1.0FL1+1.0WLX2	1145.7	144.3	567.8	0.3	1623.4	2739.3	144.3	834.8	0.3	1623.9
215 1.0DL+1.0FL1+1.0WLZ1	1131.8	17.0	1567.0	-121.8	441.5	2725.4	17.0	1598.4	-121.8	216.1
216 1.0DL+1.0FL1+1.0WLZ2	1131.6	29.0	1607.9	126.5	2849.7	2725.2	29.0	1661.4	126.5	3083.7
217 1.0DL+1.0FL2+1.0WLX1	1275.7	-95.1	2885.2	4.9	1989.2	2869.3	-95.1	2709.2	4.9	1998.3
218 1.0DL+1.0FL2+1.0WLX2	1303.7	147.3	849.4	0.8	1944.9	2897.3	147.3	1121.9	0.8	1946.3
219 1.0DL+1.0FL2+1.0WLZ1	1289.8	20.0	1848.6	-121.3	763.0	2883.4	20.0	1885.5	-121.3	538.5
220 1.0DL+1.0FL2+1.0WLZ2	1289.5	31.9	1889.4	127.0	3171.2	2883.1	31.9	1948.5	127.0	3406.1
221 1.0DL+1.0FL1+0.7SLX1	1124.4	-93.4	2266.6	5.7	1671.9	2718.0	-93.4	2093.8	5.7	1682.5
222 1.0DL+1.0FL1+0.7SLX2	1139.0	139.4	908.2	-1.0	1619.2	2732.6	139.4	1166.1	-1.0	1617.4
223 1.0DL+1.0FL1+0.7SLZ1	1131.7	11.3	1602.3	-100.5	1038.5	2725.3	11.3	1623.2	-100.5	852.6
224 1.0DL+1.0FL1+0.7SLZ2	1131.7	34.6	1572.6	105.2	2252.6	2725.3	34.6	1636.6	105.2	2447.3
225 1.0DL+1.0FL2+0.7SLX1	1282.4	-90.5	2548.2	6.2	1993.4	2875.9	-90.5	2380.8	6.2	2004.9
226 1.0DL+1.0FL2+0.7SLX2	1297.0	142.3	1189.8	-0.6	1940.8	2890.6	142.3	1453.1	-0.6	1939.7
227 1.0DL+1.0FL2+0.7SLZ1	1289.7	14.3	1883.9	-100.1	1360.1	2883.3	14.3	1910.3	-100.1	1175.0
228 1.0DL+1.0FL2+0.7SLZ2	1289.6	37.6	1854.1	105.7	2574.1	2883.2	37.6	1923.7	105.7	2769.7
229 1.0DL+1.0FL1+0.75WLX1+0.7	1223.8	-64.9	2437.4	3.6	1750.3	2817.4	-64.9	2317.4	3.6	1756.9
230 1.0DL+1.0FL1+0.75WLX2+0.7	1244.8	117.0	910.5	0.5	1717.1	2838.4	117.0	1126.8	0.5	1717.9
231 1.0DL+1.0FL1+0.75WLZ1+0.7	1234.4	21.4	1659.9	-91.1	830.6	2828.0	21.4	1699.6	-91.1	662.1
232 1.0DL+1.0FL1+0.75WLZ2+0.7	1234.2	30.4	1690.5	95.1	2636.8	2827.8	30.4	1746.8	95.1	2812.8
233 1.0DL+1.0FL2+0.75WLX1+0.7	1381.8	-61.9	2719.0	4.0	2071.8	2975.4	-61.9	2604.4	4.0	2079.3
234 1.0DL+1.0FL2+0.75WLX2+0.7	1402.8	119.9	1192.1	0.9	2038.6	2996.4	119.9	1413.9	0.9	2040.3
235 1.0DL+1.0FL2+0.75WLZ1+0.7	1392.3	24.4	1941.5	-90.6	1152.1	2985.9	24.4	1986.6	-90.6	984.5
236 1.0DL+1.0FL2+0.75WLZ2+0.7	1392.2	33.4	1972.1	95.6	2958.3	2985.7	33.4	2033.9	95.6	3135.2
237 1.0DL+1.0FL1+0.525SLX1+0.7	1228.8	-61.4	2184.6	4.6	1753.5	2822.4	-61.4	2071.1	4.6	1761.9
238 1.0DL+1.0FL1+0.525SLX2+0.7	1239.8	113.2	1165.8	-0.5	1714.0	2833.3	113.2	1375.3	-0.5	1713.0
239 1.0DL+1.0FL1+0.525SLZ1+0.7	1234.3	17.2	1686.4	-75.1	1278.4	2827.9	17.2	1718.1	-75.1	1139.4
240 1.0DL+1.0FL1+0.525SLZ2+0.7	1234.3	34.7	1664.1	79.2	2189.0	2827.8	34.7	1728.2	79.2	2335.5
241 1.0DL+1.0FL2+0.525SLX1+0.7	1386.8	-58.4	2466.2	5.0	2075.0	2980.4	-58.4	2358.1	5.0	2084.3
242 1.0DL+1.0FL2+0.525SLX2+0.7	1397.7	116.2	1447.4	-0.1	2035.5	2991.3	116.2	1662.4	-0.1	2035.4
243 1.0DL+1.0FL2+0.525SLZ1+0.7	1392.3	20.1	1967.9	-74.7	1600.0	2985.9	20.1	2005.2	-74.7	1461.8
244 1.0DL+1.0FL2+0.525SLZ2+0.7	1392.2	37.6	1945.7	79.6	2510.5	2985.8	37.6	2015.3	79.6	2657.8
245 0.6DL+1.0WLX1	263.5	-115.2	1256.9	2.3	191.9	1857.1	-115.2	1043.8	2.3	196.1
246 0.6DL+1.0WLX2	291.5	127.3	-778.9	-1.9	147.5	1885.1	127.3	-543.5	-1.9	144.1
247 0.6DL+1.0WLZ1	277.6	-0.1	220.3	-124.0	-1034.4	1871.2	-0.1	220.1	-124.0	-1263.7
248 0.6DL+1.0WLZ2	277.3	11.9	261.2	124.4	1373.8	1870.9	11.9	283.1	124.4	1603.9
249 0.6DL+0.6FL1+0.7SLX1	671.7	-102.6	1631.7	4.8	1013.7	2265.3	-102.6	1441.8	4.8	1022.6
250 0.6DL+0.6FL1+0.7SLX2	686.3	130.2	273.3	-2.0	961.0	2279.9	130.2	514.1	-2.0	957.4
251 0.6DL+0.6FL1+0.7SLZ1	679.1	2.1	967.3	-101.5	380.3	2272.6	2.1	971.2	-101.5	192.6
252 0.6DL+0.6FL1+0.7SLZ2	679.0	25.5	937.6	104.3	1594.4	2272.6	25.5	984.7	104.3	1787.3

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MA'ADEN Saudi Arabian Mining Company معادن شركة التعدين العربية السعودية

Outotec LARSEN & TOUBRO

MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

L/C	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201 1.0DL+1.0FL1	2725.3	23.0	1629.9	2.4	1649.9	6.0	5.9	47.5	461.1
202 1.0DL+1.0FL2	2883.3	25.9	1917.0	2.8	1972.3	5.4	5.3	44.5	407.8
203 1.0DL+1.0FL1+1.0LL+1.0T	2890.3	277.2	284.7	4.6	1790.4	36.5	5.8	4.2	254.1
204 1.0DL+1.0FL2+1.0LL+1.0T	3048.3	280.1	571.8	5.0	2112.8	19.2	5.2	4.4	243.2
205 1.0DL+1.0FL1+0.75LL+0.7	2849.0	213.6	621.0	4.0	1755.3	16.5	5.8	5.3	284.6
206 1.0DL+1.0FL2+0.75LL+0.7	3007.0	216.6	908.1	4.5	2077.7	11.9	5.2	5.6	269.2
207 1.0DL+1.0FL1+1.0PL	2942.7	135.2	484.6	-90.0	389.0	21.9	27.2	8.7	13.1
208 1.0DL+1.0FL2+1.0PL	3100.7	138.2	771.7	-89.5	711.4	14.5	15.7	9.0	13.9
209 1.0DL+1.0FL1+1.0LL+1.0T	3107.8	389.4	-860.6	-87.8	529.4	13.0	21.1	3.2	14.2
210 1.0DL+1.0FL2+1.0LL+1.0T	3265.7	392.4	-573.5	-87.3	851.8	20.5	13.8	3.3	15.0
211 1.0DL+1.0FL1+0.75LL+0.7	3066.5	325.8	-524.3	-88.3	494.3	21.1	22.3	3.8	13.9
212 1.0DL+1.0FL2+0.75LL+0.7	3224.5	328.8	-237.2	-87.9	816.7	48.9	14.2	3.9	14.7
213 1.0DL+1.0FL1+1.0WLX1	2711.3	-98.1	2422.1	4.4	1675.9	4.0	5.8	11.1	245.1
214 1.0DL+1.0FL1+1.0WLX2	2739.3	144.3	834.8	0.3	1623.9	11.8	6.1	7.6	3652.4
215 1.0DL+1.0FL1+1.0WLZ1	2725.4	17.0	1598.4	-121.8	216.1	6.1	45.4	64.2	9.0
216 1.0DL+1.0FL1+1.0WLZ2	2725.2	29.0	1661.4	126.5	3083.7	5.9	3.2	37.6	8.6
217 1.0DL+1.0FL2+1.0WLX1	2869.3	-95.1	2709.2	4.9	1998.3	3.8	5.2	12.1	234.8
218 1.0DL+1.0FL2+1.0WLX2	2897.3	147.3	1121.9	0.8	1946.3	9.3	5.4	7.9	1516.9
219 1.0DL+1.0FL2+1.0WLZ1	2883.4	20.0	1885.5	-121.3	538.5	5.5	19.3	57.8	9.5
220 1.0DL+1.0FL2+1.0WLZ2	2883.1	31.9	1948.5	127.0	3406.1	5.3	3.0	36.1	9.1
221 1.0DL+1.0FL1+0.7SLX1	2718.0	-93.4	2093.8	5.7	1682.5	4.7	5.8	11.6	189.2
222 1.0DL+1.0FL1+0.7SLX2	2732.6	139.4	1166.1	-1.0	1617.4	8.4	6.1	7.8	1072.7
223 1.0DL+1.0FL1+0.7SLZ1	2725.3	11.3	1623.2	-100.5	852.6	6.0	11.5	96.4	10.8
224 1.0DL+1.0FL1+0.7SLZ2	2725.3	34.6	1636.6	105.2	2447.3	6.0	4.0	31.5	10.4
225 1.0DL+1.0FL2+0.7SLX1	2875.9	-90.5	2380.8	6.2	2004.9	4.3	5.2	12.7	185.2
226 1.0DL+1.0FL2+0.7SLX2	2890.6	142.3	1453.1	-0.6	1939.7	7.2	5.4	8.1	2087.0
227 1.0DL+1.0FL2+0.7SLZ1	2883.3	14.3	1910.3	-100.1	1175.0	5.4	8.8	80.8	11.5
228 1.0DL+1.0FL2+0.7SLZ2	2883.2	37.6	1923.7	105.7	2769.7	5.4	3.7	30.7	10.9
229 1.0DL+1.0FL1+0.75WLX1+	2817.4	-64.9	2317.4	3.6	1756.9	4.4	5.8	17.4	316.5
230 1.0DL+1.0FL1+0.75WLX2+	2838.4	117.0	1126.8	0.5	1717.9	9.1	5.9	9.7	2426.0
231 1.0DL+1.0FL1+0.75WLZ1+	2828.0	21.4	1699.6	-91.1	662.1	6.0	15.4	52.8	12.4
232 1.0DL+1.0FL1+0.75WLZ2+	2827.8	30.4	1746.8	95.1	2812.8	5.8	3.6	37.2	11.9
233 1.0DL+1.0FL2+0.75WLX1+	2975.4	-61.9	2604.4	4.0	2079.3	4.1	5.2	19.2	295.7
234 1.0DL+1.0FL2+0.75WLX2+	2996.4	119.9	1413.9	0.9	2040.3	7.6	5.3	10.0	1286.0
235 1.0DL+1.0FL2+0.75WLZ1+	2985.9	24.4	1986.6	-90.6	984.5	5.4	10.9	48.9	13.2
236 1.0DL+1.0FL2+0.75WLZ2+	2985.7	33.4	2033.9	95.6	3135.2	5.3	3.4	35.8	12.5
237 1.0DL+1.0FL1+0.525SLX1-	2822.4	-61.4	2071.1	4.6	1761.9	4.9	5.8	18.4	248.0
238 1.0DL+1.0FL1+0.525SLX2-	2833.3	113.2	1375.3	-0.5	1713.0	7.4	6.0	10.0	2179.5
239 1.0DL+1.0FL1+0.525SLZ1+	2827.9	17.2	1718.1	-75.1	1139.4	5.9	8.9	65.9	15.1
240 1.0DL+1.0FL1+0.525SLZ2+	2827.8	34.7	1728.2	79.2	2335.5	5.9	4.4	32.6	14.3
241 1.0DL+1.0FL2+0.525SLX1-	2980.4	-58.4	2358.1	5.0	2084.3	4.5	5.1	20.4	237.6
242 1.0DL+1.0FL2+0.525SLX2-	2991.3	116.2	1662.4	-0.1	2035.4	6.5	5.3	10.3	20991.7
243 1.0DL+1.0FL2+0.525SLZ1+	2985.9	20.1	2005.2	-74.7	1461.8	5.4	7.4	59.3	16.0
244 1.0DL+1.0FL2+0.525SLZ2+	2985.8	37.6	2015.3	79.6	2657.8	5.3	4.0	31.7	15.0
245 0.6DL+1.0WLX1	1857.1	-115.2	1043.8	2.3	196.1	6.4	34.1	6.4	328.0
246 0.6DL+1.0WLX2	1885.1	127.3	-543.5	-1.9	144.1	12.5	47.1	5.9	405.6
247 0.6DL+1.0WLZ1	1871.2	-0.1	220.1	-124.0	1263.7	30.6	5.3	7410.5	6.0
248 0.6DL+1.0WLZ2	1870.9	11.9	283.1	124.4	1603.9	23.8	4.2	63.0	6.0
249 0.6DL+0.6FL1+0.7SLX1	2265.3	-102.6	1441.8	4.8	1022.6	5.7	8.0	8.8	188.7
250 0.6DL+0.6FL1+0.7SLX2	2279.9	130.2	514.1	-2.0	957.4	16.0	8.6	7.0	464.3
251 0.6DL+0.6FL1+0.7SLZ1	2272.6	2.1	971.2	-101.5	192.6	8.4	42.5	429.2	9.0
252 0.6DL+0.6FL1+0.7SLZ2	2272.6	25.5	984.7	104.3	1787.3	8.3	4.6	35.7	8.7
					Min.=	3.8	3.0	3.2	6.0

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Foundation Base Pressure :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	5	201	0	48.765	0
Min Px	5	201	0	48.765	0
Max Py	802	236	0	151.426	0
Min Py	892	220	0	3.642	0
Max Pz	5	201	0	48.765	0
Min Pz	5	201	0	48.765	0

Gross Maximum Base pressure, p_{max} = 151.4 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2 (Bore hole 10)
Depth of soil above footing = 1.5 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 477 KN/m^2
> 151.4 KN/m^2
Hence OK

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Design of Foundation F4:

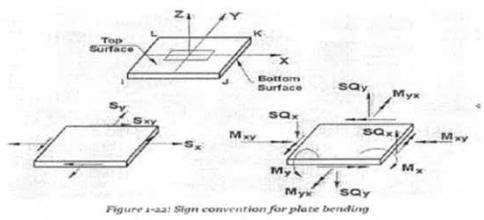
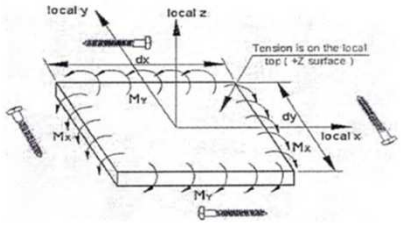


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : FDN_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	120	701.15	107.77	0	0	0	-114.6	-175.547	0.718	-115.272	-176.265	102 1.4DL+1.4FL2
Min Qx	139	-695.4	123.79	0	0	0	-123.4	-186.514	0	-123.449	-186.514	102 1.4DL+1.4FL2
Max Qy	107	1.034	671.86	0	0	0	-194.9	-120.653	-31.784	-226.719	-152.437	102 1.4DL+1.4FL2
Min Qy	72	284.06	-900.1	0	0	0	-147.9	-197.556	-2.514	-150.447	-200.07	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Sx	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Min Sx	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Max Sy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Min Sy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Max Sxy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Min Sxy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Max Mx	181	188.64	-18.31	0	0	0	78.333	37.039	-24.008	102.341	61.047	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Mx	106	48.819	664.22	0	0	0	-197.2	-117.186	4.265	-201.467	-121.451	102 1.4DL+1.4FL2
Max My	23	98.538	-48.35	0	0	0	-41.56	116.84	-48.937	-90.492	165.777	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min My	72	284.06	-900.1	0	0	0	-147.9	-197.556	-2.514	-150.447	-200.07	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Max Mxy	80	-55.01	213.48	0	0	0	-35.69	16.772	42.984	-78.671	59.756	111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
Min Mxy	98	41.152	145.96	0	0	0	-13.09	28.694	-87.531	-100.62	116.225	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL

PLATE CENTER STRESS ; FDN_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	332	345.79	17.491	0	0	0	-23.7	-99.411	-16.059	-39.76	-115.47	102 1.4DL+1.4FL2
Min Qx	722	-293.4	22.89	0	0	0	-31.69	-121.26	0	-31.692	-121.26	102 1.4DL+1.4FL2
Max Qy	582	-51.41	310.3	0	0	0	-133.3	-35.43	-20.144	-153.478	-55.574	102 1.4DL+1.4FL2
Min Qy	214	-37.4	-344	0	0	0	-102.8	-12.606	-11.888	-114.643	-24.494	102 1.4DL+1.4FL2
Max Sx	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Min Sx	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Max Sy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Min Sy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Max Sxy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Min Sxy	880	0.53	0.018	0	0	0	-1.168	-1.143	-1.358	-2.526	-2.501	101 1.4DL+1.4FL1
Max Mx	592	5.571	-71.73	0	0	0	53.292	13.224	-14.641	67.933	27.865	112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
Min Mx	582	-51.41	310.3	0	0	0	-133.3	-35.43	-20.144	-153.478	-55.574	102 1.4DL+1.4FL2
Max My	186	8.447	13.285	0	0	0	-14.64	84.401	-19.152	-33.792	103.553	106 1.2DL+1.2FL2+1.2TL+1.6LL
Min My	722	-293.4	22.89	0	0	0	-31.69	-121.26	-16.727	-48.419	-137.987	102 1.4DL+1.4FL2
Max Mxy	719	-163.7	-189.6	0	0	0	-32.67	-60.536	17.573	-50.24	-78.109	102 1.4DL+1.4FL2
Min Mxy	339	141.69	-150.4	0	0	0	-26.1	-17.667	-63.863	-89.961	-81.53	102 1.4DL+1.4FL2

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

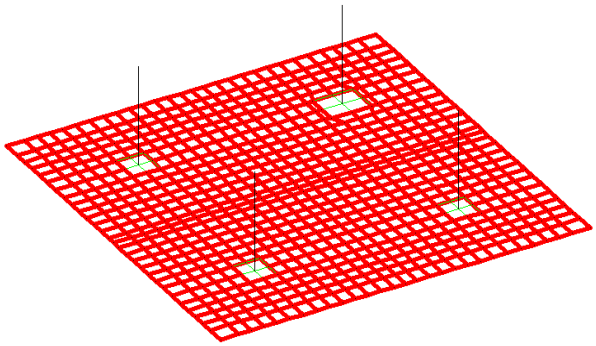
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	$\varepsilon_t = \varepsilon_{cu}/c*(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	102.34	top face	50	542	102.34	0.46	0.0011	604.2	1080	16	186.24	150	1341	18.91	23.64	0.0658	tension control
Min Mx	-201.47	bottom face	50	542	201.47	0.91	0.0022	1198.7		16	167.79	150	1341	18.91	23.64	0.0658	tension control
Max Mxy	-78.67	bottom face	50	542	78.671	0.36	0.0009	463.6		16	186.24	150	1341	18.91	23.64	0.0658	tension control
Min Mxy	-100.62	bottom face	50	542	100.62	0.46	0.0011	593.9		16	186.24	150	1341	18.91	23.64	0.0658	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	345.79	1	207473	39.76	0.002	0.542	64988.3	535.839	929.889	535.84	1989	check	check	401.9	φVc<Vu	required
	Min Qx	293.44	1	176062	31.692			69155.4			535.84	1989	check	check	401.9	φVc<Vu	required

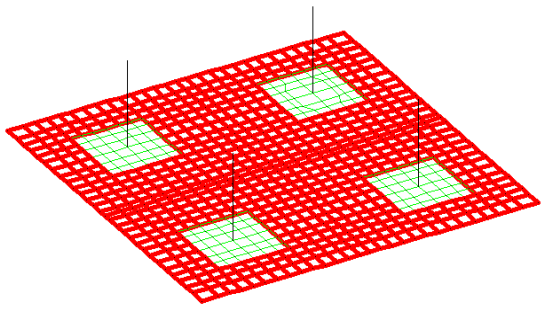
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	$\varepsilon_t = \varepsilon_{cu}/c*(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	165.78	top face	50	542	165.78	0.75	0.0018	983.6	1080	16	186.24	150	1341	18.91	23.64	0.0658	tension control
Min My	-200.07	bottom face	50	542	200.07	0.91	0.0022	1190.3		16	168.98	150	1341	18.91	23.64	0.0658	tension control
Max Mxy	59.76	top face	50	542	59.756	0.27	0.0006	351.6		16	186.24	150	1341	18.91	23.64	0.0658	tension control
Min Mxy	116.23	top face	50	542	116.23	0.53	0.0013	686.9		16	186.24	150	1341	18.91	23.64	0.0658	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	310.3	1	186.18	55.574	0.002	0.542	fc'554.4	fc'535.8	fc'929.9	535.84	561	186.17	ok	401.9	φVc>Vu	not required
	Min Qy	343.98	1	206.39	24.494			617.153			535.84	561	206.385	ok	401.9	φVc>Vu	not required



Highlighted plate for moment (at column face)



Highlighted plate for shear (at d distance from column face)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
- At Bottom **16ϕT** **150** c/c
- At Top **16ϕT** **150** c/c

Along Z Direction :
- At Bottom **16ϕT** **150** c/c
- At Top **16ϕT** **150** c/c

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F5

STAAD Node - 8036 8037 7663 7976
Calculation has been done for node no 8036

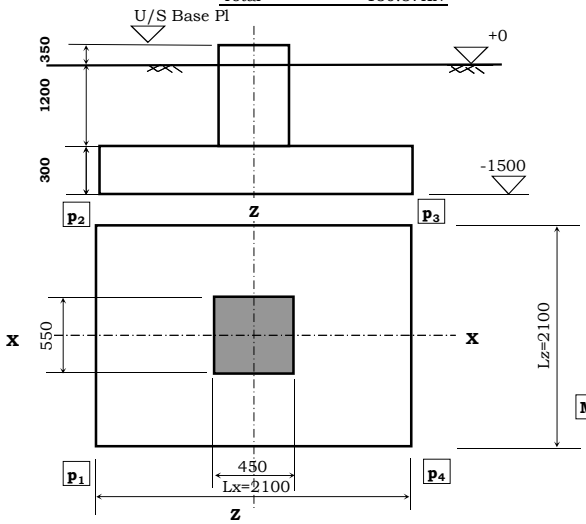
	Along-X	Along-Z
Size of column :	0.45m	0.55m
Size of fdn. raft :	2.10m	2.10m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

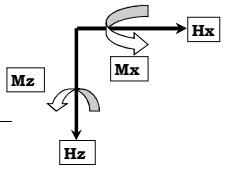
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(2.1*2.1*0.3*24)+(0.45*0.55*1.55*24) =	40.96KN
Weight of soil =	((2.1*2.1)-(0.45*0.55))*1.2*18 =	89.91KN
	Total =	130.87KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 244mm = 0.244m
Effective depth of slab along Z direction = 232mm = 0.232m
Clear projection along Z direction = 0.775m
Clear projection along X direction = 0.825m

H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F5

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	43.8	-6.3	0.0	-6.7	0.0	174.64	-6.33	-11.70	-6.67	-12.33
202	1.0DL+1.0FL2	43.7	-6.3	0.0	-6.7	0.0	174.62	-6.31	-11.68	-6.68	-12.35
203	1.0DL+1.0FL1+1.0LL+1.0TL	206.7	-27.7	0.0	-35.0	0.0	337.59	-27.75	-51.34	-35.02	-64.79
204	1.0DL+1.0FL2+1.0LL+1.0TL	206.7	-27.7	0.0	-35.0	0.0	337.57	-27.74	-51.31	-35.03	-64.81
205	1.0DL+1.0FL1+0.75LL+0.75TL	166.0	-22.4	0.0	-27.9	0.0	296.86	-22.39	-41.43	-27.93	-51.68
206	1.0DL+1.0FL2+0.75LL+0.75TL	166.0	-22.4	0.0	-27.9	0.0	296.83	-22.38	-41.40	-27.94	-51.70
207	1.0DL+1.0FL1+1.0PL	92.8	-6.3	0.0	-6.6	0.0	223.67	-6.33	-11.72	-6.64	-12.28
208	1.0DL+1.0FL2+1.0PL	92.8	-6.3	0.0	-6.6	0.0	223.64	-6.32	-11.69	-6.65	-12.30
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	255.8	-27.8	0.0	-35.0	0.0	386.62	-27.76	-51.35	-34.99	-64.74
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	255.7	-27.7	0.0	-35.0	0.0	386.60	-27.74	-51.33	-35.00	-64.76
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	215.0	-22.4	0.0	-27.9	0.0	345.88	-22.40	-41.44	-27.90	-51.62
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	215.0	-22.4	0.0	-27.9	0.0	345.86	-22.39	-41.42	-27.91	-51.64
213	1.0DL+1.0FL1+1.0WLX1	41.6	-3.8	0.0	-5.9	0.0	172.52	-3.83	-7.08	-5.88	-10.88
214	1.0DL+1.0FL1+1.0WLX2	45.9	-8.8	0.0	-7.4	0.0	176.77	-8.82	-16.32	-7.45	-13.78
215	1.0DL+1.0FL1+1.0WLZ1	43.7	-6.3	0.0	-9.4	0.0	174.62	-6.32	-11.69	-9.44	-17.47
216	1.0DL+1.0FL1+1.0WLZ2	43.8	-6.3	0.0	-3.9	0.0	174.67	-6.33	-11.71	-3.89	-7.20
217	1.0DL+1.0FL2+1.0WLX1	41.6	-3.8	0.0	-5.9	0.0	172.49	-3.81	-7.05	-5.89	-10.90
218	1.0DL+1.0FL2+1.0WLX2	45.9	-8.8	0.0	-7.5	0.0	176.74	-8.81	-16.30	-7.46	-13.80
219	1.0DL+1.0FL2+1.0WLZ1	43.7	-6.3	0.0	-9.5	0.0	174.59	-6.31	-11.66	-9.45	-17.49
220	1.0DL+1.0FL2+1.0WLZ2	43.8	-6.3	0.0	-3.9	0.0	174.64	-6.32	-11.69	-3.90	-7.22
221	1.0DL+1.0FL1+0.7SLX1	42.3	-5.1	0.0	-6.2	0.0	173.20	-5.07	-9.38	-6.20	-11.48
222	1.0DL+1.0FL1+0.7SLX2	45.1	-7.6	0.0	-7.1	0.0	175.99	-7.58	-14.02	-7.12	-13.18
223	1.0DL+1.0FL1+0.7SLZ1	43.8	-6.4	0.0	-6.9	0.0	174.71	-6.39	-11.82	-6.91	-12.79
224	1.0DL+1.0FL1+0.7SLZ2	43.6	-6.3	0.0	-6.4	0.0	174.48	-6.26	-11.58	-6.41	-11.86
225	1.0DL+1.0FL2+0.7SLX1	42.3	-5.1	0.0	-6.2	0.0	173.17	-5.06	-9.35	-6.21	-11.49
226	1.0DL+1.0FL2+0.7SLX2	45.1	-7.6	0.0	-7.1	0.0	175.95	-7.57	-14.00	-7.13	-13.19
227	1.0DL+1.0FL2+0.7SLZ1	43.8	-6.4	0.0	-6.9	0.0	174.68	-6.38	-11.80	-6.92	-12.81
228	1.0DL+1.0FL2+0.7SLZ2	43.6	-6.2	0.0	-6.4	0.0	174.44	-6.25	-11.56	-6.42	-11.88
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	163.8	-20.6	0.0	-27.2	0.0	294.70	-20.63	-38.16	-27.16	-50.25
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	167.0	-24.4	0.0	-28.2	0.0	297.89	-24.41	-45.16	-28.21	-52.20
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	165.4	-22.5	0.0	-29.7	0.0	296.29	-22.52	-41.65	-29.67	-54.89
232	1.0DL+1.0FL1+0.75WLZ2+0.75LL	165.4	-22.5	0.0	-25.7	0.0	296.30	-22.52	-41.67	-25.71	-47.56

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MD-216-3310-EG-CV-CAL-1213				Mansourah Massarah Gold Project							
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION							Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2				AS BUILT							Rev: D01
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	163.8	-20.6	0.0	-27.2	0.0	294.66	-20.62	-38.14	-27.17	-50.27	
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	167.0	-24.4	0.0	-28.2	0.0	297.86	-24.40	-45.14	-28.22	-52.21	
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	165.4	-22.5	0.0	-29.7	0.0	296.25	-22.50	-41.63	-29.68	-54.91	
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	165.4	-22.5	0.0	-25.7	0.0	296.27	-22.51	-41.64	-25.72	-47.58	
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	164.4	-21.6	0.0	-27.3	0.0	295.25	-21.58	-39.92	-27.34	-50.59	
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	166.5	-23.5	0.0	-28.0	0.0	297.34	-23.46	-43.40	-28.03	-51.86	
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	165.5	-22.6	0.0	-27.9	0.0	296.38	-22.57	-41.75	-27.88	-51.57	
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	165.3	-22.5	0.0	-27.5	0.0	296.21	-22.47	-41.57	-27.50	-50.88	
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	164.3	-21.6	0.0	-27.4	0.0	295.22	-21.56	-39.89	-27.35	-50.60	
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	166.4	-23.4	0.0	-28.0	0.0	297.31	-23.45	-43.38	-28.04	-51.88	
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	165.5	-22.6	0.0	-27.9	0.0	296.35	-22.56	-41.73	-27.89	-51.59	
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	165.3	-22.5	0.0	-27.5	0.0	296.17	-22.46	-41.55	-27.51	-50.90	
245 0.6DL+1.0WLX1	24.2	-1.3	0.0	-3.3	0.0	155.06	-1.31	-2.42	-3.26	-6.04	
246 0.6DL+1.0WLX2	28.5	-6.4	0.0	-4.7	0.0	159.32	-6.35	-11.75	-4.66	-8.63	
247 0.6DL+1.0WLZ1	26.3	-3.8	0.0	-6.6	0.0	157.18	-3.83	-7.08	-6.60	-12.22	
248 0.6DL+1.0WLZ2	26.3	-3.8	0.0	-1.3	0.0	157.20	-3.84	-7.09	-1.32	-2.45	
249 0.6DL+0.6FL1+0.7SLX1	24.8	-2.5	0.0	-3.5	0.0	155.71	-2.54	-4.70	-3.54	-6.55	
250 0.6DL+0.6FL1+0.7SLX2	27.6	-5.1	0.0	-4.5	0.0	158.50	-5.05	-9.34	-4.46	-8.25	
251 0.6DL+0.6FL1+0.7SLZ1	26.4	-3.9	0.0	-4.2	0.0	157.22	-3.86	-7.14	-4.25	-7.86	
252 0.6DL+0.6FL1+0.7SLZ2	26.1	-3.7	0.0	-3.7	0.0	156.99	-3.73	-6.90	-3.75	-6.93	
Ultimate Conditions											
101 1.4DL+1.4FL1	61.3	-8.9	0.0	-9.3	0.0	192.15	-8.86	-16.38	-9.33	-17.27	
102 1.4DL+1.4FL2	61.2	-8.8	0.0	-9.3	0.0	192.11	-8.84	-16.35	-9.35	-17.29	
103 1.2DL+1.2FL1+1.6LL	313.3	-42.1	0.0	-52.9	0.0	444.15	-42.14	-77.96	-52.91	-97.89	
104 1.2DL+1.2FL2+1.6LL	313.2	-42.1	0.0	-52.9	0.0	444.12	-42.12	-77.93	-52.92	-97.91	
105 1.2DL+1.2FL1+1.2TL+1.6LL	313.3	-41.9	0.0	-53.3	0.0	444.13	-41.94	-77.58	-53.26	-98.52	
106 1.2DL+1.2FL2+1.2TL+1.6LL	313.2	-41.9	0.0	-53.3	0.0	444.10	-41.92	-77.55	-53.27	-98.55	
107 1.4DL+1.4FL1+1.4PL	129.9	-8.9	0.0	-9.3	0.0	260.79	-8.87	-16.40	-9.29	-17.19	
108 1.4DL+1.4FL2+1.4PL	129.9	-8.8	0.0	-9.3	0.0	260.75	-8.85	-16.37	-9.31	-17.21	
109 1.2DL+1.2FL1+1.6LL+1.2PL	372.1	-42.1	0.0	-52.9	0.0	502.98	-42.15	-77.97	-52.88	-97.82	
110 1.2DL+1.2FL2+1.6LL+1.2PL	372.1	-42.1	0.0	-52.9	0.0	502.95	-42.13	-77.94	-52.89	-97.84	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	372.1	-41.9	0.0	-53.2	0.0	502.96	-41.95	-77.60	-53.22	-98.46	

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	372.1	-41.9	0.0	-53.2	0.0	502.93	-41.93	-77.57	-53.23	-98.48
113 1.2DL+0.8WLX1	51.0	-5.7	0.0	-7.3	0.0	181.82	-5.66	-10.47	-7.30	-13.51
114 1.2DL+0.8WLX2	54.3	-9.7	0.0	-8.6	0.0	185.22	-9.66	-17.87	-8.56	-15.83
115 1.2DL+0.8WLZ1	52.6	-7.7	0.0	-10.2	0.0	183.50	-7.66	-14.16	-10.15	-18.78
116 1.2DL+0.8WLZ2	52.7	-7.7	0.0	-5.7	0.0	183.54	-7.67	-14.18	-5.71	-10.56
117 1.2DL+1.6WLX1+0.5LL	130.7	-14.5	0.0	-20.7	0.0	261.61	-14.46	-26.75	-20.71	-38.32
118 1.2DL+1.6WLX2+0.5LL	137.5	-22.5	0.0	-23.2	0.0	268.40	-22.46	-41.54	-23.22	-42.95
119 1.2DL+1.6WLZ1+0.5LL	134.1	-18.4	0.0	-26.4	0.0	264.96	-18.45	-34.13	-26.41	-48.85
120 1.2DL+1.6WLZ2+0.5LL	134.2	-18.5	0.0	-17.5	0.0	265.05	-18.47	-34.16	-17.52	-32.42
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	132.1	-16.7	0.0	-21.2	0.0	262.98	-16.68	-30.87	-21.16	-39.14
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	136.1	-20.2	0.0	-22.8	0.0	266.92	-20.17	-37.31	-22.83	-42.24
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	134.2	-18.5	0.0	-22.7	0.0	265.11	-18.52	-34.26	-22.65	-41.91
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	133.9	-18.3	0.0	-21.3	0.0	264.79	-18.34	-33.92	-21.33	-39.47
125 0.9DL+1.6WLX1	36.1	-1.7	0.0	-4.7	0.0	166.96	-1.75	-3.23	-4.69	-8.68
126 0.9DL+1.6WLX2	42.9	-9.7	0.0	-7.2	0.0	173.75	-9.74	-18.02	-7.20	-13.32
127 0.9DL+1.6WLZ1	39.4	-5.7	0.0	-10.4	0.0	170.32	-5.74	-10.61	-10.39	-19.22
128 0.9DL+1.6WLZ2	39.5	-5.8	0.0	-1.5	0.0	170.40	-5.76	-10.65	-1.51	-2.78
129 0.9DL+1.0SLX1	37.5	-4.0	0.0	-5.1	0.0	168.38	-4.00	-7.40	-5.11	-9.45
130 0.9DL+1.0SLX2	41.5	-7.5	0.0	-6.8	0.0	172.33	-7.49	-13.85	-6.78	-12.55
131 0.9DL+1.0SLZ1	39.6	-5.8	0.0	-6.6	0.0	170.52	-5.84	-10.79	-6.61	-12.22
132 0.9DL+1.0SLZ2	39.3	-5.7	0.0	-5.3	0.0	170.20	-5.66	-10.46	-5.29	-9.78

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0FL1	174.6	-6.3	-11.7	-6.7	-12.3	15.7	14.9	11.0	10.5
202	1.0DL+1.0FL2	174.6	-6.3	-11.7	-6.7	-12.4	15.7	14.8	11.1	10.5
203	1.0DL+1.0FL1+1.0LL+1.0TL	337.6	-27.7	-51.3	-35.0	-64.8	6.9	5.5	4.9	3.9
204	1.0DL+1.0FL2+1.0LL+1.0TL	337.6	-27.7	-51.3	-35.0	-64.8	6.9	5.5	4.9	3.9
205	1.0DL+1.0FL1+0.75LL+0.75TL	296.9	-22.4	-41.4	-27.9	-51.7	7.5	6.0	5.3	4.3
206	1.0DL+1.0FL2+0.75LL+0.75TL	296.8	-22.4	-41.4	-27.9	-51.7	7.5	6.0	5.3	4.2
207	1.0DL+1.0FL1+1.0PL	223.7	-6.3	-11.7	-6.6	-12.3	20.0	19.1	14.1	13.5
208	1.0DL+1.0FL2+1.0PL	223.6	-6.3	-11.7	-6.6	-12.3	20.1	19.1	14.2	13.5
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	386.6	-27.8	-51.4	-35.0	-64.7	7.9	6.3	5.6	4.4
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	386.6	-27.7	-51.3	-35.0	-64.8	7.9	6.3	5.6	4.4
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	345.9	-22.4	-41.4	-27.9	-51.6	8.8	7.0	6.2	5.0

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية									Outotec LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213				Mansourah Massarah Gold Project					
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION					Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2				AS BUILT					Rev: D01
212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	345.9	-22.4	-41.4	-27.9	-51.6	8.8	7.0	6.2	5.0
213 1.0DL+1.0FL1+1.0WLX1	172.5	-3.8	-7.1	-5.9	-10.9	25.6	16.6	18.0	11.7
214 1.0DL+1.0FL1+1.0WLX2	176.8	-8.8	-16.3	-7.4	-13.8	11.4	13.5	8.0	9.5
215 1.0DL+1.0FL1+1.0WLZ1	174.6	-6.3	-11.7	-9.4	-17.5	15.7	10.5	11.1	7.4
216 1.0DL+1.0FL1+1.0WLZ2	174.7	-6.3	-11.7	-3.9	-7.2	15.7	25.5	11.0	18.0
217 1.0DL+1.0FL2+1.0WLX1	172.5	-3.8	-7.1	-5.9	-10.9	25.7	16.6	18.1	11.7
218 1.0DL+1.0FL2+1.0WLX2	176.7	-8.8	-16.3	-7.5	-13.8	11.4	13.4	8.0	9.5
219 1.0DL+1.0FL2+1.0WLZ1	174.6	-6.3	-11.7	-9.5	-17.5	15.7	10.5	11.1	7.4
220 1.0DL+1.0FL2+1.0WLZ2	174.6	-6.3	-11.7	-3.9	-7.2	15.7	25.4	11.1	17.9
221 1.0DL+1.0FL1+0.7SLX1	173.2	-5.1	-9.4	-6.2	-11.5	19.4	15.8	13.7	11.2
222 1.0DL+1.0FL1+0.7SLX2	176.0	-7.6	-14.0	-7.1	-13.2	13.2	14.0	9.3	9.9
223 1.0DL+1.0FL1+0.7SLZ1	174.7	-6.4	-11.8	-6.9	-12.8	15.5	14.3	10.9	10.1
224 1.0DL+1.0FL1+0.7SLZ2	174.5	-6.3	-11.6	-6.4	-11.9	15.8	15.4	11.2	10.9
225 1.0DL+1.0FL2+0.7SLX1	173.2	-5.1	-9.4	-6.2	-11.5	19.4	15.8	13.7	11.1
226 1.0DL+1.0FL2+0.7SLX2	176.0	-7.6	-14.0	-7.1	-13.2	13.2	14.0	9.3	9.9
227 1.0DL+1.0FL2+0.7SLZ1	174.7	-6.4	-11.8	-6.9	-12.8	15.5	14.3	11.0	10.1
228 1.0DL+1.0FL2+0.7SLZ2	174.4	-6.2	-11.6	-6.4	-11.9	15.9	15.4	11.2	10.9
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	294.7	-20.6	-38.2	-27.2	-50.3	8.1	6.2	5.7	4.3
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	297.9	-24.4	-45.2	-28.2	-52.2	6.9	6.0	4.9	4.2
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	296.3	-22.5	-41.7	-29.7	-54.9	7.5	5.7	5.3	4.0
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	296.3	-22.5	-41.7	-25.7	-47.6	7.5	6.5	5.3	4.6
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	294.7	-20.6	-38.1	-27.2	-50.3	8.1	6.2	5.7	4.3
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	297.9	-24.4	-45.1	-28.2	-52.2	6.9	6.0	4.9	4.2
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	296.3	-22.5	-41.6	-29.7	-54.9	7.5	5.7	5.3	4.0
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	296.3	-22.5	-41.6	-25.7	-47.6	7.5	6.5	5.3	4.6
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	295.3	-21.6	-39.9	-27.3	-50.6	7.8	6.1	5.5	4.3
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	297.3	-23.5	-43.4	-28.0	-51.9	7.2	6.0	5.1	4.2
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	296.4	-22.6	-41.8	-27.9	-51.6	7.5	6.0	5.3	4.3
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	296.2	-22.5	-41.6	-27.5	-50.9	7.5	6.1	5.3	4.3
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	295.2	-21.6	-39.9	-27.4	-50.6	7.8	6.1	5.5	4.3
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	297.3	-23.4	-43.4	-28.0	-51.9	7.2	6.0	5.1	4.2
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	296.3	-22.6	-41.7	-27.9	-51.6	7.5	6.0	5.3	4.3
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	296.2	-22.5	-41.5	-27.5	-50.9	7.5	6.1	5.3	4.3
245 0.6DL+1.0WLX1	155.1	-1.3	-2.4	-3.3	-6.0	67.3	27.0	47.4	19.0
246 0.6DL+1.0WLX2	159.3	-6.4	-11.8	-4.7	-8.6	14.2	19.4	10.0	13.7
247 0.6DL+1.0WLZ1	157.2	-3.8	-7.1	-6.6	-12.2	23.3	13.5	16.4	9.5
248 0.6DL+1.0WLZ2	157.2	-3.8	-7.1	-1.3	-2.4	23.3	67.4	16.4	47.5
249 0.6DL+0.6FL1+0.7SLX1	155.7	-2.5	-4.7	-3.5	-6.5	34.8	25.0	24.5	17.6
250 0.6DL+0.6FL1+0.7SLX2	158.5	-5.1	-9.3	-4.5	-8.2	17.8	20.2	12.6	14.2
251 0.6DL+0.6FL1+0.7SLZ1	157.2	-3.9	-7.1	-4.2	-7.9	23.1	21.0	16.3	14.8
252 0.6DL+0.6FL1+0.7SLZ2	157.0	-3.7	-6.9	-3.7	-6.9	23.9	23.8	16.8	16.8
					Min.=	6.9	5.5	4.9	3.9
						OK	OK	OK	OK

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F5

Load Combination & Calculation of Base pressure

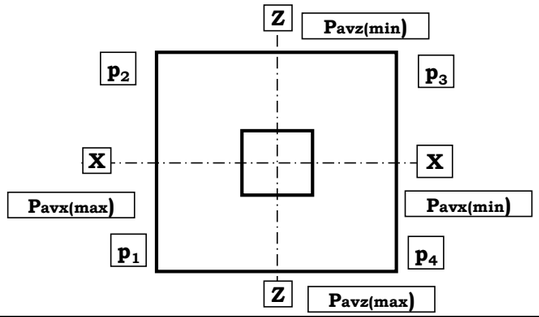
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	Base Pressure (kN/m²) p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
201	1.0DL+1.0FL1	174.64	-11.70	-12.33	39.60	7.58	7.99	55.17	39.19	40.01	24.03	28.17	47.18	32.02	47.59	31.61	O.K
202	1.0DL+1.0FL2	174.62	-11.68	-12.35	39.60	7.56	8.00	55.16	39.16	40.03	24.03	28.16	47.16	32.03	47.60	31.60	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	337.59	-51.34	-64.79	76.55	33.26	41.98	151.79	67.83	85.27	1.31	124.79	109.81	43.29	118.53	34.57	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	337.57	-51.31	-64.81	76.55	33.24	41.99	151.78	67.80	85.29	1.31	124.78	109.79	43.30	118.54	34.56	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	296.86	-41.43	-51.68	67.31	26.84	33.48	127.63	60.67	73.96	6.99	100.63	94.15	40.48	100.80	33.83	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	296.83	-41.40	-51.70	67.31	26.82	33.49	127.62	60.64	73.98	6.99	100.62	94.13	40.49	100.80	33.82	O.K
207	1.0DL+1.0FL1+1.0PL	223.67	-11.72	-12.28	50.72	7.59	7.95	66.26	50.36	51.08	35.17	39.26	58.31	43.13	58.67	42.77	O.K
208	1.0DL+1.0FL2+1.0PL	223.64	-11.69	-12.30	50.71	7.57	7.97	66.25	50.32	51.10	35.17	39.25	58.29	43.14	58.68	42.75	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	386.62	-51.35	-64.74	87.67	33.27	41.94	162.88	78.99	96.34	12.46	135.88	120.94	54.40	129.61	45.73	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	386.60	-51.33	-64.76	87.66	33.25	41.95	162.87	78.96	96.36	12.46	135.87	120.92	54.41	129.62	45.71	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+	345.88	-41.44	-51.62	78.43	26.85	33.45	138.73	71.84	85.03	18.14	111.73	105.29	51.59	111.88	44.99	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+	345.86	-41.42	-51.64	78.43	26.83	33.46	138.72	71.80	85.05	18.13	111.72	105.26	51.59	111.89	44.97	O.K
213	1.0DL+1.0FL1+1.0WLX1	172.52	-7.08	-10.88	39.12	4.59	7.05	50.76	36.65	41.59	27.48	23.76	43.71	34.54	46.18	32.07	O.K
214	1.0DL+1.0FL1+1.0WLX2	176.77	-16.32	-13.78	40.08	10.58	8.93	59.59	41.73	38.43	20.58	32.59	50.66	29.51	49.01	31.16	O.K
215	1.0DL+1.0FL1+1.0WLZ1	174.62	-11.69	-17.47	39.60	7.57	11.32	58.48	35.85	43.34	20.71	31.48	47.17	32.03	50.91	28.28	O.K
216	1.0DL+1.0FL1+1.0WLZ2	174.67	-11.71	-7.20	39.61	7.59	4.66	51.86	42.53	36.68	27.36	24.86	47.20	32.02	44.27	34.95	O.K
217	1.0DL+1.0FL2+1.0WLX1	172.49	-7.05	-10.90	39.11	4.57	7.06	50.75	36.62	41.61	27.48	23.75	43.69	34.55	46.18	32.05	O.K
218	1.0DL+1.0FL2+1.0WLX2	176.74	-16.30	-13.80	40.08	10.56	8.94	59.58	41.70	38.46	20.58	32.58	50.64	29.52	49.02	31.14	O.K
219	1.0DL+1.0FL2+1.0WLZ1	174.59	-11.66	-17.49	39.59	7.56	11.33	58.48	35.82	43.36	20.70	31.48	47.15	32.03	50.92	28.26	O.K
220	1.0DL+1.0FL2+1.0WLZ2	174.64	-11.69	-7.22	39.60	7.57	4.67	51.85	42.50	36.70	27.35	24.85	47.18	32.03	44.28	34.93	O.K

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MD-216-3310-EG-CV-CAL-1213				Mansourah Massarah Gold Project													
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION										Date: 10.11.2023			
O19045_450_C_K_ 24091 _001_R2				AS BUILT										Rev: D01			
221 1.0DL+1.0FL1+0.7SLX1	173.20	-9.38	-11.48	39.27	6.08	7.43	52.78	37.92	40.63	25.76	25.78	45.35	33.20	46.71	31.84	O.K
222 1.0DL+1.0FL1+0.7SLX2	175.99	-14.02	-13.18	39.91	9.09	8.54	57.53	40.46	39.36	22.28	30.53	49.00	30.82	48.45	31.37	O.K
223 1.0DL+1.0FL1+0.7SLZ1	174.71	-11.82	-12.79	39.62	7.66	8.28	55.56	38.99	40.24	23.67	28.56	47.28	31.96	47.90	31.33	O.K
224 1.0DL+1.0FL1+0.7SLZ2	174.48	-11.58	-11.86	39.56	7.50	7.69	54.75	39.38	39.75	24.38	27.75	47.07	32.07	47.25	31.88	O.K
225 1.0DL+1.0FL2+0.7SLX1	173.17	-9.35	-11.49	39.27	6.06	7.45	52.77	37.88	40.65	25.76	25.77	45.33	33.21	46.71	31.82	O.K
226 1.0DL+1.0FL2+0.7SLX2	175.95	-14.00	-13.19	39.90	9.07	8.55	57.52	40.42	39.38	22.28	30.52	48.97	30.83	48.45	31.35	O.K
227 1.0DL+1.0FL2+0.7SLZ1	174.68	-11.80	-12.81	39.61	7.64	8.30	55.55	38.96	40.26	23.67	28.55	47.26	31.97	47.91	31.32	O.K
228 1.0DL+1.0FL2+0.7SLZ2	174.44	-11.56	-11.88	39.56	7.49	7.70	54.74	39.34	39.77	24.37	27.74	47.04	32.07	47.26	31.86	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75I	294.70	-38.16	-50.25	66.82	24.72	32.56	124.11	58.99	74.66	9.54	97.11	91.55	42.10	99.39	34.27	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75I	297.89	-45.16	-52.20	67.55	29.26	33.82	130.63	62.99	72.11	4.47	103.63	96.81	38.29	101.37	33.73	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75I	296.29	-41.65	-54.89	67.19	26.99	35.56	129.73	58.61	75.76	4.64	102.73	94.17	40.20	102.75	31.63	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75I	296.30	-41.67	-47.56	67.19	27.00	30.81	125.00	63.37	71.01	9.38	98.00	94.19	40.20	98.01	36.38	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75I	294.66	-38.14	-50.27	66.82	24.71	32.57	124.09	58.96	74.68	9.54	97.09	91.53	42.11	99.39	34.25	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75I	297.86	-45.14	-52.21	67.54	29.24	33.83	130.61	62.96	72.13	4.47	103.61	96.79	38.30	101.37	33.72	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75I	296.25	-41.63	-54.91	67.18	26.97	35.57	129.72	58.58	75.78	4.63	102.72	94.15	40.21	102.75	31.61	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75I	296.27	-41.64	-47.58	67.18	26.98	30.82	124.99	63.34	71.03	9.38	97.99	94.17	40.21	98.01	36.36	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	295.25	-39.92	-50.59	66.95	25.86	32.77	125.59	60.04	73.86	8.31	98.59	92.82	41.09	99.73	34.18	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	297.34	-43.40	-51.86	67.42	28.12	33.60	129.14	61.95	72.90	5.70	102.14	95.55	39.30	101.02	33.83	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	296.38	-41.75	-51.57	67.21	27.05	33.41	127.67	60.85	73.57	6.75	100.67	94.26	40.16	100.62	33.80	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	296.21	-41.57	-50.88	67.17	26.93	32.96	127.06	61.14	73.20	7.27	100.06	94.10	40.24	100.13	34.21	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	295.22	-39.89	-50.60	66.94	25.85	32.79	125.57	60.00	73.88	8.31	98.57	92.79	41.10	99.73	34.16	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	297.31	-43.38	-51.88	67.42	28.10	33.61	129.13	61.91	72.92	5.70	102.13	95.52	39.31	101.03	33.81	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	296.35	-41.73	-51.59	67.20	27.04	33.42	127.66	60.81	73.59	6.74	100.66	94.24	40.17	100.63	33.78	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	296.17	-41.55	-50.90	67.16	26.92	32.97	127.05	61.10	73.22	7.27	100.05	94.08	40.25	100.14	34.19	O.K
245 0.6DL+1.0WLX1	155.06	-2.42	-6.04	35.16	1.57	3.91	40.64	32.82	37.50	29.68	13.64	36.73	33.59	39.07	31.25	O.K
246 0.6DL+1.0WLX2	159.32	-11.75	-8.63	36.13	7.61	5.59	49.33	38.15	34.10	22.92	22.33	43.74	28.51	41.72	30.54	O.K
247 0.6DL+1.0WLZ1	157.18	-7.08	-12.22	35.64	4.58	7.92	48.14	32.31	38.97	23.14	21.14	40.23	31.06	43.56	27.73	O.K
248 0.6DL+1.0WLZ2	157.20	-7.09	-2.45	35.65	4.60	1.59	41.83	38.66	32.64	29.46	14.83	40.25	31.05	37.24	34.06	O.K
249 0.6DL+0.6FL1+0.7SLX1	155.71	-4.70	-6.55	35.31	3.04	4.24	42.59	34.11	36.51	28.03	15.59	38.35	32.27	39.55	31.07	O.K
250 0.6DL+0.6FL1+0.7SLX2	158.50	-9.34	-8.25	35.94	6.05	5.34	47.34	36.65	35.23	24.54	20.34	42.00	29.89	41.29	30.60	O.K
251 0.6DL+0.6FL1+0.7SLZ1	157.22	-7.14	-7.86	35.65	4.63	5.09	45.37	35.19	36.11	25.93	18.37	40.28	31.02	40.74	30.56	O.K
252 0.6DL+0.6FL1+0.7SLZ2	156.99	-6.90	-6.93	35.60	4.47	4.49	44.56	35.57	35.62	26.64	17.56	40.07	31.13	40.09	31.11	O.K

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Ultimate Conditions

101 1.4DL+1.4FL1	192.15	-16.38	-17.27	43.57	10.61	11.19	65.37	43.00	44.14	21.77	38.37	54.19	32.96	54.76	32.39	
102 1.4DL+1.4FL2	192.11	-16.35	-17.29	43.56	10.59	11.20	65.36	42.95	44.17	21.77	38.36	54.16	32.97	54.77	32.36	
103 1.2DL+1.2FL1+1.6LL	444.15	-77.96	-97.89	100.71	50.51	63.42	214.64	87.80	113.62	-13.21	187.64	151.22	50.21	164.13	37.30	
104 1.2DL+1.2FL2+1.6LL	444.12	-77.93	-97.91	100.71	50.49	63.43	214.62	87.76	113.65	-13.21	187.62	151.19	50.22	164.14	37.28	
105 1.2DL+1.2FL1+1.2TL+1.6LL	444.13	-77.58	-98.52	100.71	50.26	63.83	214.80	87.14	114.28	-13.39	187.80	150.97	50.45	164.54	36.88	
106 1.2DL+1.2FL2+1.2TL+1.6LL	444.10	-77.55	-98.55	100.70	50.24	63.85	214.79	87.10	114.30	-13.39	187.79	150.95	50.46	164.55	36.86	
107 1.4DL+1.4FL1+1.4PL	260.79	-16.40	-17.19	59.14	10.63	11.14	80.90	58.63	59.64	37.37	53.90	69.77	48.51	70.27	48.00	
108 1.4DL+1.4FL2+1.4PL	260.75	-16.37	-17.21	59.13	10.60	11.15	80.88	58.58	59.68	37.37	53.88	69.73	48.53	70.28	47.98	
109 1.2DL+1.2FL1+1.6LL+1.2PL	502.98	-77.97	-97.82	114.05	50.52	63.37	227.95	101.20	126.91	0.16	200.95	164.58	63.54	177.43	50.68	
110 1.2DL+1.2FL2+1.6LL+1.2PL	502.95	-77.94	-97.84	114.05	50.50	63.39	227.93	101.16	126.94	0.16	200.93	164.55	63.55	177.44	50.66	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	502.96	-77.60	-98.46	114.05	50.28	63.79	228.11	100.54	127.56	-0.01	201.11	164.33	63.78	177.84	50.27	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	502.93	-77.57	-98.48	114.04	50.26	63.80	228.10	100.50	127.59	-0.02	201.10	164.30	63.79	177.85	50.24	
113 1.2DL+0.8WLX1	181.82	-10.47	-13.51	41.23	6.79	8.75	56.77	39.26	43.20	25.69	29.77	48.02	34.45	49.99	32.48	
114 1.2DL+0.8WLX2	185.22	-17.87	-15.83	42.00	11.58	10.26	63.83	43.32	40.68	20.17	36.83	53.58	30.43	52.26	31.75	
115 1.2DL+0.8WLZ1	183.50	-14.16	-18.78	41.61	9.18	12.17	62.95	38.62	44.60	20.27	35.95	50.79	32.44	53.78	29.45	
116 1.2DL+0.8WLZ2	183.54	-14.18	-10.56	41.62	9.19	6.84	57.65	43.96	39.27	25.59	30.65	50.81	32.43	48.46	34.78	
117 1.2DL+1.6WLX1+0.5LL	261.61	-26.75	-38.32	59.32	17.33	24.82	101.47	51.83	66.82	17.17	74.47	76.65	42.00	84.15	34.50	
118 1.2DL+1.6WLX2+0.5LL	268.40	-41.54	-42.95	60.86	26.91	27.83	115.60	59.95	61.78	6.12	88.60	87.78	33.95	88.69	33.04	
119 1.2DL+1.6WLZ1+0.5LL	264.96	-34.13	-48.85	60.08	22.11	31.65	113.84	50.54	69.62	6.32	86.84	82.19	37.97	91.73	28.43	
120 1.2DL+1.6WLZ2+0.5LL	265.05	-34.16	-32.42	60.10	22.13	21.00	103.24	61.23	58.97	16.97	76.24	82.24	37.97	81.11	39.10	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	262.98	-30.87	-39.14	59.63	20.00	25.36	104.99	54.27	64.99	14.28	77.99	79.63	39.64	84.99	34.28	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	266.92	-37.31	-42.24	60.53	24.18	27.36	112.07	57.34	63.72	8.99	85.07	84.71	36.36	87.90	33.17	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	265.11	-34.26	-41.91	60.12	22.19	27.15	109.46	55.16	65.07	10.77	82.46	82.31	37.92	87.27	32.97	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	264.79	-33.92	-39.47	60.04	21.98	25.57	107.59	56.45	63.63	12.50	80.59	82.02	38.07	85.61	34.48	
125 0.9DL+1.6WLX1	166.96	-3.23	-8.68	37.86	2.09	5.63	45.58	34.33	41.39	30.14	18.58	39.96	35.77	43.49	32.24	
126 0.9DL+1.6WLX2	173.75	-18.02	-13.32	39.40	11.68	8.63	59.71	42.45	36.35	19.09	32.71	51.08	27.72	48.03	30.77	
127 0.9DL+1.6WLZ1	170.32	-10.61	-19.22	38.62	6.87	12.45	57.95	33.04	44.20	19.29	30.95	45.50	31.75	51.08	26.17	
128 0.9DL+1.6WLZ2	170.40	-10.65	-2.78	38.64	6.90	1.80	47.34	43.73	33.55	29.94	20.34	45.54	31.75	40.45	36.84	
129 0.9DL+1.0SLX1	168.38	-7.40	-9.45	38.18	4.80	6.12	49.10	36.85	39.51	27.26	22.10	42.98	33.39	44.31	32.06	
130 0.9DL+1.0SLX2	172.33	-13.85	-12.55	39.08	8.97	8.13	56.18	39.92	38.23	21.97	29.18	48.05	30.10	47.21	30.95	
131 0.9DL+1.0SLZ1	170.52	-10.79	-12.22	38.67	6.99	7.92	53.58	37.74	39.59	23.75	26.58	45.66	31.67	46.59	30.75	
132 0.9DL+1.0SLZ2	170.20	-10.46	-9.78	38.59	6.78	6.34	51.71	39.04	38.15	25.48	24.71	45.38	31.82	44.93	32.26	

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Max :	503	-2	-2	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**28.17 kN/m^2**
Min :	155	-78	-99	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**38.17 kN/m^2**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**135.88 kN/m^2**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**145.88 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) = 39.60+7.56+8.00=	55.16 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(55.16+39.16)/2=	47.16 kN/m^2
p_2= 39.60+7.56-8.00=	39.16 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(24.03+40.03)/2=	32.03 kN/m^2
p_3= 39.60-(7.56+8.00)=	24.03 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(55.16+40.03)/2=	47.60 kN/m^2
p_4= 39.60-7.56+8.00=	40.03 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(24.03+39.16)/2=	31.59 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-γh)= 55.16-1.5*18=	28.16 kN/m^2			

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F5

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.45+2*0.5*0.232)+(0.55+2*0.5*0.232))*2

= 2.928 m

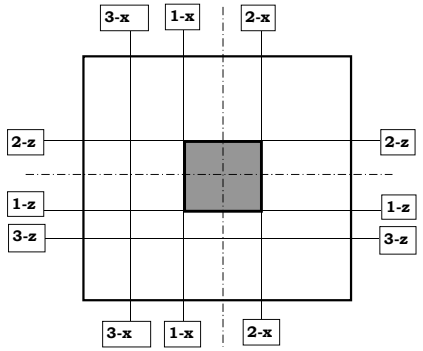
Punching Area= ((0.45+2*0.5*0.232)*(0.55+2*0.5*0.232))

= 0.53 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*24+1.2*18)

= 28.80 kN/m^2

Area of footing, A_f= 4.41 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	45.85	41.30	48.32	7.69	2.36	13.05	57.31
102	1.4DL+1.4FL2	45.84	41.29	48.30	7.69	2.36	13.03	57.28
103	1.2DL+1.2FL1+1.6LL	111.54	89.89	123.27	37.16	11.79	63.01	279.02
104	1.2DL+1.2FL2+1.6LL	111.52	89.89	123.25	37.15	11.79	62.99	279.00
105	1.2DL+1.2FL1+1.2TL+1.6LL	111.48	89.94	123.16	37.10	11.85	62.90	279.01
106	1.2DL+1.2FL2+1.2TL+1.6LL	111.47	89.94	123.15	37.09	11.85	62.89	278.98
107	1.4DL+1.4FL1+1.4PL	61.42	56.86	63.89	13.00	7.66	22.09	117.70
108	1.4DL+1.4FL2+1.4PL	61.40	56.86	63.86	12.98	7.66	22.08	117.67
109	1.2DL+1.2FL1+1.6LL+1.2PL	124.89	103.23	136.63	41.70	16.33	70.77	330.79
110	1.2DL+1.2FL2+1.6LL+1.2PL	124.87	103.23	136.61	41.70	16.33	70.75	330.76
111	1.2DL+1.2FL1+1.2TL+1.6LL+1.	124.83	103.28	136.51	41.64	16.39	70.66	330.77
112	1.2DL+1.2FL2+1.2TL+1.6LL+1.	124.81	103.28	136.49	41.63	16.39	70.65	330.74

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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113 1.2DL+0.8WLX1	42.69	39.78	44.27	5.94	2.53	10.08	48.22
114 1.2DL+0.8WLX2	44.49	39.52	47.18	7.40	1.59	12.54	51.21
115 1.2DL+0.8WLZ1	43.58	39.65	45.71	6.67	2.06	11.30	49.70
116 1.2DL+0.8WLZ2	43.59	39.65	45.72	6.67	2.05	11.31	49.74
117 1.2DL+1.6WLX1+0.5LL	63.04	55.61	67.06	14.74	6.04	25.02	118.42
118 1.2DL+1.6WLX2+0.5LL	66.63	55.10	72.89	17.67	4.15	29.94	124.40
119 1.2DL+1.6WLZ1+0.5LL	64.82	55.34	69.96	16.20	5.09	27.47	121.37
120 1.2DL+1.6WLZ2+0.5LL	64.85	55.36	69.99	16.21	5.09	27.49	121.45
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	63.92	55.35	68.57	15.52	5.47	26.32	119.63
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	65.72	55.35	71.33	16.87	4.73	28.60	123.10
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	64.87	55.36	70.03	16.23	5.08	27.52	121.51
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	64.75	55.34	69.86	16.15	5.11	27.39	121.22
125 0.9DL+1.6WLX1	38.31	37.42	38.80	3.61	2.56	6.15	35.15
126 0.9DL+1.6WLX2	41.90	36.90	44.62	6.54	0.67	11.07	41.13
127 0.9DL+1.6WLZ1	40.10	37.15	41.70	5.07	1.62	8.60	38.10
128 0.9DL+1.6WLZ2	40.12	37.17	41.72	5.08	1.62	8.62	38.18
129 0.9DL+1.0SLX1	39.21	37.16	40.33	4.40	1.99	7.47	36.40
130 0.9DL+1.0SLX2	41.00	37.15	43.08	5.75	1.24	9.74	39.87
131 0.9DL+1.0SLZ1	40.16	37.17	41.79	5.11	1.60	8.67	38.28
132 0.9DL+1.0SLZ2	40.05	37.15	41.63	5.04	1.63	8.54	38.00

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} =	54.19-(54.19-32.96)*0.825/2.1 =	45.85 kN/m^2	Overhang =	0.825 m
p_{2x} =	32.96+(54.19-32.96)*0.825/2.1 =	41.30 kN/m^2	Effective depth =	0.244 m
p_{3x} =	54.19-(54.19-32.96)*(0.825-0.244)/2.1 =	48.32 kN/m^2		
BM_{1x} =	(45.85-28.8)*0.825*0.825/2+0.5*(54.19-45.85)*0.825*0.825*0.67 =	7.69 kN-m/m		
BM_{2x} =	(41.30-28.8)*0.825*0.825/2-0.5*(-32.96+41.30)*0.825*0.825*0.67 =	2.36 kN-m/m		
SF_{3x} =	(48.32-28.8)*(0.825-0.244)+0.5*(54.19-48.32)*(0.825-0.244) =	13.05 kN/m		
$S.F_{(punching)}$ =	(4.41-0.53)*(43.57-28.8) =	57.31 kN		

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900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
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Location	Mux (kN-m)	ϕ Kn = M_u/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	41.70	0.70	0.93	0.002	551.40	439.20	12ϕT	205.11	300	200	565.49
At top	0.00	0.00	0.00	0.000	0.00	219.60	10ϕT	357.65	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 70.77 kN/m

Shear stress , τ_v= (70.77*1000)/(1000*244)

= 0.29 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 330.79 kN

Strength of Concrete in shear Vc1 = 2426.40 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1291.47 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1408.88 kN **Table 22.6.5.2_ (c)**

Design Vc= 1291.47 kN

Allowable shear stress, (ϕ Vc)/bd = 1.36 N/mm^2

Design punching shear stress, τ = (330.79*1000)/(2928*244)

= 0.46 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	46.50	40.65	48.98	6.98	1.90	12.53
102	1.4DL+1.4FL2	46.50	42.07	48.98	6.98	4.13	12.53
103	1.2DL+1.2FL1+1.6LL	117.32	45.19	131.34	36.00	4.43	64.58
104	1.2DL+1.2FL2+1.6LL	117.32	71.35	131.34	36.00	7.62	64.58
105	1.2DL+1.2FL1+1.2TL+1.6LL	117.43	71.11	131.53	36.10	7.60	64.74
106	1.2DL+1.2FL2+1.2TL+1.6LL	117.43	64.55	131.53	36.10	6.80	64.75
107	1.4DL+1.4FL1+1.4PL	62.05	71.58	64.51	11.64	7.49	20.95
108	1.4DL+1.4FL2+1.4PL	62.05	47.32	64.51	11.64	2.49	20.96
109	1.2DL+1.2FL1+1.6LL+1.2PL	130.65	50.06	144.66	40.00	2.58	71.81
110	1.2DL+1.2FL2+1.6LL+1.2PL	130.65	50.75	144.66	40.00	2.07	71.81
111	1.2DL+1.2FL1+1.2TL+1.6LL+1.	130.76	48.06	144.85	40.09	3.15	71.97
112	1.2DL+1.2FL2+1.2TL+1.6LL+1.	130.76	48.74	144.85	40.09	2.63	71.98

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113 1.2DL+0.8WLX1	43.53	38.58	45.46	5.72	1.44	10.28
114 1.2DL+0.8WLX2	44.69	38.82	46.96	6.30	0.88	11.30
115 1.2DL+0.8WLZ1	44.80	34.92	47.49	6.61	1.84	11.86
116 1.2DL+0.8WLZ2	43.41	39.18	44.92	5.40	1.64	9.72
117 1.2DL+1.6WLX1+0.5LL	65.83	39.65	71.31	14.81	1.59	26.57
118 1.2DL+1.6WLX2+0.5LL	68.15	38.52	74.30	15.95	1.47	28.61
119 1.2DL+1.6WLZ1+0.5LL	68.37	35.38	75.36	16.58	1.27	29.73
120 1.2DL+1.6WLZ2+0.5LL	65.61	41.91	70.25	14.17	1.91	25.46
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	66.28	39.51	71.88	15.02	1.57	26.95
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	67.70	38.61	73.75	15.75	1.48	28.25
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	67.23	38.24	73.23	15.57	1.55	27.94
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	66.74	58.43	72.39	15.19	4.04	27.26
125 0.9DL+1.6WLX1	39.34	57.75	40.58	4.00	3.86	7.19
126 0.9DL+1.6WLX2	41.66	57.33	43.57	5.14	3.40	9.23
127 0.9DL+1.6WLZ1	41.89	52.68	44.64	5.78	3.89	10.35
128 0.9DL+1.6WLZ2	39.12	59.92	39.52	3.37	4.18	6.07
129 0.9DL+1.0SLX1	39.79	57.64	41.14	4.21	3.85	7.56
130 0.9DL+1.0SLX2	41.21	57.45	43.01	4.93	3.40	8.86
131 0.9DL+1.0SLZ1	40.74	55.57	42.49	4.76	4.17	8.55
132 0.9DL+1.0SLZ2	40.25	57.16	41.65	4.38	3.89	7.87

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 54.76-(54.76-32.39)*0.775/2.1= 46.50 kN/m^2

p_{2z} = 32.39+(54.76-32.39)*0.775/2.1= 40.65 kN/m^2

p_{3z} = 54.76-(54.76-32.39)*(0.775-0.232)/2.1= 48.98 kN/m^2

Overhang = 0.775 m

Effective depth = 0.232 m

BM_{1z} = (46.50-28.8)*0.775*0.775/2+0.5*(54.76-46.50)*0.775*0.775*0.67 = 6.98 kN-m/m

BM_{2z} = (40.65-28.8)*0.775*0.775/2-0.5*(-32.39+40.65)*0.775*0.775*0.67 = 1.90 kN-m/m

SF_{3z} = (48.98-28.8)*(0.775-0.232)+0.5*(54.76-48.98)*(0.775-0.232) = 12.53 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{ui}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	40.09	0.673	0.898	0.0022	529.77	417.60	12ϕT	213.48	300	200	565.5
At top	0.00	0.000	0.000	0.0000	0.00	208.80	10ϕT	376.15	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 71.98 kN/m

Shear stress , τ_v= (71.98*1000)/(1000*232)

= 0.31 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom 12ϕT 200 c/c

At Top 10ϕT 200 c/c

Along Z Direction :

At Bottom 12ϕT 200 c/c

At Top 10ϕT 200 c/c

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MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
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DESIGN OF FOUNDATION MKD. F6 / F6A (STAAD Node -7730 & 7078)

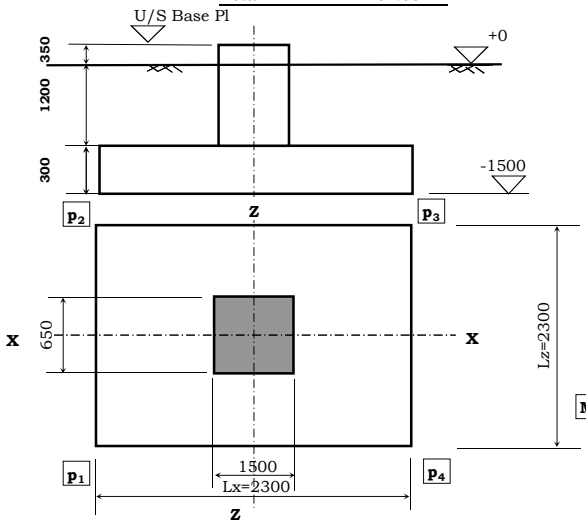
	Along-X	Along-Z
Size of column :	1.50m	0.65m
Size of fdn. raft :	2.30m	2.30m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

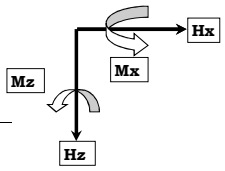
Weight of raft & Col.= (2.3*2.3*0.3*24)+(1.5*0.65*1.55*24) = 74.36KN
Weight of soil = ((2.3*2.3)-(1.5*0.65))*1.2*18 = 93.20KN
Total = 167.56KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 244mm = 0.244m
Effective depth of slab along Z direction = 232mm = 0.232m
Clear projection along Z direction = 0.825m
Clear projection along X direction = 0.4m



H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F6 / F6A

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	167.4	-2.6	-53.4	0.0	0.0	335.01	-2.61	-58.20	-0.02	-0.04
202	1.0DL+1.0FL2	169.0	-2.6	-54.1	-0.1	0.0	336.54	-2.58	-58.92	-0.11	-0.19
203	1.0DL+1.0FL1+1.0LL+1.0TL	322.0	-27.7	-118.5	-2.7	0.0	489.59	-27.72	-169.78	-2.68	-4.95
204	1.0DL+1.0FL2+1.0LL+1.0TL	323.6	-27.7	-119.3	-2.8	0.0	491.12	-27.69	-170.50	-2.76	-5.10
205	1.0DL+1.0FL1+0.75LL+0.75TL	283.4	-21.4	-102.2	-2.0	0.0	450.94	-21.45	-141.88	-2.02	-3.73
206	1.0DL+1.0FL2+0.75LL+0.75TL	284.9	-21.4	-103.0	-2.1	0.0	452.48	-21.42	-142.61	-2.10	-3.88
207	1.0DL+1.0FL1+1.0PL	143.4	-4.9	-38.7	6.3	0.0	310.94	-4.94	-47.88	6.25	11.56
208	1.0DL+1.0FL2+1.0PL	144.9	-4.9	-39.5	6.2	0.0	312.47	-4.91	-48.60	6.17	11.41
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	298.0	-30.1	-103.9	3.6	0.0	465.51	-30.05	-159.46	3.60	6.65
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	299.5	-30.0	-104.6	3.5	0.0	467.05	-30.02	-160.18	3.52	6.50
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	259.3	-23.8	-87.6	4.3	0.0	426.87	-23.77	-131.56	4.26	7.88
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	260.8	-23.7	-88.4	4.2	0.0	428.41	-23.75	-132.29	4.18	7.73
213	1.0DL+1.0FL1+1.0WLX1	185.1	-13.2	-45.8	0.2	0.0	352.62	-13.20	-70.26	0.23	0.42
214	1.0DL+1.0FL1+1.0WLX2	149.8	8.0	-60.9	-0.3	0.0	317.39	7.98	-46.13	-0.27	-0.51
215	1.0DL+1.0FL1+1.0WLZ1	136.3	-3.7	-36.5	-1.8	0.0	303.88	-3.70	-43.34	-1.78	-3.28
216	1.0DL+1.0FL1+1.0WLZ2	198.6	-1.5	-70.2	1.7	0.0	366.13	-1.51	-73.05	1.73	3.20
217	1.0DL+1.0FL2+1.0WLX1	186.6	-13.2	-46.6	0.1	0.0	354.16	-13.17	-70.98	0.15	0.27
218	1.0DL+1.0FL2+1.0WLX2	151.4	8.0	-61.7	-0.4	0.0	318.93	8.01	-46.86	-0.36	-0.66
219	1.0DL+1.0FL2+1.0WLZ1	137.9	-3.7	-37.3	-1.9	0.0	305.42	-3.68	-44.07	-1.86	-3.43
220	1.0DL+1.0FL2+1.0WLZ2	200.1	-1.5	-71.0	1.6	0.0	367.67	-1.49	-73.77	1.65	3.05
221	1.0DL+1.0FL1+0.7SLX1	171.4	-5.3	-50.9	0.1	0.0	338.98	-5.27	-60.65	0.06	0.12
222	1.0DL+1.0FL1+0.7SLX2	163.5	0.1	-55.8	-0.1	0.0	331.04	0.05	-55.74	-0.11	-0.20
223	1.0DL+1.0FL1+0.7SLZ1	157.0	-2.5	-48.4	-1.2	0.0	324.55	-2.48	-53.03	-1.21	-2.24
224	1.0DL+1.0FL1+0.7SLZ2	177.9	-2.7	-58.3	1.2	0.0	345.47	-2.74	-63.36	1.16	2.15
225	1.0DL+1.0FL2+0.7SLX1	173.0	-5.2	-51.7	0.0	0.0	340.52	-5.24	-61.38	-0.02	-0.03
226	1.0DL+1.0FL2+0.7SLX2	165.0	0.1	-56.6	-0.2	0.0	332.57	0.08	-56.46	-0.19	-0.35
227	1.0DL+1.0FL2+0.7SLZ1	158.5	-2.4	-49.2	-1.3	0.0	326.09	-2.45	-53.75	-1.29	-2.39
228	1.0DL+1.0FL2+0.7SLZ2	179.4	-2.7	-59.1	1.1	0.0	347.00	-2.71	-64.09	1.08	2.00
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	303.5	-11.5	-98.3	-2.0	0.0	471.07	-11.53	-119.66	-2.00	-3.71
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	277.1	4.4	-109.6	-2.4	0.0	444.65	4.36	-101.56	-2.38	-4.40
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	267.0	-4.4	-91.3	-3.5	0.0	434.51	-4.41	-99.47	-3.51	-6.49
232	1.0DL+1.0FL1+0.75WLZ2+0.75LL	313.6	-2.8	-116.6	-0.9	0.0	481.20	-2.76	-121.75	-0.88	-1.63

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MD-216-3310-EG-CV-CAL-1213				Mansourah Massarah Gold Project						
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION					Date:10.11.2023	
O19045_450_C_K_ 24091 _001_R2				AS BUILT					Rev: D01	
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	305.0	-11.5	-99.1	-2.1	0.0	472.61	-11.50	-120.38	-2.09	-3.86
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	278.6	4.4	-110.4	-2.5	0.0	446.18	4.39	-102.29	-2.46	-4.55
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	268.5	-4.4	-92.1	-3.6	0.0	436.05	-4.38	-100.20	-3.59	-6.64
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	315.2	-2.7	-117.4	-1.0	0.0	482.74	-2.73	-122.48	-0.96	-1.78
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	293.3	-5.6	-102.1	-2.1	0.0	460.84	-5.58	-112.45	-2.13	-3.94
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	287.3	-1.6	-105.8	-2.3	0.0	454.88	-1.59	-108.76	-2.26	-4.18
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	282.5	-3.5	-100.3	-3.1	0.0	450.01	-3.48	-106.73	-3.08	-5.70
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	298.1	-3.7	-107.7	-1.3	0.0	465.70	-3.68	-114.48	-1.30	-2.41
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	294.8	-5.5	-102.9	-2.2	0.0	462.37	-5.55	-113.18	-2.21	-4.09
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	288.9	-1.6	-106.6	-2.3	0.0	456.41	-1.56	-109.49	-2.34	-4.33
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	284.0	-3.5	-101.1	-3.2	0.0	451.55	-3.46	-107.46	-3.16	-5.85
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	299.7	-3.7	-108.4	-1.4	0.0	467.24	-3.66	-115.21	-1.38	-2.56
245 0.6DL+1.0WLX1	114.2	-12.2	-22.5	0.4	0.0	281.79	-12.23	-45.16	0.44	0.80
246 0.6DL+1.0WLX2	79.0	9.0	-37.6	-0.1	0.0	246.55	8.95	-21.04	-0.07	-0.13
247 0.6DL+1.0WLZ1	65.5	-2.7	-13.2	-1.6	0.0	233.04	-2.73	-18.25	-1.57	-2.90
248 0.6DL+1.0WLZ2	127.7	-0.5	-47.0	1.9	0.0	295.29	-0.54	-47.95	1.94	3.58
249 0.6DL+0.6FL1+0.7SLX1	104.4	-4.2	-29.6	0.1	0.0	272.00	-4.22	-37.38	0.07	0.13
250 0.6DL+0.6FL1+0.7SLX2	96.5	1.1	-34.5	-0.1	0.0	264.06	1.09	-32.46	-0.10	-0.19
251 0.6DL+0.6FL1+0.7SLZ1	90.0	-1.4	-27.1	-1.2	0.0	257.57	-1.43	-29.75	-1.20	-2.22
252 0.6DL+0.6FL1+0.7SLZ2	110.9	-1.7	-36.9	1.2	0.0	278.49	-1.70	-40.08	1.17	2.17
Ultimate Conditions										
101 1.4DL+1.4FL1	234.4	-3.7	-74.7	0.0	0.0	401.99	-3.65	-81.47	-0.03	-0.06
102 1.4DL+1.4FL2	236.6	-3.6	-75.8	-0.1	0.0	404.14	-3.61	-82.49	-0.15	-0.27
103 1.2DL+1.2FL1+1.6LL	463.0	-5.2	-172.0	-4.7	0.0	630.58	-5.21	-181.65	-4.66	-8.62
104 1.2DL+1.2FL2+1.6LL	464.9	-5.2	-172.9	-4.8	0.0	632.42	-5.18	-182.52	-4.75	-8.79
105 1.2DL+1.2FL1+1.2TL+1.6LL	452.0	-33.8	-169.2	-4.4	0.0	619.51	-33.79	-231.69	-4.37	-8.09
106 1.2DL+1.2FL2+1.2TL+1.6LL	453.8	-33.8	-170.1	-4.5	0.0	621.36	-33.75	-232.56	-4.47	-8.27
107 1.4DL+1.4FL1+1.4PL	200.7	-6.9	-54.2	8.8	0.0	368.28	-6.91	-67.02	8.75	16.19
108 1.4DL+1.4FL2+1.4PL	202.9	-6.9	-55.3	8.6	0.0	370.44	-6.87	-68.04	8.64	15.98
109 1.2DL+1.2FL1+1.6LL+1.2PL	434.1	-8.0	-154.5	2.9	0.0	601.69	-8.01	-169.27	2.87	5.32
110 1.2DL+1.2FL2+1.6LL+1.2PL	436.0	-8.0	-155.4	2.8	0.0	603.53	-7.97	-170.14	2.78	5.13
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	423.1	-36.6	-151.6	3.2	0.0	590.63	-36.58	-219.30	3.16	5.84

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	424.9	-36.6	-152.6	3.1	0.0	592.47	-36.55	-220.18	3.06	5.66
113 1.2DL+0.8WLX1	207.3	-11.7	-54.1	0.6	0.0	374.87	-11.74	-75.85	0.57	1.05
114 1.2DL+0.8WLX2	179.1	5.2	-66.2	0.2	0.0	346.69	5.20	-56.55	0.17	0.31
115 1.2DL+0.8WLZ1	168.3	-4.1	-46.6	-1.0	0.0	335.88	-4.15	-54.32	-1.04	-1.91
116 1.2DL+0.8WLZ2	218.1	-2.4	-73.6	1.8	0.0	385.68	-2.40	-78.08	1.77	3.27
117 1.2DL+1.6WLX1+0.5LL	303.3	-20.9	-81.8	-0.7	0.0	470.86	-20.87	-120.45	-0.68	-1.25
118 1.2DL+1.6WLX2+0.5LL	246.9	13.0	-105.9	-1.5	0.0	414.49	13.02	-81.84	-1.48	-2.74
119 1.2DL+1.6WLZ1+0.5LL	225.3	-5.7	-66.9	-3.9	0.0	392.88	-5.67	-77.38	-3.88	-7.18
120 1.2DL+1.6WLZ2+0.5LL	324.9	-2.2	-120.9	1.7	0.0	492.48	-2.17	-124.91	1.72	3.19
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	284.0	-7.7	-92.0	-1.1	0.0	451.57	-7.66	-106.17	-1.12	-2.07
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	272.7	-0.1	-99.0	-1.4	0.0	440.22	-0.07	-99.14	-1.37	-2.53
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	263.4	-3.7	-88.5	-2.9	0.0	430.95	-3.67	-95.28	-2.94	-5.44
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	293.3	-4.1	-102.5	0.5	0.0	460.84	-4.05	-110.04	0.45	0.83
125 0.9DL+1.6WLX1	173.1	-19.4	-33.1	0.7	0.0	340.66	-19.40	-68.95	0.68	1.25
126 0.9DL+1.6WLX2	116.7	14.5	-57.2	-0.1	0.0	284.29	14.49	-30.35	-0.13	-0.23
127 0.9DL+1.6WLZ1	95.1	-4.2	-18.1	-2.5	0.0	262.67	-4.21	-25.89	-2.53	-4.68
128 0.9DL+1.6WLZ2	194.7	-0.7	-72.1	3.1	0.0	362.27	-0.70	-73.41	3.08	5.69
129 0.9DL+1.0SLX1	150.6	-6.3	-41.6	0.4	0.0	318.15	-6.25	-53.16	0.40	0.74
130 0.9DL+1.0SLX2	139.2	1.3	-48.6	0.2	0.0	306.80	1.34	-46.14	0.15	0.28
131 0.9DL+1.0SLZ1	130.0	-2.3	-38.1	-1.4	0.0	297.53	-2.26	-42.27	-1.42	-2.63
132 0.9DL+1.0SLZ2	159.9	-2.6	-52.1	2.0	0.0	327.41	-2.64	-57.03	1.97	3.64

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1		335.0	-2.6	-58.2	0.0	0.0	6.6	8677.0	51.4	5583.5
202 1.0DL+1.0FL2		336.5	-2.6	-58.9	-0.1	-0.2	6.6	1992.4	52.2	1282.1
203 1.0DL+1.0FL1+1.0LL+1.0TL		489.6	-27.7	-169.8	-2.7	-5.0	3.3	113.6	7.1	73.1
204 1.0DL+1.0FL2+1.0LL+1.0TL		491.1	-27.7	-170.5	-2.8	-5.1	3.3	110.7	7.1	71.2
205 1.0DL+1.0FL1+0.75LL+0.75TL		450.9	-21.4	-141.9	-2.0	-3.7	3.7	139.1	8.4	89.5
206 1.0DL+1.0FL2+0.75LL+0.75TL		452.5	-21.4	-142.6	-2.1	-3.9	3.6	134.2	8.5	86.4
207 1.0DL+1.0FL1+1.0PL		310.9	-4.9	-47.9	6.3	11.6	7.5	30.9	25.2	19.9
208 1.0DL+1.0FL2+1.0PL		312.5	-4.9	-48.6	6.2	11.4	7.4	31.5	25.5	20.3
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		465.5	-30.1	-159.5	3.6	6.7	3.4	80.5	6.2	51.8
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		467.1	-30.0	-160.2	3.5	6.5	3.4	82.6	6.2	53.1
211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		426.9	-23.8	-131.6	4.3	7.9	3.7	62.3	7.2	40.1

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	428.4	-23.7	-132.3	4.2	7.7	3.7	63.7	7.2	41.0
213 1.0DL+1.0FL1+1.0WLX1	352.6	-13.2	-70.3	0.2	0.4	5.8	961.4	10.7	618.6
214 1.0DL+1.0FL1+1.0WLX2	317.4	8.0	-46.1	-0.3	-0.5	7.9	720.1	15.9	463.3
215 1.0DL+1.0FL1+1.0WLZ1	303.9	-3.7	-43.3	-1.8	-3.3	8.1	106.4	32.8	68.5
216 1.0DL+1.0FL1+1.0WLZ2	366.1	-1.5	-73.0	1.7	3.2	5.8	131.6	96.7	84.7
217 1.0DL+1.0FL2+1.0WLX1	354.2	-13.2	-71.0	0.1	0.3	5.7	1497.6	10.8	963.7
218 1.0DL+1.0FL2+1.0WLX2	318.9	8.0	-46.9	-0.4	-0.7	7.8	558.5	15.9	359.4
219 1.0DL+1.0FL2+1.0WLZ1	305.4	-3.7	-44.1	-1.9	-3.4	8.0	102.3	33.2	65.8
220 1.0DL+1.0FL2+1.0WLZ2	367.7	-1.5	-73.8	1.6	3.0	5.7	138.8	99.0	89.3
221 1.0DL+1.0FL1+0.7SLX1	339.0	-5.3	-60.7	0.1	0.1	6.4	3292.5	25.7	2118.6
222 1.0DL+1.0FL1+0.7SLX2	331.0	0.1	-55.7	-0.1	-0.2	6.8	1870.7	2648.3	1203.8
223 1.0DL+1.0FL1+0.7SLZ1	324.6	-2.5	-53.0	-1.2	-2.2	7.0	166.7	52.5	107.3
224 1.0DL+1.0FL1+0.7SLZ2	345.5	-2.7	-63.4	1.2	2.2	6.3	184.5	50.4	118.7
225 1.0DL+1.0FL2+0.7SLX1	340.5	-5.2	-61.4	0.0	0.0	6.4	11759.6	26.0	7567.0
226 1.0DL+1.0FL2+0.7SLX2	332.6	0.1	-56.5	-0.2	-0.4	6.8	1082.4	1683.9	696.5
227 1.0DL+1.0FL2+0.7SLZ1	326.1	-2.4	-53.8	-1.3	-2.4	7.0	157.0	53.3	101.0
228 1.0DL+1.0FL2+0.7SLZ2	347.0	-2.7	-64.1	1.1	2.0	6.2	199.2	51.2	128.2
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	471.1	-11.5	-119.7	-2.0	-3.7	4.5	146.1	16.3	94.0
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	444.6	4.4	-101.6	-2.4	-4.4	5.0	116.1	40.8	74.7
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	434.5	-4.4	-99.5	-3.5	-6.5	5.0	77.0	39.5	49.6
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	481.2	-2.8	-121.7	-0.9	-1.6	4.5	340.3	69.7	219.0
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	472.6	-11.5	-120.4	-2.1	-3.9	4.5	140.9	16.4	90.7
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	446.2	4.4	-102.3	-2.5	-4.6	5.0	112.7	40.7	72.5
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	436.1	-4.4	-100.2	-3.6	-6.6	5.0	75.5	39.9	48.6
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	482.7	-2.7	-122.5	-1.0	-1.8	4.5	312.6	70.6	201.1
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	460.8	-5.6	-112.5	-2.1	-3.9	4.7	134.6	33.0	86.6
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	454.9	-1.6	-108.8	-2.3	-4.2	4.8	125.3	114.4	80.6
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	450.0	-3.5	-106.7	-3.1	-5.7	4.8	90.7	51.7	58.4
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	465.7	-3.7	-114.5	-1.3	-2.4	4.7	222.2	50.6	143.0
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	462.4	-5.5	-113.2	-2.2	-4.1	4.7	130.1	33.3	83.7
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	456.4	-1.6	-109.5	-2.3	-4.3	4.8	121.3	117.0	78.1
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	451.6	-3.5	-107.5	-3.2	-5.9	4.8	88.7	52.3	57.1
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	467.2	-3.7	-115.2	-1.4	-2.6	4.7	209.9	51.1	135.0
245 0.6DL+1.0WLX1	281.8	-12.2	-45.2	0.4	0.8	7.2	402.7	9.2	259.1
246 0.6DL+1.0WLX2	246.6	9.0	-21.0	-0.1	-0.1	13.5	2253.9	11.0	1450.3
247 0.6DL+1.0WLZ1	233.0	-2.7	-18.2	-1.6	-2.9	14.7	92.4	34.1	59.5
248 0.6DL+1.0WLZ2	295.3	-0.5	-48.0	1.9	3.6	7.1	94.9	218.3	61.0
249 0.6DL+0.6FL1+0.7SLX1	272.0	-4.2	-37.4	0.1	0.1	8.4	2348.4	25.8	1511.1
250 0.6DL+0.6FL1+0.7SLX2	264.1	1.1	-32.5	-0.1	-0.2	9.4	1641.4	96.6	1056.2
251 0.6DL+0.6FL1+0.7SLZ1	257.6	-1.4	-29.8	-1.2	-2.2	10.0	133.3	71.9	85.8
252 0.6DL+0.6FL1+0.7SLZ2	278.5	-1.7	-40.1	1.2	2.2	8.0	147.6	65.6	95.0
					Min.=	3.3	30.9	6.2	19.9
						OK	OK	OK	OK

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F6 / F6A

Load Combination & Calculation of Base pressure

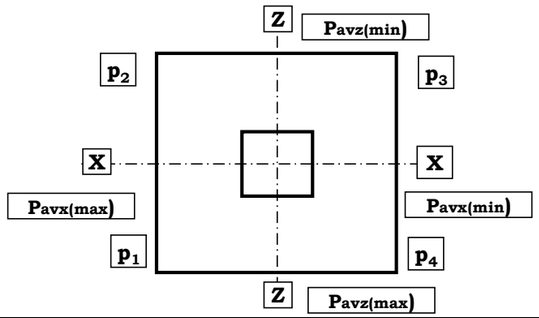
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	Base Pressure (kN/m²) p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	pavx(max)	pavx(min)	pavz(max)	pavz(min)	Remarks
201	1.0DL+1.0FL1	335.01	-58.20	-0.04	63.33	28.70	0.02	92.05	92.00	34.65	34.61	65.05	92.03	34.63	63.35	63.31	O.K
202	1.0DL+1.0FL2	336.54	-58.92	-0.19	63.62	29.06	0.10	92.77	92.58	34.66	34.47	65.77	92.68	34.57	63.72	63.53	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	489.59	-169.78	-4.95	92.55	83.72	2.44	178.72	173.83	11.27	6.38	151.72	176.28	8.83	95.00	90.11	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	491.12	-170.50	-5.10	92.84	84.08	2.52	179.44	174.40	11.28	6.24	152.44	176.92	8.76	95.36	90.32	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	450.94	-141.88	-3.73	85.24	69.97	1.84	157.05	153.37	17.12	13.44	130.05	155.21	15.28	87.09	83.41	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	452.48	-142.61	-3.88	85.53	70.33	1.91	157.77	153.95	17.12	13.30	130.77	155.86	15.21	87.45	83.63	O.K
207	1.0DL+1.0FL1+1.0PL	310.94	-47.88	11.56	58.78	23.61	5.70	88.09	76.68	40.87	29.47	61.09	82.39	35.17	64.48	53.08	O.K
208	1.0DL+1.0FL2+1.0PL	312.47	-48.60	11.41	59.07	23.97	5.63	88.66	77.41	40.73	29.47	61.66	83.04	35.10	64.70	53.44	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	465.51	-159.46	6.65	88.00	78.64	3.28	169.91	163.35	12.64	6.08	142.91	166.63	9.36	91.28	84.72	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	467.05	-160.18	6.50	88.29	78.99	3.21	170.49	164.08	12.50	6.09	143.49	167.29	9.30	91.50	85.09	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+	426.87	-131.56	7.88	80.69	64.88	3.89	149.46	141.68	19.70	11.93	122.46	145.57	15.82	84.58	76.81	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+	428.41	-132.29	7.73	80.98	65.24	3.81	150.03	142.41	19.56	11.94	123.03	146.22	15.75	84.80	77.18	O.K
213	1.0DL+1.0FL1+1.0WLX1	352.62	-70.26	0.42	66.66	34.65	0.21	101.51	101.10	32.22	31.80	74.51	101.31	32.01	66.87	66.45	O.K
214	1.0DL+1.0FL1+1.0WLX2	317.39	-46.13	-0.51	60.00	22.75	0.25	83.00	82.50	37.50	37.00	56.00	82.75	37.25	60.25	59.75	O.K
215	1.0DL+1.0FL1+1.0WLZ1	303.88	-43.34	-3.28	57.44	21.37	1.62	80.44	77.20	37.69	34.45	53.44	78.82	36.07	59.07	55.83	O.K
216	1.0DL+1.0FL1+1.0WLZ2	366.13	-73.05	3.20	69.21	36.02	1.58	106.81	103.66	34.77	31.61	79.81	105.24	33.19	70.79	67.64	O.K
217	1.0DL+1.0FL2+1.0WLX1	354.16	-70.98	0.27	66.95	35.01	0.13	102.09	101.82	32.08	31.81	75.09	101.96	31.95	67.09	66.82	O.K
218	1.0DL+1.0FL2+1.0WLX2	318.93	-46.86	-0.66	60.29	23.11	0.32	83.72	83.07	37.51	36.86	56.72	83.40	37.19	60.62	59.97	O.K
219	1.0DL+1.0FL2+1.0WLZ1	305.42	-44.07	-3.43	57.74	21.73	1.69	81.16	77.77	37.70	34.31	54.16	79.47	36.01	59.43	56.04	O.K
220	1.0DL+1.0FL2+1.0WLZ2	367.67	-73.77	3.05	69.50	36.38	1.50	107.39	104.38	34.63	31.62	80.39	105.89	33.13	71.01	68.00	O.K

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

221 1.0DL+1.0FL1+0.7SLX1	338.98	-60.65	0.12	64.08	29.91	0.06	94.05	93.93	34.23	34.11	67.05	93.99	34.17	64.14	64.02	O.K
222 1.0DL+1.0FL1+0.7SLX2	331.04	-55.74	-0.20	62.58	27.49	0.10	90.16	89.96	35.19	34.99	63.16	90.06	35.09	62.68	62.48	O.K
223 1.0DL+1.0FL1+0.7SLZ1	324.55	-53.03	-2.24	61.35	26.15	1.10	88.61	86.40	36.31	34.10	61.61	87.51	35.21	62.46	60.25	O.K
224 1.0DL+1.0FL1+0.7SLZ2	345.47	-63.36	2.15	65.31	31.25	1.06	97.61	95.49	35.12	33.00	70.61	96.55	34.06	66.37	64.25	O.K
225 1.0DL+1.0FL2+0.7SLX1	340.52	-61.38	-0.03	64.37	30.27	0.02	94.66	94.62	34.12	34.08	67.66	94.64	34.10	64.39	64.35	O.K
226 1.0DL+1.0FL2+0.7SLX2	332.57	-56.46	-0.35	62.87	27.84	0.17	90.89	90.54	35.20	34.85	63.89	90.72	35.03	63.05	62.70	O.K
227 1.0DL+1.0FL2+0.7SLZ1	326.09	-53.75	-2.39	61.64	26.51	1.18	89.33	86.97	36.31	33.96	62.33	88.15	35.14	62.82	60.47	O.K
228 1.0DL+1.0FL2+0.7SLZ2	347.00	-64.09	2.00	65.60	31.60	0.99	98.19	96.21	34.98	33.00	71.19	97.20	33.99	66.59	64.61	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75I	471.07	-119.66	-3.71	89.05	59.01	1.83	149.89	146.23	31.87	28.21	122.89	148.06	30.04	90.88	87.22	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75I	444.65	-101.56	-4.40	84.05	50.08	2.17	136.31	131.97	36.14	31.80	109.31	134.14	33.97	86.23	81.89	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75I	434.51	-99.47	-6.49	82.14	49.05	3.20	134.39	127.99	36.29	29.89	107.39	131.19	33.09	85.34	78.94	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75I	481.20	-121.75	-1.63	90.96	60.04	0.80	151.80	150.20	31.73	30.12	124.80	151.00	30.93	91.77	90.16	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75I	472.61	-120.38	-3.86	89.34	59.37	1.90	150.61	146.80	31.88	28.07	123.61	148.71	29.98	91.25	87.44	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75I	446.18	-102.29	-4.55	84.34	50.44	2.25	137.03	132.54	36.15	31.66	110.03	134.79	33.91	86.59	82.10	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75I	436.05	-100.20	-6.64	82.43	49.41	3.27	135.11	128.57	36.29	29.74	108.11	131.84	33.02	85.70	79.16	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75I	482.74	-122.48	-1.78	91.25	60.40	0.88	152.53	150.78	31.73	29.98	125.53	151.66	30.86	92.13	90.38	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	460.84	-112.45	-3.94	87.11	55.46	1.94	144.51	140.63	33.60	29.72	117.51	142.57	31.66	89.06	85.18	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	454.88	-108.76	-4.18	85.99	53.64	2.06	141.68	137.57	34.41	30.29	114.68	139.63	32.35	88.05	83.93	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	450.01	-106.73	-5.70	85.07	52.63	2.81	140.52	134.89	35.25	29.62	113.52	137.71	32.44	87.89	82.26	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	465.70	-114.48	-2.41	88.03	56.46	1.19	145.68	143.30	32.77	30.39	118.68	144.49	31.58	89.23	86.85	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	462.37	-113.18	-4.09	87.41	55.81	2.02	145.23	141.20	33.61	29.58	118.23	143.22	31.60	89.42	85.39	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	456.41	-109.49	-4.33	86.28	53.99	2.13	142.41	138.14	34.42	30.15	115.41	140.28	32.29	88.42	84.15	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	451.55	-107.46	-5.85	85.36	52.99	2.89	141.24	135.47	35.25	29.48	114.24	138.36	32.37	88.25	82.48	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	467.24	-115.21	-2.56	88.32	56.81	1.26	146.40	143.88	32.77	30.25	119.40	145.14	31.51	89.59	87.07	O.K
245 0.6DL+1.0WLX1	281.79	-45.16	0.80	53.27	22.27	0.40	75.94	75.14	31.39	30.60	48.94	75.54	31.00	53.67	52.87	O.K
246 0.6DL+1.0WLX2	246.55	-21.04	-0.13	46.61	10.37	0.06	57.04	56.92	36.30	36.17	30.04	56.98	36.24	46.67	46.55	O.K
247 0.6DL+1.0WLZ1	233.04	-18.25	-2.90	44.05	9.00	1.43	54.48	51.62	36.49	33.62	27.48	53.05	35.06	45.49	42.62	O.K
248 0.6DL+1.0WLZ2	295.29	-47.95	3.58	55.82	23.65	1.77	81.23	77.70	33.94	30.41	54.23	79.47	32.18	57.59	54.06	O.K
249 0.6DL+0.6FL1+0.7SLX1	272.00	-37.38	0.13	51.42	18.43	0.07	69.92	69.78	33.05	32.92	42.92	69.85	32.99	51.49	51.35	O.K
250 0.6DL+0.6FL1+0.7SLX2	264.06	-32.46	-0.19	49.92	16.01	0.09	66.01	65.83	34.00	33.82	39.01	65.92	33.91	50.01	49.83	O.K
251 0.6DL+0.6FL1+0.7SLZ1	257.57	-29.75	-2.22	48.69	14.67	1.10	64.46	62.27	35.11	32.92	37.46	63.37	34.02	49.79	47.60	O.K
252 0.6DL+0.6FL1+0.7SLZ2	278.49	-40.08	2.17	52.64	19.77	1.07	73.48	71.34	33.95	31.81	46.48	72.41	32.88	53.72	51.58	O.K

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Ultimate Conditions

101 1.4DL+1.4FL1	401.99	-81.47	-0.06	75.99	40.18	0.03	116.20	116.14	35.84	35.78	89.20	116.17	35.81	76.02	75.96	
102 1.4DL+1.4FL2	404.14	-82.49	-0.27	76.40	40.68	0.13	117.21	116.94	35.85	35.59	90.21	117.08	35.72	76.53	76.27	
103 1.2DL+1.2FL1+1.6LL	630.58	-181.65	-8.62	119.20	89.58	4.25	213.03	204.53	33.87	25.37	186.03	208.78	29.62	123.45	114.95	
104 1.2DL+1.2FL2+1.6LL	632.42	-182.52	-8.79	119.55	90.01	4.34	213.90	205.22	33.88	25.21	186.90	209.56	29.55	123.89	115.22	
105 1.2DL+1.2FL1+1.2TL+1.6LL	619.51	-231.69	-8.09	117.11	114.25	3.99	235.35	227.38	6.84	-1.13	208.35	231.37	2.86	121.10	113.13	
106 1.2DL+1.2FL2+1.2TL+1.6LL	621.36	-232.56	-8.27	117.46	114.68	4.08	236.22	228.07	6.85	-1.30	209.22	232.15	2.78	121.54	113.39	
107 1.4DL+1.4FL1+1.4PL	368.28	-67.02	16.19	69.62	33.05	7.98	110.65	94.69	44.55	28.58	83.65	102.67	36.57	77.60	61.64	
108 1.4DL+1.4FL2+1.4PL	370.44	-68.04	15.98	70.03	33.55	7.88	111.46	95.70	44.35	28.59	84.46	103.58	36.47	77.91	62.15	
109 1.2DL+1.2FL1+1.6LL+1.2PL	601.69	-169.27	5.32	113.74	83.47	2.62	199.84	194.59	32.89	27.65	172.84	197.22	30.27	116.37	111.12	
110 1.2DL+1.2FL2+1.6LL+1.2PL	603.53	-170.14	5.13	114.09	83.90	2.53	200.52	195.46	32.72	27.65	173.52	197.99	30.19	116.62	111.56	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	590.63	-219.30	5.84	111.65	108.15	2.88	222.68	216.92	6.38	0.62	195.68	219.80	3.50	114.53	108.77	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	592.47	-220.18	5.66	112.00	108.58	2.79	223.37	217.78	6.21	0.63	196.37	220.58	3.42	114.79	109.21	
113 1.2DL+0.8WLX1	374.87	-75.85	1.05	70.86	37.41	0.52	108.79	107.75	33.98	32.94	81.79	108.27	33.46	71.39	70.35	
114 1.2DL+0.8WLX2	346.69	-56.55	0.31	65.54	27.89	0.15	93.57	93.27	37.80	37.50	66.57	93.42	37.65	65.69	65.39	
115 1.2DL+0.8WLZ1	335.88	-54.32	-1.91	63.49	26.79	0.94	91.22	89.34	37.65	35.76	64.22	90.28	36.71	64.44	62.55	
116 1.2DL+0.8WLZ2	385.68	-78.08	3.27	72.91	38.50	1.61	113.02	109.80	36.02	32.79	86.02	111.41	34.41	74.52	71.30	
117 1.2DL+1.6WLX1+0.5LL	470.86	-120.45	-1.25	89.01	59.40	0.62	149.02	147.79	30.23	29.00	122.02	148.41	29.62	89.63	88.40	
118 1.2DL+1.6WLX2+0.5LL	414.49	-81.84	-2.74	78.35	40.36	1.35	120.06	117.36	39.35	36.64	93.06	118.71	38.00	79.71	77.00	
119 1.2DL+1.6WLZ1+0.5LL	392.88	-77.38	-7.18	74.27	38.16	3.54	115.97	108.89	39.65	32.57	88.97	112.43	36.11	77.81	70.73	
120 1.2DL+1.6WLZ2+0.5LL	492.48	-124.91	3.19	93.10	61.60	1.57	156.26	153.12	33.07	29.93	129.26	154.69	31.50	94.67	91.53	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	451.57	-106.17	-2.07	85.36	52.36	1.02	138.74	136.70	34.03	31.98	111.74	137.72	33.01	86.39	84.34	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	440.22	-99.14	-2.53	83.22	48.89	1.25	133.36	130.86	35.57	33.08	106.36	132.11	34.33	84.47	81.97	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	430.95	-95.28	-5.44	81.47	46.99	2.68	131.13	125.77	37.16	31.80	104.13	128.45	34.48	84.15	78.79	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	460.84	-110.04	0.83	87.11	54.26	0.41	141.79	140.97	33.26	32.44	114.79	141.38	32.85	87.53	86.71	
125 0.9DL+1.6WLX1	340.66	-68.95	1.25	64.40	34.00	0.62	99.02	97.78	31.01	29.78	72.02	98.40	30.40	65.02	63.78	
126 0.9DL+1.6WLX2	284.29	-30.35	-0.23	53.74	14.97	0.12	68.82	68.59	38.89	38.66	41.82	68.71	38.78	53.86	53.63	
127 0.9DL+1.6WLZ1	262.67	-25.89	-4.68	49.65	12.77	2.31	64.73	60.11	39.20	34.58	37.73	62.42	36.89	51.97	47.35	
128 0.9DL+1.6WLZ2	362.27	-73.41	5.69	68.48	36.20	2.81	107.49	101.88	35.09	29.47	80.49	104.69	32.28	71.29	65.68	
129 0.9DL+1.0SLX1	318.15	-53.16	0.74	60.14	26.22	0.36	86.72	85.99	34.29	33.56	59.72	86.36	33.93	60.51	59.78	
130 0.9DL+1.0SLX2	306.80	-46.14	0.28	58.00	22.75	0.14	80.89	80.61	35.38	35.10	53.89	80.75	35.24	58.14	57.86	
131 0.9DL+1.0SLZ1	297.53	-42.27	-2.63	56.24	20.84	1.30	78.38	75.79	36.70	34.10	51.38	77.09	35.40	57.54	54.95	
132 0.9DL+1.0SLZ2	327.41	-57.03	2.64	61.89	28.12	1.80	91.81	88.02	35.57	31.97	64.81	90.02	33.77	63.69	60.10	

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900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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Max :	632	-18	16	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**65.05 kN/m^2**
Min :	233	-233	-9	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**75.05 kN/m^2**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**152.44 kN/m^2**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**162.44 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) =	63.62+29.06+0.10=	92.77 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(92.77+92.58)/2= 92.68 kN/m^2
p_2=	63.62+29.06-0.10=	92.58 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(34.47+34.66)/2= 34.56 kN/m^2
p_3=	63.62-(29.06+0.10)=	34.47 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(92.77+34.66)/2= 63.71 kN/m^2
p_4=	63.62-29.06+0.10=	34.66 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(34.47+92.58)/2= 63.52 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-γh)=	92.77-1.5*18=	65.77 kN/m^2		

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DESIGN OF FOUNDATION MKD. F6 / F6A

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((1.5+2*0.5*0.232)+(0.65+2*0.5*0.232))*2

= 5.228 m

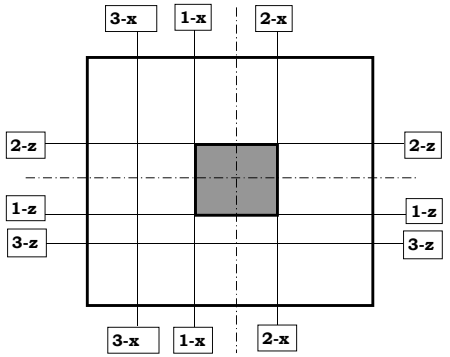
Punching Area= ((1.5+2*0.5*0.232)*(0.65+2*0.5*0.232))

= 1.53 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*24+1.2*18)

= 28.80 kN/m^2

Area of footing, A_f= 5.29 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M$_{1x}$ kNm	B.M$_{2x}$ kNm	S.F$_{3x}$ kN	S.F$_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	102.19	49.79	110.72	6.62	0.93	13.20	177.43
102	1.4DL+1.4FL2	102.93	49.87	111.56	6.69	0.93	13.34	178.96
103	1.2DL+1.2FL1+1.6LL	177.62	60.78	196.63	13.57	0.90	27.13	339.91
104	1.2DL+1.2FL2+1.6LL	178.25	60.86	197.35	13.63	0.89	27.25	341.22
105	1.2DL+1.2FL1+1.2TL+1.6LL	191.63	42.60	215.87	15.15	-1.02	30.39	332.05
106	1.2DL+1.2FL2+1.2TL+1.6LL	192.26	42.67	216.59	15.20	-1.02	30.51	333.36
107	1.4DL+1.4FL1+1.4PL	91.17	48.07	98.19	5.60	0.93	11.17	153.48
108	1.4DL+1.4FL2+1.4PL	91.91	48.14	99.03	5.67	0.92	11.31	155.01
109	1.2DL+1.2FL1+1.6LL+1.2PL	168.19	59.30	185.90	12.70	0.89	25.39	319.38
110	1.2DL+1.2FL2+1.6LL+1.2PL	168.81	59.37	186.61	12.76	0.89	25.51	320.69
111	1.2DL+1.2FL1+1.2TL+1.6LL+	182.18	41.12	205.13	14.28	-1.02	28.65	311.51
112	1.2DL+1.2FL2+1.2TL+1.6LL+	182.81	41.19	205.85	14.34	-1.02	28.77	312.82

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113 1.2DL+0.8WLX1	95.26	46.47	103.20	6.01	0.72	12.00	158.16
114 1.2DL+0.8WLX2	83.72	47.35	89.64	4.91	0.97	9.79	138.13
115 1.2DL+0.8WLZ1	80.96	46.03	86.65	4.67	0.88	9.31	130.45
116 1.2DL+0.8WLZ2	98.02	47.80	106.19	6.25	0.81	12.48	165.84
117 1.2DL+1.6WLX1+0.5LL	127.75	50.28	140.35	9.02	0.62	18.03	226.39
118 1.2DL+1.6WLX2+0.5LL	104.67	52.04	113.24	6.82	1.11	13.60	186.32
119 1.2DL+1.6WLZ1+0.5LL	99.16	49.38	107.25	6.34	0.94	12.64	170.96
120 1.2DL+1.6WLZ2+0.5LL	133.27	52.92	146.33	9.50	0.79	18.99	241.75
121 1.2DL+1.0SLX1+0.5LL+0.5FL	119.51	51.22	130.62	8.23	0.82	16.44	212.68
122 1.2DL+1.0SLX2+0.5LL+0.5FL	115.10	51.34	125.48	7.81	0.90	15.60	204.61
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	112.11	50.82	122.08	7.54	0.89	15.05	198.02
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	122.51	51.72	134.02	8.50	0.83	16.99	219.26
125 0.9DL+1.6WLX1	86.57	42.23	93.79	5.25	0.44	10.50	133.84
126 0.9DL+1.6WLX2	63.50	43.99	66.68	3.05	0.94	6.07	93.78
127 0.9DL+1.6WLZ1	57.98	41.33	60.69	2.57	0.77	5.11	78.41
128 0.9DL+1.6WLZ2	92.10	44.87	99.78	5.74	0.61	11.46	149.21
129 0.9DL+1.0SLX1	77.24	43.05	82.80	4.36	0.65	8.70	117.85
130 0.9DL+1.0SLX2	72.84	43.15	77.66	3.94	0.73	7.86	109.78
131 0.9DL+1.0SLZ1	69.84	42.65	74.26	3.67	0.72	7.31	103.19
132 0.9DL+1.0SLZ2	80.24	43.55	86.20	4.64	0.66	9.25	124.43

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} = 116.17-(116.17-35.81)*0.4/2.3 = 102.19 kN/m^2

p_{2x} = 35.81+(116.17-35.81)*0.4/2.3 = 49.79 kN/m^2

p_{3x} = 116.17-(116.17-35.81)*(0.4-0.244)/2.3 = 110.72 kN/m^2

Overhang = 0.4 m

Effective depth = 0.244 m

BM_{1x} = (102.19-28.8)*0.4*0.4/2+0.5*(116.17-102.19)*0.4*0.4*0.67 = 6.62 kN-m/m

BM_{2x} = (49.79-28.8)*0.4*0.4/2-0.5*(-35.81+49.79)*0.4*0.4*0.67 = 0.93 kN-m/m

SF_{3x} = (110.72-28.8)*(0.4-0.244)+0.5*(116.17-110.72)*(0.4-0.244) = 13.20 kN/m

$S.F_{(punching)}$ = (5.29-1.53)*(75.99-28.8) = 177.43 kN

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	15.20	0.26	0.34	0.001	198.96	439.20	12ϕT	257.51	300	200	565.49
	-1.02	-0.02	0.02	0.000	13.32	439.20	12ϕT	257.51	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 30.51 kN/m

Shear stress , τ_v= (30.51*1000)/(1000*244)

= 0.13 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 341.22 kN

Strength of Concrete in shear Vc1 = 2347.87 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1844.82 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2515.58 kN **Table 22.6.5.2_ (c)**

Design Vc= 1844.82 kN

Allowable shear stress, (ϕ Vc)/bd = 1.08 N/mm^2

Design punching shear stress, τ = (341.22*1000)/(5228*244)

= 0.27 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M$_{1z}$ kNm	B.M$_{2z}$ kNm	S.F$_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	76.00	75.98	76.00	16.07	16.05	28.00
102	1.4DL+1.4FL2	76.44	72.04	76.46	16.23	10.39	28.28
103	1.2DL+1.2FL1+1.6LL	120.40	96.93	121.26	31.87	13.27	55.48
104	1.2DL+1.2FL2+1.6LL	120.78	106.63	121.65	32.01	21.49	55.73
105	1.2DL+1.2FL1+1.2TL+1.6LL	118.24	105.37	119.05	31.09	21.43	54.12
106	1.2DL+1.2FL2+1.2TL+1.6LL	118.62	103.06	119.44	31.23	19.29	54.37
107	1.4DL+1.4FL1+1.4PL	71.88	69.95	73.49	15.96	15.65	27.72
108	1.4DL+1.4FL2+1.4PL	72.26	63.84	73.85	16.08	12.52	27.92
109	1.2DL+1.2FL1+1.6LL+1.2PL	114.49	92.87	115.02	29.59	14.25	51.53
110	1.2DL+1.2FL2+1.6LL+1.2PL	114.81	92.73	115.32	29.68	13.34	51.69
111	1.2DL+1.2FL1+1.2TL+1.6LL+	112.46	95.15	113.04	28.94	16.31	50.40
112	1.2DL+1.2FL2+1.2TL+1.6LL+	112.79	94.10	113.35	29.04	16.00	50.57

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113 1.2DL+0.8WLX1	71.02	66.86	71.12	14.45	11.38	25.18
114 1.2DL+0.8WLX2	65.58	63.25	65.61	12.54	10.08	21.85
115 1.2DL+0.8WLZ1	63.76	65.58	63.95	12.05	13.07	20.99
116 1.2DL+0.8WLZ2	73.37	68.73	73.69	15.43	12.51	26.87
117 1.2DL+1.6WLX1+0.5LL	89.19	79.17	89.31	20.65	13.34	35.98
118 1.2DL+1.6WLX2+0.5LL	78.74	71.78	79.01	17.22	12.00	29.98
119 1.2DL+1.6WLZ1+0.5LL	75.27	69.17	75.98	16.39	12.62	28.52
120 1.2DL+1.6WLZ2+0.5LL	93.54	81.80	93.86	22.29	14.06	38.82
121 1.2DL+1.0SLX1+0.5LL+0.5FL	85.65	76.70	85.86	19.52	13.11	33.99
122 1.2DL+1.0SLX2+0.5LL+0.5FL	83.57	75.10	83.83	18.84	12.42	32.82
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	82.23	74.41	82.77	18.62	13.29	32.41
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	87.24	88.21	87.32	19.95	19.99	34.76
125 0.9DL+1.6WLX1	64.58	71.83	64.70	12.28	16.94	21.38
126 0.9DL+1.6WLX2	53.78	65.00	53.80	8.52	15.50	14.84
127 0.9DL+1.6WLZ1	50.31	63.28	50.78	7.70	17.86	13.39
128 0.9DL+1.6WLZ2	69.28	74.85	69.84	14.23	18.54	24.77
129 0.9DL+1.0SLX1	60.25	69.40	60.32	10.76	16.71	18.75
130 0.9DL+1.0SLX2	58.04	67.85	58.07	9.97	15.87	17.38
131 0.9DL+1.0SLZ1	56.61	68.29	56.87	9.68	18.48	16.84
132 0.9DL+1.0SLZ2	62.40	70.49	62.76	11.73	17.54	20.42

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 76.02-(76.02-75.96)*0.825/2.3= 76.00 kN/m^2

p_{2z} = 75.96+(76.02-75.96)*0.825/2.3= 75.98 kN/m^2 Overhang = 0.825 m

p_{3z} = 76.02-(76.02-75.96)*(0.825-0.232)/2.3= 76.00 kN/m^2 Effective depth = 0.232 m

BM_{1z} = (76.00-28.8)*0.825*0.825/2+0.5*(76.02-76.00)*0.825*0.825*0.67 = 16.07 kN-m/m

BM_{2z} = (75.98-28.8)*0.825*0.825/2-0.5*(-75.96+75.98)*0.825*0.825*0.67 = 16.05 kN-m/m

SF_{3z} = (76.00-28.8)*(0.825-0.232)+0.5*(76.02-76.00)*(0.825-0.232) = 28.00 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	32.01	0.595	0.793	0.0019	466.99	417.60	12ϕT	242.18	300	200	565.5
At top	0.00	0.000	0.000	0.0000	0.00	139.20	12ϕT	812.48	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 55.73 kN/m

Shear stress , τ_v= (55.73*1000)/(1000*232)

= 0.24 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **200** c/c

At Top **12ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **200** c/c

At Top **12ϕT** **200** c/c

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MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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DESIGN OF FOUNDATION MKD. F7 / 7A

STAAD Node - 7769 7731

Calculation has been done for node no 7769

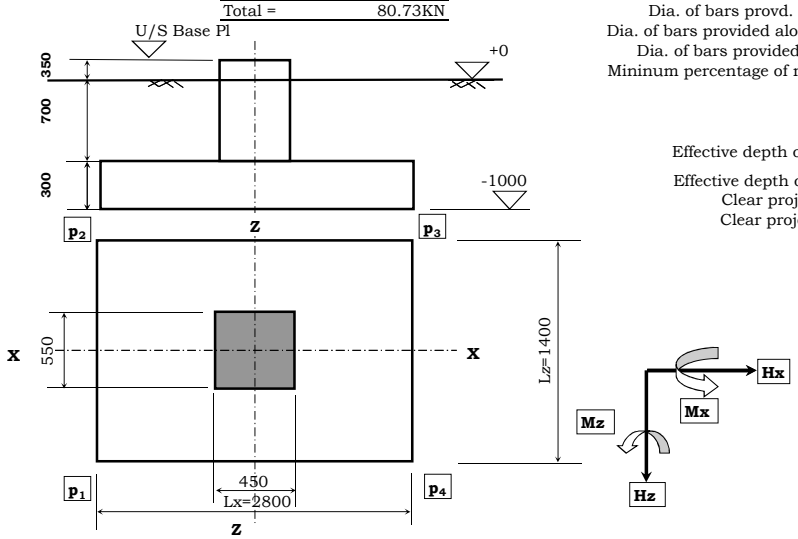
	Along-X	Along-Z
Size of column :	0.45m	0.55m
Size of fdn. raft :	2.80m	1.40m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.00 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(2.8*1.4*0.3*24)+(0.45*0.55*1.05*24) =	34.46KN
Weight of soil =	((2.8*1.4)-(0.45*0.55))*0.7*18 =	46.27KN
	Total =	80.73KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 244mm = 0.244m

Effective depth of slab along Z direction = 232mm = 0.232m

Clear projection along Z direction = 0.425m

Clear projection along X direction = 1.175m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F7 / 7A

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	9.7	-3.3	0.0	-1.4	0.0	90.41	-3.34	-4.51	-1.42	-1.91
202	1.0DL+1.0FL2	9.7	-3.3	0.0	-1.5	0.0	90.43	-3.34	-4.50	-1.46	-1.97
203	1.0DL+1.0FL1+1.0LL+1.0TL	46.3	-19.4	0.0	-9.5	0.0	126.98	-19.44	-26.24	-9.50	-12.83
204	1.0DL+1.0FL2+1.0LL+1.0TL	46.3	-19.4	0.0	-9.5	0.0	127.00	-19.44	-26.24	-9.55	-12.89
205	1.0DL+1.0FL1+0.75LL+0.75TL	37.1	-15.4	0.0	-7.5	0.0	117.84	-15.42	-20.81	-7.48	-10.10
206	1.0DL+1.0FL2+0.75LL+0.75TL	37.1	-15.4	0.0	-7.5	0.0	117.86	-15.41	-20.81	-7.52	-10.16
207	1.0DL+1.0FL1+1.0PL	57.0	-3.6	0.0	0.1	0.0	137.72	-3.64	-4.91	0.08	0.10
208	1.0DL+1.0FL2+1.0PL	57.0	-3.6	0.0	0.0	0.0	137.74	-3.64	-4.91	0.03	0.04
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	93.6	-19.7	0.0	-8.0	0.0	174.30	-19.74	-26.65	-8.01	-10.81
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	93.6	-19.7	0.0	-8.1	0.0	174.32	-19.74	-26.64	-8.05	-10.87
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	84.4	-15.7	0.0	-6.0	0.0	165.15	-15.71	-21.21	-5.99	-8.08
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	84.4	-15.7	0.0	-6.0	0.0	165.17	-15.71	-21.21	-6.03	-8.14
213	1.0DL+1.0FL1+1.0WLX1	9.9	-3.6	0.0	-1.5	0.0	90.63	-3.60	-4.86	-1.48	-1.99
214	1.0DL+1.0FL1+1.0WLX2	9.5	-3.1	0.0	-1.4	0.0	90.19	-3.07	-4.15	-1.36	-1.83
215	1.0DL+1.0FL1+1.0WLZ1	11.0	-3.5	0.0	-3.6	0.0	91.77	-3.46	-4.67	-3.56	-4.80
216	1.0DL+1.0FL1+1.0WLZ2	8.3	-3.2	0.0	0.7	0.0	89.04	-3.22	-4.34	0.72	0.98
217	1.0DL+1.0FL2+1.0WLX1	9.9	-3.6	0.0	-1.5	0.0	90.65	-3.60	-4.86	-1.52	-2.05
218	1.0DL+1.0FL2+1.0WLX2	9.5	-3.1	0.0	-1.4	0.0	90.21	-3.07	-4.14	-1.40	-1.89
219	1.0DL+1.0FL2+1.0WLZ1	11.1	-3.5	0.0	-3.6	0.0	91.79	-3.46	-4.67	-3.60	-4.86
220	1.0DL+1.0FL2+1.0WLZ2	8.3	-3.2	0.0	0.7	0.0	89.06	-3.21	-4.34	0.68	0.92
221	1.0DL+1.0FL1+0.7SLX1	10.4	-4.3	0.0	-1.5	0.0	91.12	-4.34	-5.86	-1.54	-2.07
222	1.0DL+1.0FL1+0.7SLX2	9.0	-2.3	0.0	-1.3	0.0	89.70	-2.34	-3.15	-1.30	-1.75
223	1.0DL+1.0FL1+0.7SLZ1	10.0	-3.4	0.0	-1.7	0.0	90.71	-3.45	-4.65	-1.65	-2.23
224	1.0DL+1.0FL1+0.7SLZ2	9.4	-3.2	0.0	-1.2	0.0	90.11	-3.23	-4.36	-1.18	-1.59
225	1.0DL+1.0FL2+0.7SLX1	10.4	-4.3	0.0	-1.6	0.0	91.14	-4.34	-5.85	-1.58	-2.14
226	1.0DL+1.0FL2+0.7SLX2	9.0	-2.3	0.0	-1.3	0.0	89.72	-2.33	-3.15	-1.34	-1.81
227	1.0DL+1.0FL2+0.7SLZ1	10.0	-3.4	0.0	-1.7	0.0	90.73	-3.44	-4.65	-1.70	-2.29
228	1.0DL+1.0FL2+0.7SLZ2	9.4	-3.2	0.0	-1.2	0.0	90.13	-3.23	-4.36	-1.23	-1.65
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	36.0	-14.3	0.0	-5.8	0.0	116.75	-14.30	-19.31	-5.76	-7.78
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	35.7	-13.9	0.0	-5.7	0.0	116.42	-13.91	-18.77	-5.67	-7.66
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	36.9	-14.2	0.0	-7.3	0.0	117.61	-14.20	-19.16	-7.32	-9.89
232	1.0DL+1.0FL1+0.75WLZ2+0.75LL	34.8	-14.0	0.0	-4.1	0.0	115.56	-14.01	-18.92	-4.11	-5.55

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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233 1.0DL+1.0FL2+0.75WLX1+0.75LL	36.0	-14.3	0.0	-5.8	0.0	116.77	-14.30	-19.31	-5.81	-7.84	
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	35.7	-13.9	0.0	-5.7	0.0	116.44	-13.90	-18.77	-5.72	-7.72	
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	36.9	-14.2	0.0	-7.4	0.0	117.63	-14.19	-19.16	-7.37	-9.95	
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	34.9	-14.0	0.0	-4.2	0.0	115.58	-14.01	-18.91	-4.16	-5.61	
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	36.4	-14.9	0.0	-5.8	0.0	117.12	-14.86	-20.06	-5.81	-7.84	
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	35.3	-13.4	0.0	-5.6	0.0	116.06	-13.35	-18.03	-5.63	-7.60	
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	36.1	-14.2	0.0	-5.9	0.0	116.81	-14.19	-19.15	-5.90	-7.96	
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	35.6	-14.0	0.0	-5.5	0.0	116.36	-14.02	-18.93	-5.54	-7.48	
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	36.4	-14.9	0.0	-5.9	0.0	117.14	-14.85	-20.05	-5.85	-7.90	
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	35.3	-13.4	0.0	-5.7	0.0	116.08	-13.35	-18.02	-5.67	-7.66	
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	36.1	-14.2	0.0	-5.9	0.0	116.83	-14.18	-19.14	-5.94	-8.02	
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	35.7	-14.0	0.0	-5.6	0.0	116.38	-14.02	-18.93	-5.59	-7.54	
245 0.6DL+1.0WLX1	6.0	-2.3	0.0	-0.8	0.0	86.69	-2.27	-3.07	-0.79	-1.07	
246 0.6DL+1.0WLX2	5.5	-1.7	0.0	-0.7	0.0	86.25	-1.74	-2.35	-0.67	-0.90	
247 0.6DL+1.0WLZ1	7.1	-2.1	0.0	-2.9	0.0	87.83	-2.13	-2.87	-2.87	-3.87	
248 0.6DL+1.0WLZ2	4.4	-1.9	0.0	1.4	0.0	85.10	-1.89	-2.55	1.41	1.90	
249 0.6DL+0.6FL1+0.7SLX1	6.5	-3.0	0.0	-1.0	0.0	87.25	-3.01	-4.06	-0.97	-1.31	
250 0.6DL+0.6FL1+0.7SLX2	5.1	-1.0	0.0	-0.7	0.0	85.83	-1.00	-1.35	-0.73	-0.99	
251 0.6DL+0.6FL1+0.7SLZ1	6.1	-2.1	0.0	-1.1	0.0	86.84	-2.11	-2.85	-1.09	-1.47	
252 0.6DL+0.6FL1+0.7SLZ2	5.5	-1.9	0.0	-0.6	0.0	86.24	-1.90	-2.56	-0.61	-0.83	
Ultimate Conditions											
101 1.4DL+1.4FL1	13.5	-4.7	0.0	-2.0	0.0	94.28	-4.67	-6.31	-1.98	-2.68	
102 1.4DL+1.4FL2	13.6	-4.7	0.0	-2.0	0.0	94.31	-4.67	-6.30	-2.05	-2.76	
103 1.2DL+1.2FL1+1.6LL	67.5	-27.0	0.0	-10.9	0.0	148.20	-26.97	-36.41	-10.88	-14.69	
104 1.2DL+1.2FL2+1.6LL	67.5	-27.0	0.0	-10.9	0.0	148.22	-26.97	-36.41	-10.93	-14.76	
105 1.2DL+1.2FL1+1.2TL+1.6LL	69.5	-29.1	0.0	-13.7	0.0	150.20	-29.07	-39.24	-13.69	-18.49	
106 1.2DL+1.2FL2+1.2TL+1.6LL	69.5	-29.1	0.0	-13.7	0.0	150.22	-29.07	-39.24	-13.75	-18.56	
107 1.4DL+1.4FL1+1.4PL	79.8	-5.1	0.0	0.1	0.0	160.52	-5.09	-6.88	0.10	0.14	
108 1.4DL+1.4FL2+1.4PL	79.8	-5.1	0.0	0.0	0.0	160.55	-5.09	-6.87	0.04	0.06	
109 1.2DL+1.2FL1+1.6LL+1.2PL	124.2	-27.3	0.0	-9.1	0.0	204.97	-27.33	-36.90	-9.09	-12.27	
110 1.2DL+1.2FL2+1.6LL+1.2PL	124.3	-27.3	0.0	-9.1	0.0	205.00	-27.33	-36.89	-9.14	-12.34	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	126.2	-29.4	0.0	-11.9	0.0	206.97	-29.43	-39.73	-11.90	-16.07	

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112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	126.3	-29.4	0.0	-12.0	0.0	207.00	-29.43	-39.73	-11.96	-16.14
113 1.2DL+0.8WLX1	11.7	-4.2	0.0	-1.5	0.0	92.38	-4.23	-5.71	-1.51	-2.03
114 1.2DL+0.8WLX2	11.3	-3.8	0.0	-1.4	0.0	92.03	-3.80	-5.14	-1.41	-1.90
115 1.2DL+0.8WLZ1	12.6	-4.1	0.0	-3.2	0.0	93.30	-4.11	-5.55	-3.17	-4.28
116 1.2DL+0.8WLZ2	10.4	-3.9	0.0	0.3	0.0	91.11	-3.92	-5.29	0.25	0.34
117 1.2DL+1.6WLX1+0.5LL	29.3	-11.6	0.0	-4.4	0.0	110.01	-11.62	-15.68	-4.42	-5.97
118 1.2DL+1.6WLX2+0.5LL	28.6	-10.8	0.0	-4.2	0.0	109.31	-10.77	-14.54	-4.23	-5.71
119 1.2DL+1.6WLZ1+0.5LL	31.1	-11.4	0.0	-7.8	0.0	111.85	-11.39	-15.37	-7.75	-10.47
120 1.2DL+1.6WLZ2+0.5LL	26.7	-11.0	0.0	-0.9	0.0	107.47	-11.00	-14.85	-0.90	-1.22
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	30.0	-12.6	0.0	-4.6	0.0	110.73	-12.62	-17.04	-4.60	-6.21
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	28.0	-9.8	0.0	-4.3	0.0	108.70	-9.76	-13.17	-4.26	-5.75
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	29.4	-11.3	0.0	-4.8	0.0	110.15	-11.34	-15.31	-4.77	-6.43
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	28.6	-11.0	0.0	-4.1	0.0	109.29	-11.04	-14.90	-4.09	-5.52
125 0.9DL+1.6WLX1	9.0	-3.4	0.0	-1.2	0.0	89.69	-3.44	-4.64	-1.19	-1.61
126 0.9DL+1.6WLX2	8.3	-2.6	0.0	-1.0	0.0	88.99	-2.59	-3.49	-1.00	-1.35
127 0.9DL+1.6WLZ1	10.8	-3.2	0.0	-4.5	0.0	91.52	-3.21	-4.33	-4.52	-6.10
128 0.9DL+1.6WLZ2	6.4	-2.8	0.0	2.3	0.0	87.15	-2.82	-3.80	2.33	3.15
129 0.9DL+1.0SLX1	9.6	-4.4	0.0	-1.3	0.0	90.35	-4.44	-6.00	-1.27	-1.71
130 0.9DL+1.0SLX2	7.6	-1.6	0.0	-0.9	0.0	88.32	-1.58	-2.13	-0.92	-1.25
131 0.9DL+1.0SLZ1	9.0	-3.2	0.0	-1.4	0.0	89.77	-3.16	-4.27	-1.43	-1.93
132 0.9DL+1.0SLZ2	8.2	-2.9	0.0	-0.8	0.0	88.91	-2.86	-3.86	-0.76	-1.02

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0FL1		90.4	-3.3	-4.5	-1.4	-1.9	28.1	33.1	10.8	25.5
202 1.0DL+1.0FL2		90.4	-3.3	-4.5	-1.5	-2.0	28.1	32.1	10.8	24.7
203 1.0DL+1.0FL1+1.0LL+1.0TL		127.0	-19.4	-26.2	-9.5	-12.8	6.8	6.9	2.6	5.3
204 1.0DL+1.0FL2+1.0LL+1.0TL		127.0	-19.4	-26.2	-9.5	-12.9	6.8	6.9	2.6	5.3
205 1.0DL+1.0FL1+0.75LL+0.75TL		117.8	-15.4	-20.8	-7.5	-10.1	7.9	8.2	3.1	6.3
206 1.0DL+1.0FL2+0.75LL+0.75TL		117.9	-15.4	-20.8	-7.5	-10.2	7.9	8.1	3.1	6.3
207 1.0DL+1.0FL1+1.0PL		137.7	-3.6	-4.9	0.1	0.1	39.3	952.1	15.1	734.5
208 1.0DL+1.0FL2+1.0PL		137.7	-3.6	-4.9	0.0	0.0	39.3	2462.8	15.2	1899.9
209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		174.3	-19.7	-26.6	-8.0	-10.8	9.2	11.3	3.5	8.7
210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		174.3	-19.7	-26.6	-8.1	-10.9	9.2	11.2	3.5	8.7
211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		165.2	-15.7	-21.2	-6.0	-8.1	10.9	14.3	4.2	11.0

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	165.2	-15.7	-21.2	-6.0	-8.1	10.9	14.2	4.2	11.0
213 1.0DL+1.0FL1+1.0WLX1	90.6	-3.6	-4.9	-1.5	-2.0	26.1	31.8	10.1	24.5
214 1.0DL+1.0FL1+1.0WLX2	90.2	-3.1	-4.1	-1.4	-1.8	30.4	34.5	11.7	26.6
215 1.0DL+1.0FL1+1.0WLZ1	91.8	-3.5	-4.7	-3.6	-4.8	27.5	13.4	10.6	10.3
216 1.0DL+1.0FL1+1.0WLZ2	89.0	-3.2	-4.3	0.7	1.0	28.7	63.8	11.1	49.2
217 1.0DL+1.0FL2+1.0WLX1	90.6	-3.6	-4.9	-1.5	-2.1	26.1	30.9	10.1	23.8
218 1.0DL+1.0FL2+1.0WLX2	90.2	-3.1	-4.1	-1.4	-1.9	30.5	33.3	11.8	25.7
219 1.0DL+1.0FL2+1.0WLZ1	91.8	-3.5	-4.7	-3.6	-4.9	27.5	13.2	10.6	10.2
220 1.0DL+1.0FL2+1.0WLZ2	89.1	-3.2	-4.3	0.7	0.9	28.7	68.1	11.1	52.5
221 1.0DL+1.0FL1+0.7SLX1	91.1	-4.3	-5.9	-1.5	-2.1	21.8	30.7	8.4	23.7
222 1.0DL+1.0FL1+0.7SLX2	89.7	-2.3	-3.2	-1.3	-1.8	39.8	35.9	15.4	27.7
223 1.0DL+1.0FL1+0.7SLZ1	90.7	-3.4	-4.7	-1.7	-2.2	27.3	28.4	10.5	21.9
224 1.0DL+1.0FL1+0.7SLZ2	90.1	-3.2	-4.4	-1.2	-1.6	28.9	39.6	11.2	30.5
225 1.0DL+1.0FL2+0.7SLX1	91.1	-4.3	-5.9	-1.6	-2.1	21.8	29.9	8.4	23.0
226 1.0DL+1.0FL2+0.7SLX2	89.7	-2.3	-3.1	-1.3	-1.8	39.9	34.6	15.4	26.7
227 1.0DL+1.0FL2+0.7SLZ1	90.7	-3.4	-4.6	-1.7	-2.3	27.3	27.7	10.5	21.4
228 1.0DL+1.0FL2+0.7SLZ2	90.1	-3.2	-4.4	-1.2	-1.7	29.0	38.1	11.2	29.4
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	116.8	-14.3	-19.3	-5.8	-7.8	8.5	10.5	3.3	8.1
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	116.4	-13.9	-18.8	-5.7	-7.7	8.7	10.6	3.3	8.2
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	117.6	-14.2	-19.2	-7.3	-9.9	8.6	8.3	3.3	6.4
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	115.6	-14.0	-18.9	-4.1	-5.6	8.6	14.6	3.3	11.2
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	116.8	-14.3	-19.3	-5.8	-7.8	8.5	10.4	3.3	8.0
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	116.4	-13.9	-18.8	-5.7	-7.7	8.7	10.6	3.4	8.1
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	117.6	-14.2	-19.2	-7.4	-9.9	8.6	8.3	3.3	6.4
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	115.6	-14.0	-18.9	-4.2	-5.6	8.6	14.4	3.3	11.1
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	117.1	-14.9	-20.1	-5.8	-7.8	8.2	10.5	3.2	8.1
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	116.1	-13.4	-18.0	-5.6	-7.6	9.0	10.7	3.5	8.2
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	116.8	-14.2	-19.1	-5.9	-8.0	8.5	10.3	3.3	7.9
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	116.4	-14.0	-18.9	-5.5	-7.5	8.6	10.9	3.3	8.4
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	117.1	-14.9	-20.1	-5.9	-7.9	8.2	10.4	3.2	8.0
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	116.1	-13.4	-18.0	-5.7	-7.7	9.0	10.6	3.5	8.2
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	116.8	-14.2	-19.1	-5.9	-8.0	8.5	10.2	3.3	7.9
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	116.4	-14.0	-18.9	-5.6	-7.5	8.6	10.8	3.3	8.3
245 0.6DL+1.0WLX1	86.7	-2.3	-3.1	-0.8	-1.1	39.5	57.0	15.3	43.9
246 0.6DL+1.0WLX2	86.2	-1.7	-2.4	-0.7	-0.9	51.3	66.7	19.8	51.5
247 0.6DL+1.0WLZ1	87.8	-2.1	-2.9	-2.9	-3.9	42.8	15.9	16.5	12.2
248 0.6DL+1.0WLZ2	85.1	-1.9	-2.5	1.4	1.9	46.8	31.3	18.0	24.1
249 0.6DL+0.6FL1+0.7SLX1	87.2	-3.0	-4.1	-1.0	-1.3	30.1	46.6	11.6	36.0
250 0.6DL+0.6FL1+0.7SLX2	85.8	-1.0	-1.4	-0.7	-1.0	88.9	60.9	34.3	47.0
251 0.6DL+0.6FL1+0.7SLZ1	86.8	-2.1	-2.8	-1.1	-1.5	42.7	41.4	16.5	32.0
252 0.6DL+0.6FL1+0.7SLZ2	86.2	-1.9	-2.6	-0.6	-0.8	47.2	72.9	18.2	56.3
					Min.=	6.8	6.9	2.6	5.3
						OK	OK	OK	OK

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F7 / 7A

Load Combination & Calculation of Base pressure

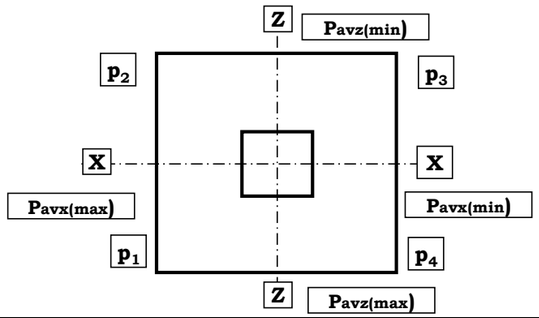
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 468 kN/m² For WL/SL case = 580.5 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2.Lz$	6.Mx/ $Lx.(Lz)^2$	Base Pressure (kN/m²) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0FL1	90.41	-4.51	-1.91	23.06	2.46	2.09	27.62	23.43	22.69	18.51	9.62	25.53	20.60	25.16	20.97	O.K
202	1.0DL+1.0FL2	90.43	-4.50	-1.97	23.07	2.46	2.16	27.69	23.37	22.76	18.45	9.69	25.53	20.61	25.23	20.91	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	126.98	-26.24	-12.83	32.39	14.35	14.02	60.76	32.72	32.07	4.03	42.76	46.74	18.05	46.42	18.38	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	127.00	-26.24	-12.89	32.40	14.34	14.09	60.83	32.66	32.14	3.97	42.83	46.75	18.06	46.49	18.32	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	117.84	-20.81	-10.10	30.06	11.38	11.04	52.48	30.40	29.72	7.65	34.48	41.44	18.69	41.10	19.03	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	117.86	-20.81	-10.16	30.07	11.37	11.11	52.55	30.33	29.80	7.59	34.55	41.44	18.70	41.18	18.96	O.K
207	1.0DL+1.0FL1+1.0PL	137.72	-4.91	0.10	35.13	2.68	0.11	37.93	37.71	32.56	32.34	19.93	37.82	32.45	35.25	35.03	O.K
208	1.0DL+1.0FL2+1.0PL	137.74	-4.91	0.04	35.14	2.68	0.04	37.86	37.78	32.50	32.41	19.86	37.82	32.46	35.18	35.10	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	174.30	-26.65	-10.81	44.46	14.57	11.82	70.85	47.21	41.72	18.08	52.85	59.03	29.90	56.29	32.65	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	174.32	-26.64	-10.87	44.47	14.57	11.89	70.92	47.15	41.79	18.02	52.92	59.04	29.91	56.36	32.59	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+	165.15	-21.21	-8.08	42.13	11.60	8.84	62.57	44.89	39.37	21.70	44.57	53.73	30.54	50.97	33.30	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+	165.17	-21.21	-8.14	42.14	11.59	8.90	62.63	44.83	39.45	21.64	44.63	53.73	30.55	51.04	33.24	O.K
213	1.0DL+1.0FL1+1.0WLX1	90.63	-4.86	-1.99	23.12	2.66	2.18	27.96	23.60	22.64	18.28	9.96	25.78	20.46	25.30	20.94	O.K
214	1.0DL+1.0FL1+1.0WLX2	90.19	-4.15	-1.83	23.01	2.27	2.00	27.28	23.27	22.74	18.74	9.28	25.28	20.74	25.01	21.01	O.K
215	1.0DL+1.0FL1+1.0WLZ1	91.77	-4.67	-4.80	23.41	2.55	5.25	31.22	20.71	26.11	15.61	13.22	25.97	20.86	28.67	18.16	O.K
216	1.0DL+1.0FL1+1.0WLZ2	89.04	-4.34	0.98	22.71	2.37	1.07	26.16	24.02	21.41	19.27	8.16	25.09	20.34	23.79	21.65	O.K
217	1.0DL+1.0FL2+1.0WLX1	90.65	-4.86	-2.05	23.12	2.66	2.25	28.03	23.53	22.71	18.22	10.03	25.78	20.47	25.37	20.88	O.K
218	1.0DL+1.0FL2+1.0WLX2	90.21	-4.14	-1.89	23.01	2.27	2.07	27.35	23.21	22.82	18.68	9.35	25.28	20.75	25.09	20.95	O.K
219	1.0DL+1.0FL2+1.0WLZ1	91.79	-4.67	-4.86	23.42	2.55	5.32	31.29	20.65	26.18	15.55	13.29	25.97	20.87	28.74	18.10	O.K
220	1.0DL+1.0FL2+1.0WLZ2	89.06	-4.34	0.92	22.72	2.37	1.00	26.09	24.09	21.35	19.35	8.09	25.09	20.35	23.72	21.72	O.K

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MD-216-3310-EG-CV-CAL-1213				Mansourah Massarah Gold Project													
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2				DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION										Date: 10.11.2023			
O19045_450_C_K_ 24091 _001_R2				AS BUILT										Rev: D01			
221 1.0DL+1.0FL1+0.7SLX1	91.12	-5.86	-2.07	23.24	3.20	2.27	28.72	24.18	22.31	17.77	10.72	26.45		20.04	25.52	20.98	O.K
222 1.0DL+1.0FL1+0.7SLX2	89.70	-3.15	-1.75	22.88	1.72	1.91	26.52	22.69	23.07	19.24	8.52	24.61		21.16	24.80	20.97	O.K
223 1.0DL+1.0FL1+0.7SLZ1	90.71	-4.65	-2.23	23.14	2.54	2.44	28.12	23.24	23.04	18.16	10.12	25.68		20.60	25.58	20.70	O.K
224 1.0DL+1.0FL1+0.7SLZ2	90.11	-4.36	-1.59	22.99	2.38	1.74	27.11	23.63	22.34	18.86	9.11	25.37		20.60	24.73	21.25	O.K
225 1.0DL+1.0FL2+0.7SLX1	91.14	-5.85	-2.14	23.25	3.20	2.33	28.79	24.12	22.38	17.71	10.79	26.46		20.05	25.59	20.92	O.K
226 1.0DL+1.0FL2+0.7SLX2	89.72	-3.15	-1.81	22.89	1.72	1.98	26.59	22.63	23.15	19.18	8.59	24.61		21.17	24.87	20.91	O.K
227 1.0DL+1.0FL2+0.7SLZ1	90.73	-4.65	-2.29	23.15	2.54	2.51	28.19	23.18	23.11	18.10	10.19	25.69		20.61	25.65	20.64	O.K
228 1.0DL+1.0FL2+0.7SLZ2	90.13	-4.36	-1.65	22.99	2.38	1.81	27.18	23.57	22.42	18.80	9.18	25.38		20.61	24.80	21.19	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75I	116.75	-19.31	-7.78	29.78	10.56	8.51	48.85	31.83	27.74	10.72	30.85	40.34		19.23	38.30	21.28	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75I	116.42	-18.77	-7.66	29.70	10.26	8.37	48.34	31.59	27.81	11.06	30.34	39.97		19.44	38.08	21.33	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75I	117.61	-19.16	-9.89	30.00	10.48	10.81	51.29	29.67	30.34	8.72	33.29	40.48		19.53	40.82	19.20	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75I	115.56	-18.92	-5.55	29.48	10.34	6.07	45.89	33.75	25.21	13.07	27.89	39.82		19.14	35.55	23.41	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75I	116.77	-19.31	-7.84	29.79	10.55	8.57	48.92	31.77	27.81	10.66	30.92	40.35		19.24	38.37	21.22	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75I	116.44	-18.77	-7.72	29.71	10.26	8.44	48.41	31.52	27.89	11.00	30.41	39.97		19.45	38.15	21.26	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75I	117.63	-19.16	-9.95	30.01	10.47	10.88	51.36	29.61	30.41	8.66	33.36	40.49		19.54	40.89	19.14	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75I	115.58	-18.91	-5.61	29.49	10.34	6.14	45.96	33.69	25.28	13.01	27.96	39.83		19.15	35.62	23.35	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	117.12	-20.06	-7.84	29.88	10.96	8.57	49.41	32.27	27.49	10.34	31.41	40.84		18.92	38.45	21.31	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	116.06	-18.03	-7.60	29.61	9.85	8.31	47.77	31.15	28.06	11.44	29.77	39.46		19.75	37.92	21.30	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	116.81	-19.15	-7.96	29.80	10.47	8.70	48.97	31.56	28.03	10.63	30.97	40.27		19.33	38.50	21.10	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	116.36	-18.93	-7.48	29.68	10.35	8.18	48.21	31.86	27.51	11.16	30.21	40.04		19.34	37.86	21.51	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	117.14	-20.05	-7.90	29.88	10.96	8.64	49.48	32.20	27.56	10.28	31.48	40.84		18.92	38.52	21.24	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	116.08	-18.02	-7.66	29.61	9.85	8.37	47.84	31.09	28.13	11.38	29.84	39.47		19.76	37.99	21.24	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	116.83	-19.14	-8.02	29.80	10.47	8.77	49.04	31.50	28.11	10.57	31.04	40.27		19.34	38.58	21.04	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	116.38	-18.93	-7.54	29.69	10.35	8.24	48.28	31.79	27.59	11.10	30.28	40.04		19.35	37.94	21.45	O.K
245 0.6DL+1.0WLX1	86.69	-3.07	-1.07	22.11	1.68	1.16	24.96	22.63	21.60	19.27	6.96	23.80		20.44	23.28	20.95	O.K
246 0.6DL+1.0WLX2	86.25	-2.35	-0.90	22.00	1.29	0.99	24.28	22.30	21.70	19.73	6.28	23.29		20.72	22.99	21.02	O.K
247 0.6DL+1.0WLZ1	87.83	-2.87	-3.87	22.41	1.57	4.24	28.21	19.74	25.07	16.60	10.21	23.98		20.84	26.64	18.17	O.K
248 0.6DL+1.0WLZ2	85.10	-2.55	1.90	21.71	1.39	2.08	25.18	21.02	22.40	18.24	7.18	23.10		20.32	23.79	19.63	O.K
249 0.6DL+0.6FL1+0.7SLX1	87.25	-4.06	-1.31	22.26	2.22	1.43	25.91	23.04	21.47	18.61	7.91	24.48		20.04	23.69	20.83	O.K
250 0.6DL+0.6FL1+0.7SLX2	85.83	-1.35	-0.99	21.89	0.74	1.08	23.71	21.55	22.23	20.08	5.71	22.63		21.16	22.97	20.82	O.K
251 0.6DL+0.6FL1+0.7SLZ1	86.84	-2.85	-1.47	22.15	1.56	1.60	25.31	22.11	22.20	18.99	7.31	23.71		20.60	23.76	20.55	O.K
252 0.6DL+0.6FL1+0.7SLZ2	86.24	-2.56	-0.83	22.00	1.40	0.90	24.30	22.49	21.50	19.69	6.30	23.40		20.60	22.90	21.09	O.K

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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Ultimate Conditions

101 1.4DL+1.4FL1	94.28	-6.31	-2.68	24.05	3.45	2.93	30.43	24.57	23.53	17.67	12.43	27.50	20.60	26.98	21.12	
102 1.4DL+1.4FL2	94.31	-6.30	-2.76	24.06	3.45	3.02	30.52	24.48	23.63	17.59	12.52	27.50	20.61	27.08	21.04	
103 1.2DL+1.2FL1+1.6LL	148.20	-36.41	-14.69	37.80	19.91	16.06	73.77	41.66	33.95	1.84	55.77	57.72	17.90	53.86	21.75	
104 1.2DL+1.2FL2+1.6LL	148.22	-36.41	-14.76	37.81	19.90	16.14	73.85	41.58	34.04	1.77	55.85	57.72	17.91	53.95	21.68	
105 1.2DL+1.2FL1+1.2TL+1.6LL	150.20	-39.24	-18.49	38.32	21.45	20.21	79.98	39.56	37.07	-3.35	61.98	59.77	16.86	58.53	18.11	
106 1.2DL+1.2FL2+1.2TL+1.6LL	150.22	-39.24	-18.56	38.32	21.45	20.29	80.06	39.48	37.16	-3.42	62.06	59.77	16.87	58.61	18.03	
107 1.4DL+1.4FL1+1.4PL	160.52	-6.88	0.14	40.95	3.76	0.15	44.86	44.55	37.34	37.04	26.86	44.71	37.19	41.10	40.80	
108 1.4DL+1.4FL2+1.4PL	160.55	-6.87	0.06	40.96	3.76	0.06	44.77	44.65	37.26	37.14	26.77	44.71	37.20	41.02	40.90	
109 1.2DL+1.2FL1+1.6LL+1.2PL	204.97	-36.90	-12.27	52.29	20.17	13.41	85.87	59.05	45.53	18.70	67.87	72.46	32.12	65.70	38.88	
110 1.2DL+1.2FL2+1.6LL+1.2PL	205.00	-36.89	-12.34	52.29	20.17	13.49	85.96	58.97	45.62	18.63	67.96	72.47	32.13	65.79	38.80	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	206.97	-39.73	-16.07	52.80	21.72	17.57	92.09	56.95	48.65	13.51	74.09	74.52	31.08	70.37	35.23	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	207.00	-39.73	-16.14	52.81	21.72	17.65	92.17	56.87	48.74	13.44	74.17	74.52	31.09	70.46	35.16	
113 1.2DL+0.8WLX1	92.38	-5.71	-2.03	23.57	3.12	2.22	28.91	24.46	22.67	18.22	10.91	26.69	20.45	25.79	21.34	
114 1.2DL+0.8WLX2	92.03	-5.14	-1.90	23.48	2.81	2.08	28.37	24.20	22.75	18.59	10.37	26.29	20.67	25.56	21.40	
115 1.2DL+0.8WLZ1	93.30	-5.55	-4.28	23.80	3.04	4.68	31.52	22.16	25.45	16.08	13.52	26.84	20.77	28.49	19.12	
116 1.2DL+0.8WLZ2	91.11	-5.29	0.34	23.24	2.89	0.37	26.51	25.76	20.73	19.98	8.51	26.14	20.36	23.62	22.87	
117 1.2DL+1.6WLX1+0.5LL	110.01	-15.68	-5.97	28.06	8.57	6.53	43.16	30.11	26.02	12.96	25.16	36.64	19.49	34.59	21.54	
118 1.2DL+1.6WLX2+0.5LL	109.31	-14.54	-5.71	27.88	7.95	6.24	42.08	29.59	26.18	13.69	24.08	35.84	19.94	34.13	21.64	
119 1.2DL+1.6WLZ1+0.5LL	111.85	-15.37	-10.47	28.53	8.40	11.44	48.38	25.49	31.57	8.69	30.38	36.94	20.13	39.98	17.09	
120 1.2DL+1.6WLZ2+0.5LL	107.47	-14.85	-1.22	27.42	8.12	1.33	36.86	34.20	20.63	17.97	18.86	35.53	19.30	28.75	26.09	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	110.73	-17.04	-6.21	28.25	9.31	6.79	44.35	30.77	25.72	12.15	26.35	37.56	18.94	35.04	21.46	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	108.70	-13.17	-5.75	27.73	7.20	6.28	41.21	28.65	26.81	14.25	23.21	34.93	20.53	34.01	21.45	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	110.15	-15.31	-6.43	28.10	8.37	7.03	43.50	29.43	26.76	12.69	25.50	36.47	19.73	35.13	21.06	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	109.29	-14.90	-5.52	27.88	8.14	6.04	42.06	29.99	25.77	13.70	24.06	36.03	19.74	33.92	21.85	
125 0.9DL+1.6WLX1	89.69	-4.64	-1.61	22.88	2.54	1.76	27.17	23.66	22.10	18.59	9.17	25.42	20.35	24.64	21.13	
126 0.9DL+1.6WLX2	88.99	-3.49	-1.35	22.70	1.91	1.47	26.08	23.14	22.27	19.32	8.08	24.61	20.80	24.18	21.23	
127 0.9DL+1.6WLZ1	91.52	-4.33	-6.10	23.35	2.37	6.67	32.38	19.04	27.65	14.31	14.38	25.71	20.98	30.02	16.68	
128 0.9DL+1.6WLZ2	87.15	-3.80	3.15	22.23	2.08	3.44	27.75	20.87	23.59	16.71	9.75	24.31	20.15	25.67	18.79	
129 0.9DL+1.0SLX1	90.35	-6.00	-1.71	23.05	3.28	1.87	28.19	24.46	21.64	17.90	10.19	26.33	19.77	24.92	21.18	
130 0.9DL+1.0SLX2	88.32	-2.13	-1.25	22.53	1.17	1.36	25.06	22.33	22.73	20.00	7.06	23.70	21.37	23.90	21.17	
131 0.9DL+1.0SLZ1	89.77	-4.27	-1.93	22.90	2.33	2.12	27.35	23.12	22.68	18.45	9.35	25.24	20.57	25.02	20.79	
132 0.9DL+1.0SLZ2	88.91	-3.86	-1.02	22.68	2.11	1.11	25.90	23.68	21.68	19.46	7.90	24.79	20.57	23.79	21.57	

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Max :	207	-1	3	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**9.62 kN/m^2**
Min :	85	-40	-19	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**19.62 kN/m^2**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**52.92 kN/m^2**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**62.92 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) =	23.07+2.46+2.16=	27.69 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(27.69+23.37)/2= 25.53 kN/m^2
p_2=	23.07+2.46-2.16=	23.37 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(18.45+22.76)/2= 20.61 kN/m^2
p_3=	23.07-(2.46+2.16)=	18.45 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(27.69+22.76)/2= 25.23 kN/m^2
p_4=	23.07-2.46+2.16=	22.76 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(18.45+23.37)/2= 20.91 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-γh)=	27.69-1*18=	9.69 kN/m^2		

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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DESIGN OF FOUNDATION MKD. F7 / 7A

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.45+2*0.5*0.232)+(0.55+2*0.5*0.232))*2

= 2.928 m

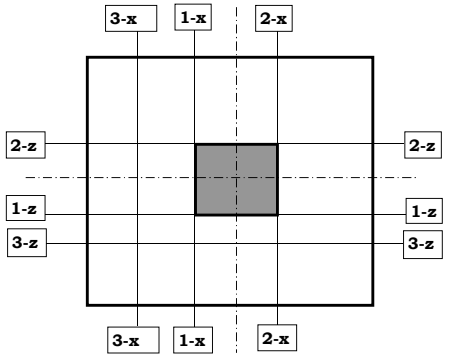
Punching Area= ((0.45+2*0.5*0.232)*(0.55+2*0.5*0.232))

= 0.53 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*24+0.7*18)

= 19.80 kN/m^2

Area of footing, A_f= 3.92 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M$_{1x}$ kNm	B.M$_{2x}$ kNm	S.F$_{3x}$ kN	S.F$_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	24.60	23.50	25.21	4.65	1.22	6.10	14.41
102	1.4DL+1.4FL2	24.61	23.50	25.21	4.65	1.22	6.10	14.43
103	1.2DL+1.2FL1+1.6LL	41.01	34.61	44.48	22.33	2.53	29.14	61.04
104	1.2DL+1.2FL2+1.6LL	41.01	34.62	44.48	22.33	2.54	29.14	61.06
105	1.2DL+1.2FL1+1.2TL+1.6LL	41.76	34.87	45.50	23.45	2.11	30.57	62.77
106	1.2DL+1.2FL2+1.2TL+1.6LL	41.77	34.87	45.51	23.45	2.12	30.57	62.79
107	1.4DL+1.4FL1+1.4PL	41.55	40.35	42.21	16.47	12.73	22.03	71.69
108	1.4DL+1.4FL2+1.4PL	41.56	40.35	42.21	16.47	12.74	22.03	71.72
109	1.2DL+1.2FL1+1.6LL+1.2PL	55.53	49.05	59.05	32.46	12.40	42.78	110.14
110	1.2DL+1.2FL2+1.6LL+1.2PL	55.54	49.06	59.06	32.46	12.41	42.79	110.16
111	1.2DL+1.2FL1+1.2TL+1.6LL+	56.29	49.31	60.08	33.58	11.98	44.22	111.87
112	1.2DL+1.2FL2+1.2TL+1.6LL+	56.29	49.32	60.08	33.58	11.99	44.22	111.89

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MA'ADEN Saudi Arabian Mining Company			Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project		
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION		Date: 10.11.2023
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113 1.2DL+0.8WLX1	24.07	23.07	24.62	4.15	1.05	5.45	12.77
114 1.2DL+0.8WLX2	23.93	23.03	24.42	3.94	1.14	5.17	12.46
115 1.2DL+0.8WLZ1	24.29	23.32	24.82	4.27	1.26	5.61	13.56
116 1.2DL+0.8WLZ2	23.71	22.79	24.22	3.82	0.94	5.01	11.67
117 1.2DL+1.6WLX1+0.5LL	29.44	26.69	30.94	9.97	1.44	13.02	28.01
118 1.2DL+1.6WLX2+0.5LL	29.17	26.61	30.55	9.54	1.63	12.47	27.41
119 1.2DL+1.6WLZ1+0.5LL	29.89	27.18	31.35	10.21	1.85	13.36	29.60
120 1.2DL+1.6WLZ2+0.5LL	28.72	26.11	30.13	9.29	1.22	12.13	25.82
121 1.2DL+1.0SLX1+0.5LL+0.5FL	29.75	26.75	31.37	10.46	1.20	13.65	28.64
122 1.2DL+1.0SLX2+0.5LL+0.5FL	28.89	26.57	30.14	9.05	1.89	11.86	26.88
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	29.45	26.75	30.90	9.89	1.57	12.93	28.13
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	29.19	26.58	30.61	9.63	1.53	12.59	27.39
125 0.9DL+1.6WLX1	23.29	22.48	23.73	3.39	0.87	4.45	10.44
126 0.9DL+1.6WLX2	23.01	22.40	23.34	2.95	1.06	3.89	9.83
127 0.9DL+1.6WLZ1	23.73	22.96	24.14	3.62	1.27	4.77	12.03
128 0.9DL+1.6WLZ2	22.56	21.90	22.93	2.71	0.64	3.55	8.24
129 0.9DL+1.0SLX1	23.58	22.52	24.15	3.87	0.61	5.06	11.01
130 0.9DL+1.0SLX2	22.72	22.35	22.93	2.47	1.31	3.27	9.26
131 0.9DL+1.0SLZ1	23.28	22.53	23.69	3.30	0.98	4.34	10.51
132 0.9DL+1.0SLZ2	23.02	22.34	23.39	3.04	0.94	3.99	9.76

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} =	27.5-(27.5-20.6)*1.175/2.8 =	24.60 kN/m^2	Overhang =	1.175 m
p_{2x} =	20.6+(27.5-20.6)*1.175/2.8 =	23.50 kN/m^2	Effective depth =	0.244 m
p_{3x} =	27.5-(27.5-20.6)*(1.175-0.244)/2.8 =	25.21 kN/m^2		
BM_{1x} =	(24.60-19.8)*1.175*1.175/2+0.5*(27.5-24.60)*1.175*1.175*0.67 =	4.65 kN-m/m		
BM_{2x} =	(23.50-19.8)*1.175*1.175/2-0.5*(-20.6+23.50)*1.175*1.175*0.67 =	1.22 kN-m/m		
SF_{3x} =	(25.21-19.8)*(1.175-0.244)+0.5*(27.5-25.21)*(1.175-0.244) =	6.10 kN/m		
$S.F_{(punching)}$ =	(3.92-0.53)*(24.05-19.8) =	14.41 kN		

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	33.58	0.56	0.75	0.002	442.57	439.20	12ϕT	255.55	300	200	565.49
At top	0.00	0.00	0.00	0.000	0.00	219.60	10ϕT	357.65	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 44.22 kN/m

Shear stress , τ_v= (44.22*1000)/(1000*244)

= 0.18 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 111.89 kN

Strength of Concrete in shear Vc1 = 2426.40 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1291.47 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1408.88 kN **Table 22.6.5.2_ (c)**

Design Vc= 1291.47 kN

Allowable shear stress, (ϕ Vc)/bd = 1.36 N/mm^2

Design punching shear stress, τ = (111.89*1000)/(2928*244)

= 0.16 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL+1.4FL1	25.20	22.90	26.17	0.60	0.17	1.31
102	1.4DL+1.4FL2	25.25	25.35	26.25	0.60	1.09	1.32
103	1.2DL+1.2FL1+1.6LL	44.11	25.83	49.43	2.79	1.11	6.15
104	1.2DL+1.2FL2+1.6LL	44.15	32.19	49.50	2.79	1.15	6.16
105	1.2DL+1.2FL1+1.2TL+1.6LL	46.26	29.72	52.96	3.13	1.07	6.94
106	1.2DL+1.2FL2+1.2TL+1.6LL	46.29	28.03	53.02	3.14	1.06	6.95
107	1.4DL+1.4FL1+1.4PL	41.01	43.91	41.06	1.92	1.53	4.11
108	1.4DL+1.4FL2+1.4PL	40.98	36.16	41.00	1.92	0.56	4.09
109	1.2DL+1.2FL1+1.6LL+1.2PL	57.56	34.67	62.00	3.90	0.52	8.50
110	1.2DL+1.2FL2+1.6LL+1.2PL	57.60	35.72	62.07	3.91	0.38	8.52
111	1.2DL+1.2FL1+1.2TL+1.6LL+	59.70	31.76	65.53	4.25	0.47	9.29
112	1.2DL+1.2FL2+1.2TL+1.6LL+	59.74	32.19	65.59	4.26	0.43	9.31

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
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113 1.2DL+0.8WLX1	24.44	22.48	25.18	0.50	0.15	1.10
114 1.2DL+0.8WLX2	24.30	23.63	24.99	0.48	0.01	1.06
115 1.2DL+0.8WLZ1	25.65	20.52	27.20	0.70	0.14	1.55
116 1.2DL+0.8WLZ2	23.39	23.67	23.52	0.34	0.19	0.73
117 1.2DL+1.6WLX1+0.5LL	30.63	22.53	32.79	1.22	0.15	2.68
118 1.2DL+1.6WLX2+0.5LL	30.34	22.84	32.41	1.18	0.14	2.60
119 1.2DL+1.6WLZ1+0.5LL	33.03	19.41	36.82	1.62	0.08	3.59
120 1.2DL+1.6WLZ2+0.5LL	27.94	25.94	28.38	0.78	0.25	1.69
121 1.2DL+1.0SLX1+0.5LL+0.5FL	30.92	22.50	33.17	1.25	0.15	2.76
122 1.2DL+1.0SLX2+0.5LL+0.5FL	30.20	22.73	32.28	1.17	0.14	2.58
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	30.86	22.20	33.19	1.26	0.16	2.77
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	30.26	26.84	32.26	1.17	0.30	2.56
125 0.9DL+1.6WLX1	23.57	26.28	24.16	0.41	0.29	0.89
126 0.9DL+1.6WLX2	23.28	27.18	23.77	0.37	0.18	0.81
127 0.9DL+1.6WLZ1	25.97	22.41	28.18	0.80	0.30	1.79
128 0.9DL+1.6WLZ2	23.58	24.73	24.72	0.47	0.23	1.04
129 0.9DL+1.0SLX1	23.78	26.33	24.40	0.43	0.28	0.94
130 0.9DL+1.0SLX2	23.07	27.16	23.52	0.35	0.18	0.75
131 0.9DL+1.0SLZ1	23.74	25.29	24.44	0.43	0.38	0.95
132 0.9DL+1.0SLZ2	23.12	26.69	23.48	0.34	0.30	0.74

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 26.98-(26.98-21.12)*0.425/1.4= 25.20 kN/m^2

p_{2z} = 21.12+(26.98-21.12)*0.425/1.4= 22.90 kN/m^2 Overhang = 0.425 m

p_{3z} = 26.98-(26.98-21.12)*(0.425-0.232)/1.4= 26.17 kN/m^2 Effective depth = 0.232 m

BM_{1z} = (25.20-19.8)*0.425*0.425/2+0.5*(26.98-25.20)*0.425*0.425*0.67 = 0.60 kN-m/m

BM_{2z} = (22.90-19.8)*0.425*0.425/2-0.5*(-21.12+22.90)*0.425*0.425*0.67 = 0.17 kN-m/m

SF_{3z} = (26.17-19.8)*(0.425-0.232)+0.5*(26.98-26.17)*(0.425-0.232) = 1.31 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	4.26	0.079	0.105	0.0003	61.36	417.60	12ϕT	270.83	300	200	565.5
At top	0.00	0.000	0.000	0.0000	0.00	208.80	12ϕT	541.65	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 9.31 kN/m

Shear stress , τ_v= (9.31*1000)/(1000*232)

= 0.04 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **200** c/c

At Top **10ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **200** c/c

At Top **12ϕT** **200** c/c

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MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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DESIGN OF FOUNDATION MKD. F11 STAAD Node - 7763

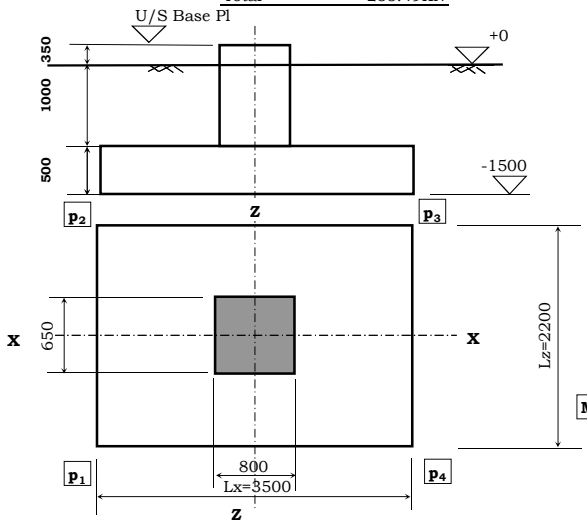
	Along-X	Along-Z
Size of column :	0.80m	0.65m
Size of fdn. raft :	3.50m	2.20m
Thickness of raft :	0.50 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

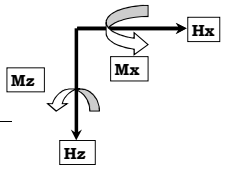
SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3.5*2.2*0.5*24)+(0.8*0.65*1.35*24) =	109.25KN
Weight of soil =	((3.5*2.2)-(0.8*0.65))*1*18 =	129.24KN
	Total =	238.49KN





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 442mm = 0.442m

Effective depth of slab along Z direction = 426mm = 0.426m

Clear projection along Z direction = 0.775m

Clear projection along X direction = 1.35m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. **F11**

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0FL1	445.5	77.6	0.0	0.0	0.0	684.00	77.62	143.59	0.00	0.00
202	1.0DL+1.0FL2	494.9	86.3	0.0	0.0	0.0	733.35	86.32	159.69	0.00	0.00
203	1.0DL+1.0FL1+1.0LL+1.0TL	424.1	113.2	0.0	0.0	0.0	662.56	113.16	209.35	0.00	-0.01
204	1.0DL+1.0FL2+1.0LL+1.0TL	473.4	121.9	0.0	0.0	0.0	711.91	121.86	225.44	0.00	-0.01
205	1.0DL+1.0FL1+0.75LL+0.75TL	429.4	104.3	0.0	0.0	0.0	667.92	104.28	192.91	0.00	0.00
206	1.0DL+1.0FL2+0.75LL+0.75TL	478.8	113.0	0.0	0.0	0.0	717.27	112.98	209.01	0.00	0.00
207	1.0DL+1.0FL1+1.0PL	479.4	81.6	0.0	0.0	0.0	717.94	81.65	151.05	0.01	0.01
208	1.0DL+1.0FL2+1.0PL	528.8	90.3	0.0	0.0	0.0	767.29	90.35	167.14	0.01	0.01
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	458.0	117.2	0.0	0.0	0.0	696.50	117.19	216.80	0.00	0.01
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	507.4	125.9	0.0	0.0	0.0	745.85	125.89	232.90	0.00	0.01
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	463.4	108.3	0.0	0.0	0.0	701.86	108.31	200.36	0.00	0.01
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	512.7	117.0	0.0	0.0	0.0	751.21	117.00	216.46	0.00	0.01
213	1.0DL+1.0FL1+1.0WLX1	368.2	54.5	0.0	0.0	0.0	606.64	54.53	100.87	0.00	0.00
214	1.0DL+1.0FL1+1.0WLX2	522.8	100.7	0.0	0.0	0.0	761.32	100.70	186.29	0.00	0.00
215	1.0DL+1.0FL1+1.0WLZ1	451.8	82.2	0.0	-5.4	0.0	690.30	82.15	151.98	-5.43	-10.04
216	1.0DL+1.0FL1+1.0WLZ2	439.2	73.1	0.0	5.4	0.0	677.70	73.09	135.21	5.43	10.04
217	1.0DL+1.0FL2+1.0WLX1	417.5	63.2	0.0	0.0	0.0	656.00	63.23	116.97	0.00	0.00
218	1.0DL+1.0FL2+1.0WLX2	572.2	109.4	0.0	0.0	0.0	810.67	109.40	202.38	0.00	0.00
219	1.0DL+1.0FL2+1.0WLZ1	501.2	90.9	0.0	-5.4	0.0	739.66	90.85	168.08	-5.43	-10.04
220	1.0DL+1.0FL2+1.0WLZ2	488.6	81.8	0.0	5.4	0.0	727.06	81.78	151.30	5.43	10.04
221	1.0DL+1.0FL1+0.7SLX1	343.2	49.2	0.0	0.0	0.0	581.70	49.24	91.09	0.00	0.00
222	1.0DL+1.0FL1+0.7SLX2	547.8	106.0	0.0	0.0	0.0	786.30	106.00	196.10	0.00	0.00
223	1.0DL+1.0FL1+0.7SLZ1	466.6	87.0	0.0	0.0	0.0	705.06	87.01	160.96	-0.01	-0.01
224	1.0DL+1.0FL1+0.7SLZ2	424.4	68.2	0.0	0.0	0.0	662.94	68.23	126.23	0.00	0.01
225	1.0DL+1.0FL2+0.7SLX1	392.6	57.9	0.0	0.0	0.0	631.05	57.94	107.18	0.00	0.00
226	1.0DL+1.0FL2+0.7SLX2	597.2	114.7	0.0	0.0	0.0	835.66	114.70	212.19	0.00	0.00
227	1.0DL+1.0FL2+0.7SLZ1	515.9	95.7	0.0	0.0	0.0	754.42	95.71	177.05	-0.01	-0.01
228	1.0DL+1.0FL2+0.7SLZ2	473.8	76.9	0.0	0.0	0.0	712.29	76.93	142.32	0.00	0.01
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	391.6	62.1	0.0	0.0	0.0	630.11	62.15	114.97	0.00	0.00
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	507.6	96.8	0.0	0.0	0.0	746.12	96.78	179.04	0.00	0.00
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	454.4	82.9	0.0	-4.1	0.0	692.86	82.87	153.31	-4.07	-7.53
232	1.0DL+1.0FL1+0.75WLZ2+0.75LL	444.9	76.1	0.0	4.1	0.0	683.41	76.07	140.72	4.07	7.53

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date:10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

233 1.0DL+1.0FL2+0.75WLX1+0.75LL	441.0	70.8	0.0	0.0	0.0	679.47	70.85	131.07	0.00	0.00
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	557.0	105.5	0.0	0.0	0.0	795.48	105.48	195.13	0.00	0.00
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	503.7	91.6	0.0	-4.1	0.0	742.21	91.57	169.40	-4.07	-7.53
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	494.3	84.8	0.0	4.1	0.0	732.77	84.77	156.82	4.07	7.53
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	372.9	58.2	0.0	0.0	0.0	611.41	58.18	107.63	0.00	0.00
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	526.4	100.8	0.0	0.0	0.0	764.86	100.75	186.39	0.00	0.00
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	465.4	86.5	0.0	0.0	0.0	703.93	86.51	160.04	0.00	-0.01
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	433.8	72.4	0.0	0.0	0.0	672.34	72.43	133.99	0.00	0.00
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	422.3	66.9	0.0	0.0	0.0	660.76	66.88	123.73	0.00	0.00
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	575.7	109.5	0.0	0.0	0.0	814.21	109.45	202.48	0.00	0.00
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	514.8	95.2	0.0	0.0	0.0	753.28	95.21	176.13	0.00	-0.01
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	483.2	81.1	0.0	0.0	0.0	721.69	81.12	150.08	0.00	0.00
245 0.6DL+1.0WLX1	-24.0	-13.6	0.0	0.0	0.0	214.47	-13.57	-25.10	0.00	0.00
246 0.6DL+1.0WLX2	130.7	32.6	0.0	0.0	0.0	369.15	32.61	60.32	0.00	0.00
247 0.6DL+1.0WLZ1	59.6	14.1	0.0	-5.4	0.0	298.13	14.06	26.01	-5.43	-10.04
248 0.6DL+1.0WLZ2	47.0	5.0	0.0	5.4	0.0	285.53	4.99	9.24	5.43	10.04
249 0.6DL+0.6FL1+0.7SLX1	165.0	18.2	0.0	0.0	0.0	403.49	18.19	33.65	0.00	0.00
250 0.6DL+0.6FL1+0.7SLX2	369.6	75.0	0.0	0.0	0.0	608.10	74.95	138.66	0.00	0.00
251 0.6DL+0.6FL1+0.7SLZ1	288.4	56.0	0.0	0.0	0.0	526.86	55.96	103.52	-0.01	-0.01
252 0.6DL+0.6FL1+0.7SLZ2	246.2	37.2	0.0	0.0	0.0	484.73	37.18	68.79	0.00	0.01
Ultimate Conditions										
101 1.4DL+1.4FL1	623.7	108.7	0.0	0.0	0.0	862.20	108.67	201.03	0.00	0.00
102 1.4DL+1.4FL2	692.8	120.8	0.0	0.0	0.0	931.30	120.84	223.56	0.00	0.00
103 1.2DL+1.2FL1+1.6LL	543.4	97.1	0.0	0.0	0.0	781.92	97.09	179.61	0.00	0.00
104 1.2DL+1.2FL2+1.6LL	602.7	107.5	0.0	0.0	0.0	841.14	107.52	198.92	0.00	0.00
105 1.2DL+1.2FL1+1.2TL+1.6LL	511.1	136.8	0.0	0.0	0.0	749.58	136.78	253.05	0.00	-0.01
106 1.2DL+1.2FL2+1.2TL+1.6LL	570.3	147.2	0.0	0.0	0.0	808.80	147.22	272.36	0.00	-0.01
107 1.4DL+1.4FL1+1.4PL	671.2	114.3	0.0	0.0	0.0	909.72	114.31	211.46	0.01	0.01
108 1.4DL+1.4FL2+1.4PL	740.3	126.5	0.0	0.0	0.0	978.82	126.48	233.99	0.01	0.01
109 1.2DL+1.2FL1+1.6LL+1.2PL	584.2	101.9	0.0	0.0	0.0	822.65	101.92	188.55	0.01	0.01
110 1.2DL+1.2FL2+1.6LL+1.2PL	643.4	112.4	0.0	0.0	0.0	881.87	112.36	207.86	0.01	0.01
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	551.8	141.6	0.0	0.0	0.0	790.30	141.62	261.99	0.00	0.01

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	611.0	152.1	0.0	0.0	0.0	849.53	152.05	281.30	0.00	0.01
113 1.2DL+0.8WLX1	44.8	0.6	0.0	0.0	0.0	283.28	0.58	1.07	0.00	0.00
114 1.2DL+0.8WLX2	168.5	37.5	0.0	0.0	0.0	407.02	37.52	69.40	0.00	0.00
115 1.2DL+0.8WLZ1	111.7	22.7	0.0	-4.3	0.0	350.20	22.68	41.96	-4.34	-8.04
116 1.2DL+0.8WLZ2	101.6	15.4	0.0	4.3	0.0	340.12	15.42	28.53	4.34	8.03
117 1.2DL+1.6WLX1+0.5LL	-14.3	-16.7	0.0	0.0	0.0	224.15	-16.66	-30.83	0.00	-0.01
118 1.2DL+1.6WLX2+0.5LL	233.1	57.2	0.0	0.0	0.0	471.63	57.21	105.84	0.00	0.00
119 1.2DL+1.6WLZ1+0.5LL	119.5	27.5	0.0	-8.7	0.0	358.00	27.54	50.95	-8.69	-16.07
120 1.2DL+1.6WLZ2+0.5LL	99.4	13.0	0.0	8.7	0.0	337.84	13.03	24.11	8.69	16.07
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	141.6	10.6	0.0	0.0	0.0	380.08	10.61	19.63	0.00	0.00
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	433.9	91.7	0.0	0.0	0.0	672.37	91.70	169.64	0.00	0.00
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	317.8	64.6	0.0	0.0	0.0	556.31	64.57	119.45	-0.01	-0.01
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	257.6	37.7	0.0	0.0	0.0	496.14	37.74	69.82	0.01	0.01
125 0.9DL+1.6WLX1	-43.8	-22.7	0.0	0.0	0.0	194.72	-22.66	-41.92	0.00	0.00
126 0.9DL+1.6WLX2	203.7	51.2	0.0	0.0	0.0	442.21	51.22	94.75	0.00	0.00
127 0.9DL+1.6WLZ1	90.1	21.5	0.0	-8.7	0.0	328.58	21.55	39.86	-8.69	-16.07
128 0.9DL+1.6WLZ2	69.9	7.0	0.0	8.7	0.0	308.42	7.04	13.01	8.69	16.07
129 0.9DL+1.0SLX1	-66.1	-26.3	0.0	0.0	0.0	172.35	-26.26	-48.57	0.00	0.00
130 0.9DL+1.0SLX2	226.1	54.8	0.0	0.0	0.0	464.64	54.83	101.44	0.00	0.00
131 0.9DL+1.0SLZ1	110.1	27.7	0.0	0.0	0.0	348.58	27.70	51.25	-0.01	-0.01
132 0.9DL+1.0SLZ2	49.9	0.9	0.0	0.0	0.0	288.40	0.88	1.62	0.01	0.01

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0FL1	684.0	77.6	143.6	0.0	0.0	8.3	406701.5	3.5	273599.2
202	1.0DL+1.0FL2	733.4	86.3	159.7	0.0	0.0	8.0	4360483	3.4	2933416
203	1.0DL+1.0FL1+1.0LL+1.0TL	662.6	113.2	209.4	0.0	0.0	5.5	131317.8	2.3	88341.1
204	1.0DL+1.0FL2+1.0LL+1.0TL	711.9	121.9	225.4	0.0	0.0	5.5	141100.1	2.3	94921.9
205	1.0DL+1.0FL1+0.75LL+0.75TL	667.9	104.3	192.9	0.0	0.0	6.1	198570.2	2.6	133583.6
206	1.0DL+1.0FL2+0.75LL+0.75TL	717.3	113.0	209.0	0.0	0.0	6.0	213243.6	2.5	143454.8
207	1.0DL+1.0FL1+1.0PL	717.9	81.6	151.0	0.0	0.0	8.3	71147.0	3.5	47862.5
208	1.0DL+1.0FL2+1.0PL	767.3	90.3	167.1	0.0	0.0	8.0	76038.1	3.4	51152.9
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	696.5	117.2	216.8	0.0	0.0	5.6	103533.5	2.4	69649.8
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	745.9	125.9	232.9	0.0	0.0	5.6	110870.2	2.4	74585.4
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	701.9	108.3	200.4	0.0	0.0	6.1	104330.2	2.6	70185.8

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212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	751.2	117.0	216.5	0.0	0.0	6.1	111666.9	2.6	75121.4
213 1.0DL+1.0FL1+1.0WLX1	606.6	54.5	100.9	0.0	0.0	10.5	180353.0	4.5	121328.4
214 1.0DL+1.0FL1+1.0WLX2	761.3	100.7	186.3	0.0	0.0	7.2	452675.0	3.0	304526.8
215 1.0DL+1.0FL1+1.0WLZ1	690.3	82.2	152.0	-5.4	-10.0	7.9	75.6	3.4	50.9
216 1.0DL+1.0FL1+1.0WLZ2	677.7	73.1	135.2	5.4	10.0	8.8	74.2	3.7	49.9
217 1.0DL+1.0FL2+1.0WLX1	656.0	63.2	117.0	0.0	0.0	9.8	195026.1	4.2	131199.4
218 1.0DL+1.0FL2+1.0WLX2	810.7	109.4	202.4	0.0	0.0	7.0	482021.8	3.0	324269.2
219 1.0DL+1.0FL2+1.0WLZ1	739.7	90.9	168.1	-5.4	-10.0	7.7	81.0	3.3	54.5
220 1.0DL+1.0FL2+1.0WLZ2	727.1	81.8	151.3	5.4	10.0	8.4	79.6	3.6	53.6
221 1.0DL+1.0FL1+0.7SLX1	581.7	49.2	91.1	0.0	0.0	11.2	345873.3	4.7	232678.4
222 1.0DL+1.0FL1+0.7SLX2	786.3	106.0	196.1	0.0	0.0	7.0	4675291.4	3.0	3145196.0
223 1.0DL+1.0FL1+0.7SLZ1	705.1	87.0	161.0	0.0	0.0	7.7	83845.0	3.2	56404.8
224 1.0DL+1.0FL1+0.7SLZ2	662.9	68.2	126.2	0.0	0.0	9.2	98544.5	3.9	66293.6
225 1.0DL+1.0FL2+0.7SLX1	631.1	57.9	107.2	0.0	0.0	10.3	375220.1	4.4	252420.8
226 1.0DL+1.0FL2+0.7SLX2	835.7	114.7	212.2	0.0	0.0	6.9	4968759	2.9	3342620
227 1.0DL+1.0FL2+0.7SLZ1	754.4	95.7	177.1	0.0	0.0	7.5	89714.3	3.2	60353.3
228 1.0DL+1.0FL2+0.7SLZ2	712.3	76.9	142.3	0.0	0.0	8.8	105881.1	3.7	71229.1
229 1.0DL+1.0FL1+0.75WLX1+0.75LL	630.1	62.1	115.0	0.0	0.0	9.6	187331.2	4.1	126022.8
230 1.0DL+1.0FL1+0.75WLX2+0.75LL	746.1	96.8	179.0	0.0	0.0	7.3	4436395.1	3.1	2984484
231 1.0DL+1.0FL1+0.75WLZ1+0.75LL	692.9	82.9	153.3	-4.1	-7.5	7.9	101.2	3.3	68.1
232 1.0DL+1.0FL1+0.75WLZ2+0.75LL	683.4	76.1	140.7	4.1	7.5	8.5	99.8	3.6	67.1
233 1.0DL+1.0FL2+0.75WLX1+0.75LL	679.5	70.8	131.1	0.0	0.0	9.1	202004.6	3.8	135894.0
234 1.0DL+1.0FL2+0.75WLX2+0.75LL	795.5	105.5	195.1	0.0	0.0	7.1	4729863	3.0	3181908.0
235 1.0DL+1.0FL2+0.75WLZ1+0.75LL	742.2	91.6	169.4	-4.1	-7.5	7.7	108.4	3.2	72.9
236 1.0DL+1.0FL2+0.75WLZ2+0.75LL	732.8	84.8	156.8	4.1	7.5	8.2	107.0	3.5	72.0
237 1.0DL+1.0FL1+0.525SLX1+0.75LL	611.4	58.2	107.6	0.0	0.0	9.9	363538.1	4.2	244562.0
238 1.0DL+1.0FL1+0.525SLX2+0.75LL	764.9	100.8	186.4	0.0	0.0	7.2	4547798	3.0	3059428
239 1.0DL+1.0FL1+0.525SLZ1+0.75LL	703.9	86.5	160.0	0.0	0.0	7.7	104637.9	3.3	70392.8
240 1.0DL+1.0FL1+0.525SLZ2+0.75LL	672.3	72.4	134.0	0.0	0.0	8.8	199883.4	3.7	134467.0
241 1.0DL+1.0FL2+0.525SLX1+0.75LL	660.8	66.9	123.7	0.0	0.0	9.3	392884.9	4.0	264304.4
242 1.0DL+1.0FL2+0.525SLX2+0.75LL	814.2	109.5	202.5	0.0	0.0	7.0	4841266.5	3.0	3256852
243 1.0DL+1.0FL2+0.525SLZ1+0.75LL	753.3	95.2	176.1	0.0	0.0	7.5	111974.6	3.2	75328.4
244 1.0DL+1.0FL2+0.525SLZ2+0.75LL	721.7	81.1	150.1	0.0	0.0	8.4	214556.5	3.6	144338.0
245 0.6DL+1.0WLX1	214.5	-13.6	-25.1	0.0	0.0	15.0	63761.1	6.3	42893.8
246 0.6DL+1.0WLX2	369.1	32.6	60.3	0.0	0.0	10.7	219491.6	4.5	147658.0
247 0.6DL+1.0WLZ1	298.1	14.1	26.0	-5.4	-10.0	20.1	32.7	8.5	22.0
248 0.6DL+1.0WLZ2	285.5	5.0	9.2	5.4	10.0	54.1	31.3	22.9	21.0
249 0.6DL+0.6FL1+0.7SLX1	403.5	18.2	33.7	0.0	0.0	21.0	239914.8	8.9	161397.2
250 0.6DL+0.6FL1+0.7SLX2	608.1	75.0	138.7	0.0	0.0	7.7	3615705.9	3.2	2432384
251 0.6DL+0.6FL1+0.7SLZ1	526.9	56.0	103.5	0.0	0.0	8.9	62653.3	3.8	42148.6
252 0.6DL+0.6FL1+0.7SLZ2	484.7	37.2	68.8	0.0	0.0	12.3	72054.8	5.2	48473.2
					Min.=	5.5	31.3	2.3	21.0
						OK	OK	OK	OK

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DESIGN OF FOUNDATION MKD. F11

Load Combination & Calculation of Base pressure

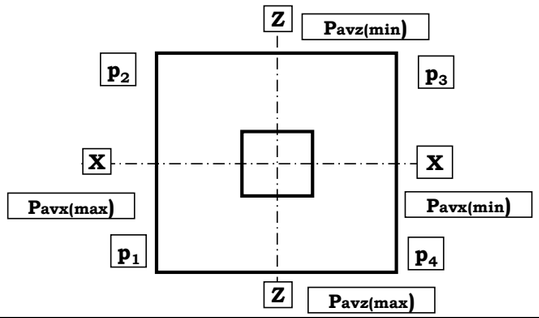
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 477 kN/m^2 For WL/SL case = 589.5 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)^2.Lz	6.Mx/Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0FL1	684.00	143.59	0.00	88.83	31.97	0.00	120.80	120.80	56.86	56.86	93.80	120.80	56.86	88.83	88.83	O.K
202	1.0DL+1.0FL2	733.35	159.69	0.00	95.24	35.55	0.00	130.79	130.79	59.69	59.69	103.79	130.79	59.69	95.24	95.24	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	662.56	209.35	-0.01	86.05	46.61	0.00	132.66	132.65	39.44	39.44	105.66	132.66	39.44	86.05	86.05	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	711.91	225.44	-0.01	92.46	50.19	0.00	142.65	142.65	42.27	42.26	115.65	142.65	42.27	92.46	92.46	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	667.92	192.91	0.00	86.74	42.95	0.00	129.69	129.69	43.79	43.79	102.69	129.69	43.79	86.74	86.74	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	717.27	209.01	0.00	93.15	46.53	0.00	139.69	139.68	46.62	46.62	112.69	139.69	46.62	93.16	93.15	O.K
207	1.0DL+1.0FL1+1.0PL	717.94	151.05	0.01	93.24	33.63	0.00	126.87	126.86	59.61	59.61	99.87	126.87	59.61	93.24	93.24	O.K
208	1.0DL+1.0FL2+1.0PL	767.29	167.14	0.01	99.65	37.21	0.00	136.86	136.86	62.44	62.43	109.86	136.86	62.44	99.65	99.65	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	696.50	216.80	0.01	90.45	48.27	0.00	138.72	138.72	42.19	42.18	111.72	138.72	42.19	90.46	90.45	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	745.85	232.90	0.01	96.86	51.85	0.00	148.72	148.71	45.02	45.01	121.72	148.72	45.02	96.87	96.86	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+	701.86	200.36	0.01	91.15	44.61	0.00	135.76	135.76	46.55	46.54	108.76	135.76	46.55	91.16	91.15	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+	751.21	216.46	0.01	97.56	48.19	0.00	145.75	145.75	49.37	49.37	118.75	145.75	49.37	97.56	97.56	O.K
213	1.0DL+1.0FL1+1.0WLX1	606.64	100.87	0.00	78.78	22.46	0.00	101.24	101.24	56.33	56.33	74.24	101.24	56.33	78.79	78.79	O.K
214	1.0DL+1.0FL1+1.0WLX2	761.32	186.29	0.00	98.87	41.48	0.00	140.35	140.35	57.40	57.40	113.35	140.35	57.40	98.88	98.88	O.K
215	1.0DL+1.0FL1+1.0WLZ1	690.30	151.98	-10.04	89.65	33.84	3.56	127.04	119.93	59.37	52.25	100.04	123.49	55.81	93.21	86.09	O.K
216	1.0DL+1.0FL1+1.0WLZ2	677.70	135.21	10.04	88.01	30.10	3.56	121.67	114.56	61.47	54.35	94.67	118.12	57.91	91.57	84.46	O.K
217	1.0DL+1.0FL2+1.0WLX1	656.00	116.97	0.00	85.19	26.04	0.00	111.24	111.23	59.16	59.15	84.24	111.24	59.16	85.20	85.19	O.K
218	1.0DL+1.0FL2+1.0WLX2	810.67	202.38	0.00	105.28	45.06	0.00	150.34	150.34	60.23	60.22	123.34	150.34	60.23	105.29	105.28	O.K
219	1.0DL+1.0FL2+1.0WLZ1	739.66	168.08	-10.04	96.06	37.42	3.56	137.04	129.92	62.20	55.08	110.04	133.48	58.64	99.62	92.50	O.K
220	1.0DL+1.0FL2+1.0WLZ2	727.06	151.30	10.04	94.42	33.68	3.56	131.66	124.55	64.30	57.18	104.66	128.11	60.74	97.98	90.87	O.K

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

221 1.0DL+1.0FL1+0.7SLX1	581.70	91.09	0.00	75.54	20.28	0.00	95.83	95.82	55.27	55.26	68.83	95.83	55.27	75.55	75.54	O.K
222 1.0DL+1.0FL1+0.7SLX2	786.30	196.10	0.00	102.12	43.66	0.00	145.78	145.78	58.46	58.46	118.78	145.78	58.46	102.12	102.12	O.K
223 1.0DL+1.0FL1+0.7SLZ1	705.06	160.96	-0.01	91.57	35.84	0.00	127.41	127.40	55.73	55.73	100.41	127.41	55.73	91.57	91.57	O.K
224 1.0DL+1.0FL1+0.7SLZ2	662.94	126.23	0.01	86.10	28.10	0.00	114.20	114.20	58.00	57.99	87.20	114.20	58.00	86.10	86.10	O.K
225 1.0DL+1.0FL2+0.7SLX1	631.05	107.18	0.00	81.95	23.86	0.00	105.82	105.82	58.09	58.09	78.82	105.82	58.09	81.96	81.96	O.K
226 1.0DL+1.0FL2+0.7SLX2	835.66	212.19	0.00	108.53	47.24	0.00	155.77	155.77	61.29	61.29	128.77	155.77	61.29	108.53	108.53	O.K
227 1.0DL+1.0FL2+0.7SLZ1	754.42	177.05	-0.01	97.98	39.42	0.00	137.40	137.39	58.56	58.55	110.40	137.40	58.56	97.98	97.97	O.K
228 1.0DL+1.0FL2+0.7SLZ2	712.29	142.32	0.01	92.51	31.69	0.00	124.19	124.19	60.82	60.82	97.19	124.19	60.82	92.51	92.51	O.K
229 1.0DL+1.0FL1+0.75WLX1+0.75I	630.11	114.97	0.00	81.83	25.60	0.00	107.43	107.43	56.24	56.23	80.43	107.43	56.24	81.84	81.83	O.K
230 1.0DL+1.0FL1+0.75WLX2+0.75I	746.12	179.04	0.00	96.90	39.86	0.00	136.76	136.76	57.04	57.04	109.76	136.76	57.04	96.90	96.90	O.K
231 1.0DL+1.0FL1+0.75WLZ1+0.75I	692.86	153.31	-7.53	89.98	34.13	2.67	126.78	121.44	58.52	53.18	99.78	124.11	55.85	92.65	87.31	O.K
232 1.0DL+1.0FL1+0.75WLZ2+0.75I	683.41	140.72	7.53	88.75	31.33	2.67	122.75	117.42	60.09	54.76	95.75	120.09	57.43	91.42	86.09	O.K
233 1.0DL+1.0FL2+0.75WLX1+0.75I	679.47	131.07	0.00	88.24	29.18	0.00	117.42	117.42	59.06	59.06	90.42	117.42	59.06	88.24	88.24	O.K
234 1.0DL+1.0FL2+0.75WLX2+0.75I	795.48	195.13	0.00	103.31	43.44	0.00	146.75	146.75	59.87	59.87	119.75	146.75	59.87	103.31	103.31	O.K
235 1.0DL+1.0FL2+0.75WLZ1+0.75I	742.21	169.40	-7.53	96.39	37.71	2.67	136.77	131.44	61.35	56.01	109.77	134.11	58.68	99.06	93.73	O.K
236 1.0DL+1.0FL2+0.75WLZ2+0.75I	732.77	156.82	7.53	95.16	34.91	2.67	132.74	127.41	62.92	57.58	105.74	130.08	60.25	97.83	92.50	O.K
237 1.0DL+1.0FL1+0.525SLX1+0.75	611.41	107.63	0.00	79.40	23.96	0.00	103.37	103.37	55.44	55.44	76.37	103.37	55.44	79.41	79.41	O.K
238 1.0DL+1.0FL1+0.525SLX2+0.75	764.86	186.39	0.00	99.33	41.50	0.00	140.83	140.83	57.83	57.83	113.83	140.83	57.83	99.33	99.33	O.K
239 1.0DL+1.0FL1+0.525SLZ1+0.75	703.93	160.04	-0.01	91.42	35.63	0.00	127.05	127.05	55.79	55.79	100.05	127.05	55.79	91.42	91.42	O.K
240 1.0DL+1.0FL1+0.525SLZ2+0.75	672.34	133.99	0.00	87.32	29.83	0.00	117.15	117.15	57.49	57.48	90.15	117.15	57.49	87.32	87.32	O.K
241 1.0DL+1.0FL2+0.525SLX1+0.75	660.76	123.73	0.00	85.81	27.55	0.00	113.36	113.36	58.27	58.27	86.36	113.36	58.27	85.82	85.82	O.K
242 1.0DL+1.0FL2+0.525SLX2+0.75	814.21	202.48	0.00	105.74	45.08	0.00	150.82	150.82	60.66	60.66	123.82	150.82	60.66	105.74	105.74	O.K
243 1.0DL+1.0FL2+0.525SLZ1+0.75	753.28	176.13	-0.01	97.83	39.21	0.00	137.04	137.04	58.62	58.61	110.04	137.04	58.62	97.83	97.83	O.K
244 1.0DL+1.0FL2+0.525SLZ2+0.75	721.69	150.08	0.00	93.73	33.41	0.00	127.14	127.14	60.31	60.31	100.14	127.14	60.31	93.73	93.73	O.K
245 0.6DL+1.0WLX1	214.47	-25.10	0.00	27.85	5.59	0.00	33.44	33.44	22.27	22.26	6.44	33.44	22.27	27.86	27.85	O.K
246 0.6DL+1.0WLX2	369.15	60.32	0.00	47.94	13.43	0.00	61.37	61.37	34.51	34.51	34.37	61.37	34.51	47.94	47.94	O.K
247 0.6DL+1.0WLZ1	298.13	26.01	-10.04	38.72	5.79	3.56	48.07	40.95	36.48	29.37	21.07	44.51	32.93	42.28	35.16	O.K
248 0.6DL+1.0WLZ2	285.53	9.24	10.04	37.08	2.06	3.56	42.69	35.58	38.58	31.47	15.69	39.14	35.03	40.64	33.53	O.K
249 0.6DL+0.6FL1+0.7SLX1	403.49	33.65	0.00	52.40	7.49	0.00	59.89	59.89	44.91	44.91	32.89	59.89	44.91	52.40	52.40	O.K
250 0.6DL+0.6FL1+0.7SLX2	608.10	138.66	0.00	78.97	30.87	0.00	109.84	109.84	48.10	48.10	82.84	109.84	48.10	78.97	78.97	O.K
251 0.6DL+0.6FL1+0.7SLZ1	526.86	103.52	-0.01	68.42	23.05	0.00	91.47	91.47	45.38	45.37	64.47	91.47	45.38	68.43	68.42	O.K
252 0.6DL+0.6FL1+0.7SLZ2	484.73	68.79	0.01	62.95	15.31	0.00	78.27	78.26	47.64	47.63	51.27	78.27	47.64	62.96	62.95	O.K

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Ultimate Conditions

101 1.4DL+1.4FL1	862.20	201.03	0.00	111.97	44.76	0.00	156.73	156.73	67.22	67.22	129.73	156.73	67.22	111.98	111.98	
102 1.4DL+1.4FL2	931.30	223.56	0.00	120.95	49.77	0.00	170.72	170.72	71.18	71.17	143.72	170.72	71.18	120.95	120.95	
103 1.2DL+1.2FL1+1.6LL	781.92	179.61	0.00	101.55	39.99	0.00	141.54	141.53	61.56	61.56	114.54	141.54	61.56	101.55	101.55	
104 1.2DL+1.2FL2+1.6LL	841.14	198.92	0.00	109.24	44.29	0.00	153.53	153.53	64.95	64.95	126.53	153.53	64.95	109.24	109.24	
105 1.2DL+1.2FL1+1.2TL+1.6LL	749.58	253.05	-0.01	97.35	56.34	0.00	153.69	153.68	41.01	41.01	126.69	153.69	41.01	97.35	97.35	
106 1.2DL+1.2FL2+1.2TL+1.6LL	808.80	272.36	-0.01	105.04	60.64	0.00	165.68	165.67	44.41	44.40	138.68	165.68	44.41	105.05	105.04	
107 1.4DL+1.4FL1+1.4PL	909.72	211.46	0.01	118.15	47.08	0.01	165.23	165.22	71.07	71.06	138.23	165.23	71.07	118.15	118.14	
108 1.4DL+1.4FL2+1.4PL	978.82	233.99	0.01	127.12	52.10	0.01	179.22	179.21	75.03	75.02	152.22	179.22	75.03	127.13	127.12	
109 1.2DL+1.2FL1+1.6LL+1.2PL	822.65	188.55	0.01	106.84	41.98	0.00	148.82	148.81	64.86	64.85	121.82	148.82	64.86	106.84	106.83	
110 1.2DL+1.2FL2+1.6LL+1.2PL	881.87	207.86	0.01	114.53	46.28	0.00	160.81	160.80	68.26	68.25	133.81	160.81	68.26	114.54	114.53	
111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2	790.30	261.99	0.01	102.64	58.33	0.00	160.97	160.96	44.31	44.31	133.97	160.97	44.31	102.64	102.64	
112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2	849.53	281.30	0.01	110.33	62.63	0.00	172.96	172.95	47.70	47.70	145.96	172.96	47.70	110.33	110.33	
113 1.2DL+0.8WLX1	283.28	1.07	0.00	36.79	0.24	0.00	37.03	37.03	36.55	36.55	10.03	37.03	36.55	36.79	36.79	
114 1.2DL+0.8WLX2	407.02	69.40	0.00	52.86	15.45	0.00	68.31	68.31	37.41	37.41	41.31	68.31	37.41	52.86	52.86	
115 1.2DL+0.8WLZ1	350.20	41.96	-8.04	45.48	9.34	2.85	57.67	51.98	38.99	33.29	30.67	54.83	36.14	48.33	42.64	
116 1.2DL+0.8WLZ2	340.12	28.53	8.03	44.17	6.35	2.85	53.37	47.68	40.66	34.97	26.37	50.53	37.82	47.02	41.33	
117 1.2DL+1.6WLX1+0.5LL	224.15	-30.83	-0.01	29.11	6.86	0.00	35.98	35.97	22.25	22.24	8.98	35.98	22.25	29.12	29.11	
118 1.2DL+1.6WLX2+0.5LL	471.63	105.84	0.00	61.25	23.56	0.00	84.82	84.81	37.69	37.69	57.82	84.82	37.69	61.26	61.25	
119 1.2DL+1.6WLZ1+0.5LL	358.00	50.95	-16.07	46.49	11.34	5.69	63.53	52.14	40.84	29.46	36.53	57.84	35.15	52.19	40.80	
120 1.2DL+1.6WLZ2+0.5LL	337.84	24.11	16.07	43.88	5.37	5.69	54.93	43.55	44.20	32.82	27.93	49.24	38.51	49.57	38.19	
121 1.2DL+1.0SLX1+0.5LL+0.5FL1	380.08	19.63	0.00	49.36	4.37	0.00	53.73	53.73	44.99	44.99	26.73	53.73	44.99	49.36	49.36	
122 1.2DL+1.0SLX2+0.5LL+0.5FL1	672.37	169.64	0.00	87.32	37.77	0.00	125.09	125.09	49.55	49.55	98.09	125.09	49.55	87.32	87.32	
123 1.2DL+1.0SLZ1+0.5LL+0.5FL1	556.31	119.45	-0.01	72.25	26.59	0.00	98.85	98.84	45.66	45.65	71.85	98.85	45.66	72.26	72.25	
124 1.2DL+1.0SLZ2+0.5LL+0.5FL1	496.14	69.82	0.01	64.43	15.55	0.00	79.98	79.98	48.89	48.88	52.98	79.98	48.89	64.44	64.43	
125 0.9DL+1.6WLX1	194.72	-41.92	0.00	25.29	9.33	0.00	34.62	34.62	15.96	15.95	7.62	34.62	15.96	25.29	25.29	
126 0.9DL+1.6WLX2	442.21	94.75	0.00	57.43	21.09	0.00	78.52	78.52	36.33	36.33	51.52	78.52	36.33	57.43	57.43	
127 0.9DL+1.6WLZ1	328.58	39.86	-16.07	42.67	8.87	5.69	57.24	45.85	39.49	28.11	30.24	51.55	33.80	48.37	36.98	
128 0.9DL+1.6WLZ2	308.42	13.01	16.07	40.05	2.90	5.69	48.64	37.26	42.85	31.47	21.64	42.95	37.16	45.75	34.37	
129 0.9DL+1.0SLX1	172.35	-48.57	0.00	22.38	10.81	0.00	33.20	33.20	11.57	11.57	6.20	33.20	11.57	22.39	22.39	
130 0.9DL+1.0SLX2	464.64	101.44	0.00	60.34	22.58	0.00	82.93	82.93	37.76	37.76	55.93	82.93	37.76	60.35	60.35	
131 0.9DL+1.0SLZ1	348.58	51.25	-0.01	45.27	11.41	0.00	56.68	56.67	33.87	33.86	29.68	56.68	33.87	45.28	45.27	
132 0.9DL+1.0SLZ2	288.40	1.62	0.01	37.46	0.36	0.00	37.82	37.81	37.10	37.09	10.82	37.82	37.10	37.46	37.45	

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

Max :	979	281	16	**Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =**	**93.80 kN/m^2**
Min :	172	-49	-16	**Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =**	**103.80 kN/m^2**
				Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**128.77 kN/m^2**
				Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**138.77 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) =	95.24+35.55+0.00=	130.79 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(130.79+130.79)/2= 130.79 kN/m^2
p_2=	95.24+35.55-0.00=	130.79 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(59.69+59.69)/2= 59.69 kN/m^2
p_3=	95.24-(35.55+0.00)=	59.69 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(130.79+59.69)/2= 95.24 kN/m^2
p_4=	95.24-35.55+0.00=	59.69 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(59.69+130.79)/2= 95.24 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-γh)=	130.79-1.5*18=	103.79 kN/m^2		

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF FOUNDATION MKD. F11

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.8+2*0.5*0.426)+(0.65+2*0.5*0.426))*2

= 4.604 m

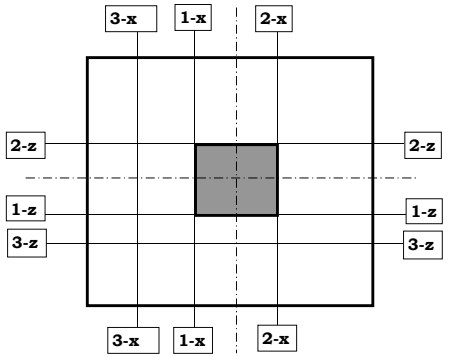
Punching Area= ((0.8+2*0.5*0.426)*(0.65+2*0.5*0.426))

= 1.32 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*24+1*18)

= 30.00 kN/m^2

Area of footing, A_f= 7.70 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M$_{1x}$ kNm	B.M$_{2x}$ kNm	S.F$_{3x}$ kN	S.F$_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	122.20	101.75	133.51	105.00	44.40	104.53	523.00
102	1.4DL+1.4FL2	132.33	109.57	144.90	116.57	49.19	116.05	580.25
103	1.2DL+1.2FL1+1.6LL	110.69	92.41	120.79	92.27	38.13	91.86	456.47
104	1.2DL+1.2FL2+1.6LL	119.36	99.12	130.55	102.19	42.23	101.73	505.55
105	1.2DL+1.2FL1+1.2TL+1.6LL	110.23	84.47	124.46	99.51	23.23	99.04	429.68
106	1.2DL+1.2FL2+1.2TL+1.6LL	118.90	91.19	134.22	109.43	27.34	108.91	478.75
107	1.4DL+1.4FL1+1.4PL	128.91	107.39	140.80	112.20	48.46	111.70	562.37
108	1.4DL+1.4FL2+1.4PL	139.03	115.22	152.19	123.77	53.24	123.22	619.62
109	1.2DL+1.2FL1+1.6LL+1.2PL	116.44	97.24	127.04	98.44	41.60	98.00	490.22
110	1.2DL+1.2FL2+1.6LL+1.2PL	125.11	103.96	136.80	108.36	45.71	107.87	539.29
111	1.2DL+1.2FL1+1.2TL+1.6LL+	115.97	89.31	130.71	105.68	26.71	105.18	463.42
112	1.2DL+1.2FL2+1.2TL+1.6LL+	124.65	96.01	140.46	115.60	30.80	115.05	512.50

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113 1.2DL+0.8WLX1	36.84	36.74	36.91	6.35	6.02	6.33	43.31
114 1.2DL+0.8WLX2	56.39	49.33	60.29	31.29	10.37	31.15	145.84
115 1.2DL+0.8WLZ1	47.62	43.35	49.98	20.44	7.78	20.34	98.77
116 1.2DL+0.8WLZ2	45.63	42.72	47.23	17.22	8.62	17.14	90.42
117 1.2DL+1.6WLX1+0.5LL	30.68	27.55	32.42	3.84	-5.45	3.81	-5.68
118 1.2DL+1.6WLX2+0.5LL	66.64	55.87	72.59	44.43	12.53	44.23	199.38
119 1.2DL+1.6WLZ1+0.5LL	49.09	43.90	51.95	22.71	7.35	22.61	105.23
120 1.2DL+1.6WLZ2+0.5LL	45.10	42.65	46.46	16.28	9.01	16.21	88.53
121 1.2DL+1.0SLX1+0.5LL+0.5FL	50.36	48.36	51.46	20.60	14.68	20.52	123.52
122 1.2DL+1.0SLX2+0.5LL+0.5FL	95.95	78.69	105.49	77.80	26.67	77.44	365.71
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	78.33	66.18	85.05	56.51	20.50	56.25	269.55
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	67.99	60.88	71.91	41.90	20.86	41.72	219.68
125 0.9DL+1.6WLX1	27.42	23.16	29.78	2.02	-10.61	2.00	-30.06
126 0.9DL+1.6WLX2	62.25	52.60	67.57	39.27	10.71	39.09	175.00
127 0.9DL+1.6WLZ1	44.70	40.65	46.95	17.56	5.54	17.48	80.85
128 0.9DL+1.6WLZ2	40.72	39.39	41.45	11.12	7.20	11.08	64.15
129 0.9DL+1.0SLX1	24.86	19.91	27.59	0.38	-14.26	0.36	-48.60
130 0.9DL+1.0SLX2	65.51	55.18	71.21	42.94	12.36	42.74	193.59
131 0.9DL+1.0SLZ1	47.88	42.67	50.76	21.64	6.20	21.54	97.43
132 0.9DL+1.0SLZ2	37.54	37.38	37.63	7.04	6.55	7.02	47.56

Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1x} =	156.73-(156.73-67.22)*1.35/3.5 =	122.20 kN/m^2	Overhang =	1.35 m
p_{2x} =	67.22+(156.73-67.22)*1.35/3.5 =	101.75 kN/m^2	Effective depth =	0.442 m
p_{3x} =	156.73-(156.73-67.22)*(1.35-0.442)/3.5 =	133.51 kN/m^2		
BM_{1x} =	(122.20-30)*1.35*1.35/2+0.5*(156.73-122.20)*1.35*1.35*0.67 =	105.00 kN-m/m		
BM_{2x} =	(101.75-30)*1.35*1.35/2-0.5*(-67.22+101.75)*1.35*1.35*0.67 =	44.40 kN-m/m		
SF_{3x} =	(133.51-30)*(1.35-0.442)+0.5*(156.73-133.51)*(1.35-0.442) =	104.53 kN/m		
$S.F_{(punching)}$ =	(7.7-1.32)*(111.97-30) =	523.00 kN		

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	123.77	0.63	0.84	0.002	901.95	795.60	16ϕT	222.92	300	200	1005.31
At Top	-14.26	-0.07	0.10	0.000	102.59	397.80	12ϕT	284.31	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 123.22 kN/m

Shear stress , τ_v= (123.22*1000)/(1000*442)

= 0.28 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 619.62 kN

Strength of Concrete in shear Vc1 = 5267.08 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3932.82 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 4013.01 kN **Table 22.6.5.2_ (c)**

Design Vc= 3932.82 kN

Allowable shear stress, (ϕ Vc)/bd = 1.45 N/mm^2

Design punching shear stress, τ = (619.62*1000)/(4604*442)

= 0.30 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M_{1z} kNm	B.M_{2z} kNm	S.F_{3z} kN
Ultimate Conditions							
101	1.4DL+1.4FL1	111.98	111.98	111.98	24.62	24.62	28.61
102	1.4DL+1.4FL2	120.95	111.19	120.95	27.31	20.77	31.74
103	1.2DL+1.2FL1+1.6LL	101.55	100.88	101.55	21.49	21.04	24.97
104	1.2DL+1.2FL2+1.6LL	109.24	102.62	109.24	23.80	19.36	27.65
105	1.2DL+1.2FL1+1.2TL+1.6LL	97.35	97.18	97.35	20.23	20.11	23.51
106	1.2DL+1.2FL2+1.2TL+1.6LL	105.05	100.15	105.05	22.54	19.26	26.19
107	1.4DL+1.4FL1+1.4PL	118.15	110.89	118.15	26.47	21.61	30.76
108	1.4DL+1.4FL2+1.4PL	127.13	110.09	127.13	29.17	17.75	33.90
109	1.2DL+1.2FL1+1.6LL+1.2PL	106.84	104.03	106.84	23.08	21.20	26.82
110	1.2DL+1.2FL2+1.6LL+1.2PL	114.54	107.02	114.54	25.39	18.92	29.50
111	1.2DL+1.2FL1+1.2TL+1.6LL+	102.64	98.74	102.64	21.81	17.77	25.35
112	1.2DL+1.2FL2+1.2TL+1.6LL+	110.33	101.48	110.33	24.12	18.19	28.04

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113 1.2DL+0.8WLX1	36.79	60.92	36.79	2.04	18.21	2.37
114 1.2DL+0.8WLX2	52.86	69.33	52.86	6.87	16.47	7.98
115 1.2DL+0.8WLZ1	46.33	62.13	47.43	5.31	15.43	6.24
116 1.2DL+0.8WLZ2	45.02	53.38	46.12	4.91	11.48	5.78
117 1.2DL+1.6WLX1+0.5LL	29.12	54.83	29.12	-0.26	16.97	-0.31
118 1.2DL+1.6WLX2+0.5LL	61.26	71.93	61.26	9.39	16.54	10.91
119 1.2DL+1.6WLZ1+0.5LL	48.18	56.76	50.38	6.27	13.94	7.43
120 1.2DL+1.6WLZ2+0.5LL	45.56	53.61	47.76	5.48	12.79	6.51
121 1.2DL+1.0SLX1+0.5LL+0.5FL	49.36	70.20	49.36	5.81	19.79	6.76
122 1.2DL+1.0SLX2+0.5LL+0.5FL	87.32	91.08	87.32	17.21	19.73	20.00
123 1.2DL+1.0SLZ1+0.5LL+0.5FL	72.26	79.39	72.26	12.69	17.47	14.75
124 1.2DL+1.0SLZ2+0.5LL+0.5FL	64.44	70.56	64.44	10.34	14.45	12.02
125 0.9DL+1.6WLX1	25.29	50.52	25.29	-1.41	15.49	-1.64
126 0.9DL+1.6WLX2	57.43	69.84	57.43	8.24	15.48	9.57
127 0.9DL+1.6WLZ1	44.36	56.16	46.56	5.12	13.88	6.10
128 0.9DL+1.6WLZ2	41.74	53.35	43.94	4.33	14.03	5.18
129 0.9DL+1.0SLX1	22.39	50.90	22.39	-2.29	16.82	-2.66
130 0.9DL+1.0SLX2	60.35	73.99	60.35	9.11	17.18	10.59
131 0.9DL+1.0SLZ1	45.28	63.79	45.28	4.59	15.92	5.33
132 0.9DL+1.0SLZ2	37.46	52.23	37.46	2.24	12.14	2.60

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Sample calculation of BM & SF

Load case : 101 1.4DL+1.4FL1

p_{1z} = 111.98-(111.98-111.98)*0.775/2.2= 111.98 kN/m^2

p_{2z} = 111.98+(111.98-111.98)*0.775/2.2= 111.98 kN/m^2 Overhang = 0.775 m

p_{3z} = 111.98-(111.98-111.98)*(0.775-0.426)/2.2= 111.98 kN/m^2 Effective depth = 0.426 m

BM_{1z} = (111.98-30)*0.775*0.775/2+0.5*(111.98-111.98)*0.775*0.775*0.67 = 24.62 kN-m/m

BM_{2z} = (111.98-30)*0.775*0.775/2-0.5*(-111.98+111.98)*0.775*0.775*0.67 = 24.62 kN-m/m

SF_{3z} = (111.98-30)*(0.775-0.426)+0.5*(111.98-111.98)*(0.775-0.426) = 28.61 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	29.17	0.161	0.214	0.0005	226.35	766.80	16ϕT	262.21	300	200	1005.3
Top	-2.29	-0.013	-0.017	0.0000	-0.17	383.40	12ϕT	294.99	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 33.90 kN/m

Shear stress , τ_v= (33.9*1000)/(1000*426)

= 0.08 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **16ϕT** **200** c/c

At Top **12ϕT** **200** c/c

Along Z Direction :

At Bottom **16ϕT** **200** c/c

At Top **12ϕT** **200** c/c

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APPENDIX – B

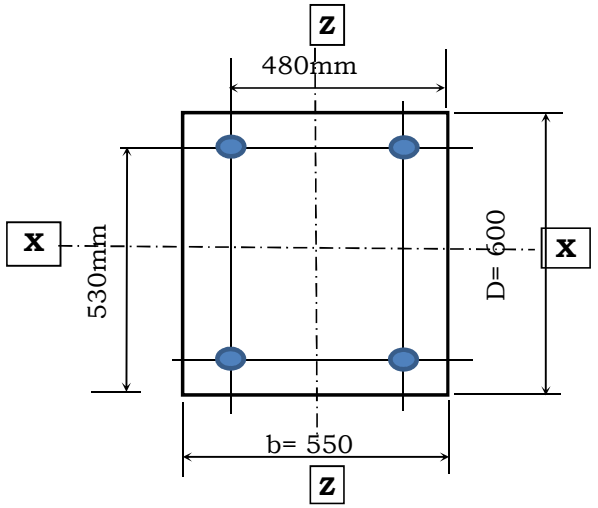
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DESIGN OF COLUMN MKD C3 STAAD Node 7851 7072 7091 7076



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	1.188kN	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1500mm
Unsupproted length of column about Z (l_{uz}) =	1500mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	10.40	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	K l_{uz} / r_z =	11.34	< 22 SECTION 6.2.5a	Radius of gyration r_z=	159	mm

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev:D01

γ_y = 0.80

γ_z = 0.78

	Node	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = $P_n/f'_c bD$ (N/mm^2)
			Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
Max Fx	7091	112 1.2DL+1.2FL2+1.2TL	39.86	1.3	0.7	39.98	2.0	1.0	61.5	3.1	1.6	0.05
Min Fx	7091	117 1.2DL+1.6WLX1+0.5I	-7.69	-1.1	0.0	-7.57	1.6	0.0	-11.7	2.5	0.1	-0.01
Max Fy	7091	108 1.4DL+1.4FL2+1.4PL	41.82	0.8	0.9	41.94	1.3	1.3	64.5	1.9	2.0	0.06
Min Fy	7091	125 0.9DL+1.6WLX1	-11.47	-1.1	0.0	-11.35	1.6	0.0	-17.5	2.4	0.0	-0.02
Max Fz	7091	107 1.4DL+1.4FL1+1.4PL	41.60	0.8	0.9	41.72	1.3	1.3	64.2	1.9	2.0	0.06
Min Fz	7851	123 1.2DL+1.0SLZ1+0.5L	19.04	0.0	-0.7	19.16	0.0	1.1	29.5	0.0	1.7	0.03
Max Mx	7072	101 1.4DL+1.4FL1	16.32	0.0	-0.1	16.44	0.0	0.1	25.3	0.0	0.2	0.02
Min Mx	7072	101 1.4DL+1.4FL1	16.32	0.0	-0.1	16.44	0.0	0.1	25.3	0.0	0.2	0.02
Max My	7072	101 1.4DL+1.4FL1	16.32	0.0	-0.1	16.44	0.0	0.1	25.3	0.0	0.2	0.02
Min My	7072	101 1.4DL+1.4FL1	16.32	0.0	-0.1	16.44	0.0	0.1	25.3	0.0	0.2	0.02
Max Mz	7072	101 1.4DL+1.4FL1	16.32	0.0	-0.1	16.44	0.0	0.1	25.3	0.0	0.2	0.02
Min Mz	7072	101 1.4DL+1.4FL1	16.32	0.0	-0.1	16.44	0.0	0.1	25.3	0.0	0.2	0.02

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
1257.14 Sq mm (Along Z axis)
1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 1.14 %

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MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1128.93 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
112 1.2D	0.07	0.07	44.5	48.5	0.55	1.16	0.06	O.K
117 1.2D	0.05	0.05	31.8	34.7	0.55	1.16	0.05	O.K
108 1.4D	0.08	0.08	50.8	55.4	0.55	1.16	0.04	O.K
125 0.9D	0.05	0.05	31.8	34.7	0.55	1.16	0.05	O.K
107 1.4D	0.08	0.08	50.8	55.4	0.55	1.16	0.04	O.K
123 1.2D	0.05	0.05	31.8	34.7	0.55	1.16	0.03	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

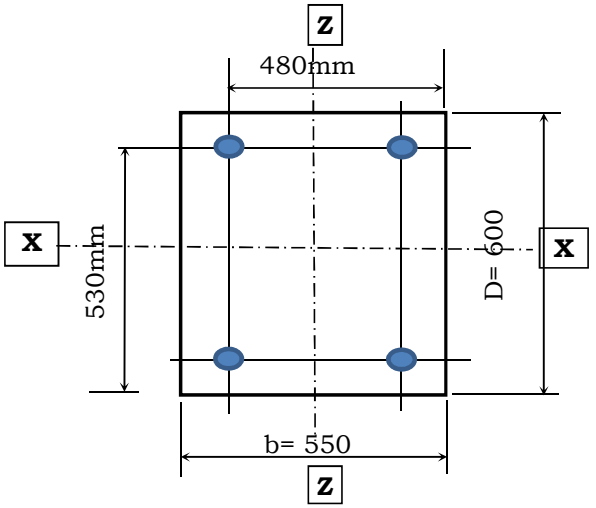
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O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C4 STAAD Node 7853 to 7855



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	1.03kN	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1300mm
Unsupproted length of column about Z (l_{uz}) =	1300mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	9.01	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	K l_{uz} / r_z =	9.83	< 22 SECTION 6.2.5a	Radius of gyration r_z=	159	mm

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O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

γ_y = 0.80

γ_z = 0.78

	Node	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
			Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
Max Fx	7855	111 1.2DL+1.2FL1+1.2TL	88.37	24.8	-2.2	88.47	32.3	2.9	136.1	49.7	4.4	0.12
Min Fx	7854	129 0.9DL+1.0SLX1	-10.58	-7.0	0.1	-10.48	9.0	0.1	-16.1	13.9	0.2	-0.01
Max Fy	7855	111 1.2DL+1.2FL1+1.2TL	88.37	24.8	-2.2	88.47	32.3	2.9	136.1	49.7	4.4	0.12
Min Fy	7855	128 0.9DL+1.6WLZ2	-23.53	-3.7	2.9	-23.43	4.8	3.8	-36.0	7.3	5.9	-0.03
Max Fz	7853	128 0.9DL+1.6WLZ2	-11.43	0.8	6.7	-11.33	1.0	8.8	-17.4	1.6	13.5	-0.02
Min Fz	7853	119 1.2DL+1.6WLZ1+0.5I	32.76	-3.1	-8.2	32.86	4.0	10.6	50.6	6.1	16.3	0.04
Max Mx	7853	101 1.4DL+1.4FL1	14.86	-1.6	-0.8	14.96	2.0	1.0	23.0	3.1	1.5	0.02
Min Mx	7853	101 1.4DL+1.4FL1	14.86	-1.6	-0.8	14.96	2.0	1.0	23.0	3.1	1.5	0.02
Max My	7853	101 1.4DL+1.4FL1	14.86	-1.6	-0.8	14.96	2.0	1.0	23.0	3.1	1.5	0.02
Min My	7853	101 1.4DL+1.4FL1	14.86	-1.6	-0.8	14.96	2.0	1.0	23.0	3.1	1.5	0.02
Max Mz	7853	101 1.4DL+1.4FL1	14.86	-1.6	-0.8	14.96	2.0	1.0	23.0	3.1	1.5	0.02
Min Mz	7853	101 1.4DL+1.4FL1	14.86	-1.6	-0.8	14.96	2.0	1.0	23.0	3.1	1.5	0.02

Reinforcement: Provide,

4-Nos	20 mm Dia
4-Nos	20 mm Dia
4-Nos	20 mm Dia

1257.14 Sq mm (At Corner)

1257.14 Sq mm (Along Z axis)

1257.14 Sq mm (Along Y axis)

A_{sc} pvd. = 3771.43 Sq mm

Total A_{sc} = 1.14 %

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900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1128.93 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
111 1.2D	0.10	0.10	60.3	65.8	0.55	1.16	0.84	O.K
129 0.9D	0.05	0.05	31.8	34.7	0.55	1.16	0.39	O.K
111 1.2D	0.10	0.10	60.3	65.8	0.55	1.16	0.84	O.K
128 0.9D	0.05	0.05	31.8	34.7	0.55	1.16	0.31	O.K
128 0.9D	0.05	0.05	31.8	34.7	0.55	1.16	0.37	O.K
119 1.2D	0.06	0.06	38.1	41.6	0.55	1.16	0.46	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.10	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.10	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.10	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.10	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.10	O.K
101 1.4D	0.05	0.05	31.8	34.7	0.55	1.16	0.10	O.K

Links, 10mm dia @ 300 C/C

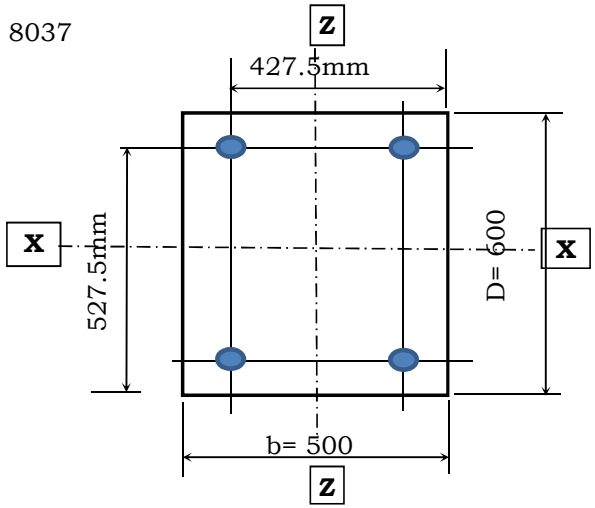
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900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C5 STAAD Node 7077 7729 7078 7429 7730 7731 8036 8037



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	25mm	
Dia. of link=	10mm	
Effective cover =	72.5mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	1.116kN	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1550mm
Unsupproted length of column about Z (l_{uz}) =	1550mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	10.74	< 22 SECTION 6.2.5a	Radius of gyration r_x=	173	mm
	K l_{uz} / r_z =	12.89	< 22 SECTION 6.2.5a	Radius of gyration r_z=	144	mm

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O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

γ_y = 0.79

γ_z = 0.75

	Node	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
			Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
Max Fx	7729	112 1.2DL+1.2FL2+1.2TL	13.22	4.2	0.6	13.33	6.5	0.9	20.5	9.9	1.4	0.02
Min Fx	8036	109 1.2DL+1.2FL1+1.6LL	37.94	-4.3	-5.4	38.06	6.7	8.4	58.5	10.2	12.9	0.06
Max Fy	7078	110 1.2DL+1.2FL2+1.6LL	41.47	-0.1	0.0	41.58	0.1	0.1	64.0	0.1	0.1	0.06
Min Fy	7429	127 0.9DL+1.6WLZ1	-7.25	0.0	-0.4	-7.14	0.1	0.6	-11.0	0.1	0.9	-0.01
Max Fz	7729	120 1.2DL+1.6WLZ2+0.5I	5.49	0.8	0.6	5.60	1.2	1.0	8.6	1.9	1.5	0.01
Min Fz	8036	106 1.2DL+1.2FL2+1.2TL	31.94	-4.3	-5.4	32.05	6.6	8.4	49.3	10.2	13.0	0.05
Max Mx	7077	101 1.4DL+1.4FL1	19.10	0.0	-0.1	19.21	0.1	0.1	29.6	0.1	0.1	0.03
Min Mx	7077	101 1.4DL+1.4FL1	19.10	0.0	-0.1	19.21	0.1	0.1	29.6	0.1	0.1	0.03
Max My	7077	101 1.4DL+1.4FL1	19.10	0.0	-0.1	19.21	0.1	0.1	29.6	0.1	0.1	0.03
Min My	7077	101 1.4DL+1.4FL1	19.10	0.0	-0.1	19.21	0.1	0.1	29.6	0.1	0.1	0.03
Max Mz	7077	101 1.4DL+1.4FL1	19.10	0.0	-0.1	19.21	0.1	0.1	29.6	0.1	0.1	0.03
Min Mz	7077	101 1.4DL+1.4FL1	19.10	0.0	-0.1	19.21	0.1	0.1	29.6	0.1	0.1	0.03

Reinforcement: Provide,

4-Nos	25 mm Dia
2-Nos	20 mm Dia
2-Nos	20 mm Dia

1964.29 Sq mm (At Corner)

628.57 Sq mm (Along Z axis)

628.57 Sq mm (Along Y axis)

A_{sc} pvd. = 3221.43 Sq mm

Total A_{sc} = 1.07 %

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O19045_450_C_K_ 24091 _001_R2	AS BUILT	Rev: D01

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1018.216 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
112 1.2D	0.05	0.05	26.3	31.5	0.55	1.16	0.35	O.K
109 1.2D	0.07	0.07	36.8	44.1	0.55	1.16	0.47	O.K
110 1.2D	0.07	0.07	36.8	44.1	0.55	1.16	0.00	O.K
127 0.9D	0.05	0.05	26.3	31.5	0.55	1.16	0.02	O.K
120 1.2D	0.05	0.05	26.3	31.5	0.55	1.16	0.08	O.K
106 1.2D	0.06	0.06	31.5	37.8	0.55	1.16	0.56	O.K
101 1.4D	0.05	0.05	26.3	31.5	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	26.3	31.5	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	26.3	31.5	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	26.3	31.5	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	26.3	31.5	0.55	1.16	0.00	O.K
101 1.4D	0.05	0.05	26.3	31.5	0.55	1.16	0.00	O.K

Links, 10mm dia @ 300 C/C

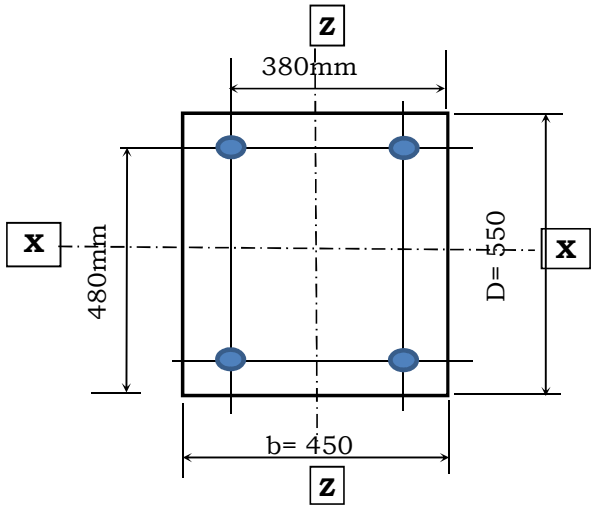
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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec LARSEN & TOUBRO
MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
O19045_450_C_K_ 24085 _001_R2	AS BUILT	Rev: D01

DESIGN OF COLUMN MKD C6 STAAD Node 7976 7663



Input:	Clear cover to reinf.=	50mm	Table 20.6.1.3.1
	Dia. of main reinf.=	20mm	
	Dia. of link=	10mm	
	Effective cover =	70mm	
	Grade of Concrete (f_c') =	M35	
	Grade of Steel (f_y) =	Fy420	
	Grade of Reinf(links).=	Fy420	
	Min. Steel Ratio	0.50%	
	Max. Steel Ratio	4.00%	
	Self weight of Column =	0.921kN	

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1550mm
Unsupproted length of column about Z (l_{uz}) =	1550mm

For Columns not braced against sidesway, Where, Effective length factor K= 1.2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	11.72	< 22 SECTION 6.2.5a	Radius of gyration r_x=	159	mm
	$K\,l_{uz} / r_z$ =	14.32	< 22 SECTION 6.2.5a	Radius of gyration r_z=	130	mm

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900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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γ_y = 0.78

γ_z = 0.73

	Node	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
			Pu (T)	Hux (T)	Huz (T)	Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
Max Fx	7976	126 0.9DL+1.6WLX2	1.13	0.2	-0.1	1.22	0.3	0.2	1.9	0.4	0.3	0.00
Min Fx	7976	117 1.2DL+1.6WLX1+0.5L	5.02	-0.2	0.8	5.12	0.3	1.2	7.9	0.4	1.9	0.01
Max Fy	7663	111 1.2DL+1.2FL1+1.2TL	20.47	0.0	1.3	20.56	0.0	2.1	31.6	0.0	3.2	0.04
Min Fy	7663	251 0.6DL+0.6FL1+0.7SL	0.03	0.0	-1.5	0.12	0.0	2.4	0.2	0.0	3.6	0.00
Max Fz	7663	124 1.2DL+1.0SLZ2+0.5L	7.68	0.0	2.1	7.77	0.0	3.3	12.0	0.0	5.1	0.01
Min Fz	7663	131 0.9DL+1.0SLZ1	0.23	0.0	-2.1	0.32	0.0	3.2	0.5	0.0	4.9	0.00
Max Mx	7663	101 1.4DL+1.4FL1	2.80	0.0	-0.2	2.89	0.0	0.3	4.4	0.0	0.5	0.01
Min Mx	7663	101 1.4DL+1.4FL1	2.80	0.0	-0.2	2.89	0.0	0.3	4.4	0.0	0.5	0.01
Max My	7663	101 1.4DL+1.4FL1	2.80	0.0	-0.2	2.89	0.0	0.3	4.4	0.0	0.5	0.01
Min My	7663	101 1.4DL+1.4FL1	2.80	0.0	-0.2	2.89	0.0	0.3	4.4	0.0	0.5	0.01
Max Mz	7663	101 1.4DL+1.4FL1	2.80	0.0	-0.2	2.89	0.0	0.3	4.4	0.0	0.5	0.01
Min Mz	7663	101 1.4DL+1.4FL1	2.80	0.0	-0.2	2.89	0.0	0.3	4.4	0.0	0.5	0.01

Reinforcement: Provide,

4-Nos	20 mm Dia
2-Nos	20 mm Dia
2-Nos	20 mm Dia

1257.14 Sq mm (At Corner)

628.57 Sq mm (Along Z axis)

628.57 Sq mm (Along Y axis)

A_{sc} pvd. = 2514.29 Sq mm

Total A_{sc} = 1.02 %

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MD-216-3310-EG-CV-CAL-1213	Mansourah Massarah Gold Project	
900707_GAU01_ZF01_CED01_ 33104_OU100675147_R2	DESIGN CALCULATION 3310 ROUGHER AND SCAVENGER FLOTATION DESIGN OF FOUNDATION	Date: 10.11.2023
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β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 834.4325 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
126 0.9D	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K
117 1.2D	0.05	0.05	19.5	23.8	0.55	1.16	0.06	O.K
111 1.2D	0.05	0.05	19.5	23.8	0.55	1.16	0.10	O.K
251 0.6D	0.05	0.05	19.5	23.8	0.55	1.16	0.11	O.K
124 1.2D	0.05	0.05	19.5	23.8	0.55	1.16	0.17	O.K
131 0.9D	0.05	0.05	19.5	23.8	0.55	1.16	0.16	O.K
101 1.4D	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
101 1.4D	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
101 1.4D	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
101 1.4D	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
101 1.4D	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
101 1.4D	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K

Links, 10mm dia @ 300 C/C

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REFERENCE

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT & OUTPUT FILE

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  2, 2020                *
*           Time=    10:20:25                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
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```
 1. STAAD SPACE
 2. START JOB INFORMATION
 3. ENGINEER DATE 15-FEB-20
 4. *** FOUNDATION MKD F4
 5. END JOB INFORMATION
 6. INPUT WIDTH 79
 7. UNIT METER KN
 8. JOINT COORDINATES
 9. 1 0 0.35 3.5; 2 4 0.35 3.5; 3 0 0.35 7.46; 4 4 0.35 7.46; 5 0 -1.5 3.5
10. 6 4 -1.5 3.5; 7 0 -1.5 7.46; 8 4 -1.5 7.46; 9 0.275 -1.5 3.5
11. 11 -0.275 -1.5 3.5; 13 0 -1.5 3.8; 14 0.275 -1.5 3.8; 15 -0.275 -1.5 3.8
12. 16 0 -1.5 3.2; 17 0.275 -1.5 3.2; 18 -0.275 -1.5 3.2; 19 0.525 -1.5 3.5
13. 20 0.525 -1.5 3.8; 21 0.525 -1.5 3.2; 22 0.775 -1.5 3.5; 23 0.775 -1.5 3.8
14. 24 0.775 -1.5 3.2; 25 -0.525 -1.5 3.5; 26 -0.525 -1.5 3.8; 27 -0.525 -1.5 3.2
15. 28 -0.775 -1.5 3.5; 29 -0.775 -1.5 3.8; 30 -0.775 -1.5 3.2; 31 0 -1.5 4.05
16. 32 0.275 -1.5 4.05; 33 -0.275 -1.5 4.05; 34 0.525 -1.5 4.05
17. 35 0.775 -1.5 4.05; 36 -0.525 -1.5 4.05; 37 -0.775 -1.5 4.05
18. 38 -0.775 -1.5 4.3; 39 -0.525 -1.5 4.3; 40 -0.275 -1.5 4.3; 41 0 -1.5 4.3
19. 42 0.275 -1.5 4.3; 43 0.525 -1.5 4.3; 44 0.775 -1.5 4.3; 45 -0.775 -1.5 2.7
20. 46 -0.775 -1.5 2.95; 47 -0.525 -1.5 2.95; 48 -0.525 -1.5 2.7
21. 49 -0.275 -1.5 2.95; 50 -0.275 -1.5 2.7; 51 0 -1.5 2.95; 52 0 -1.5 2.7
22. 53 0.275 -1.5 2.95; 54 0.275 -1.5 2.7; 55 0.525 -1.5 2.95; 56 0.525 -1.5 2.7
23. 57 0.775 -1.5 2.95; 58 0.775 -1.5 2.7; 59 0 -1.5 7.76; 60 0.275 -1.5 7.76
24. 61 0.275 -1.5 7.46; 62 -0.275 -1.5 7.46; 63 -0.275 -1.5 7.76; 64 0 -1.5 7.16
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31. 85 0.775 -1.5 8.01; 86 -0.775 -1.5 8.26; 87 -0.525 -1.5 8.26
32. 88 -0.275 -1.5 8.26; 89 0 -1.5 8.26; 90 0.275 -1.5 8.26; 91 0.525 -1.5 8.26
33. 92 0.775 -1.5 8.26; 93 -0.775 -1.5 6.66; 94 -0.775 -1.5 6.91
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36. 102 0.275 -1.5 6.66; 103 0.525 -1.5 6.91; 104 0.525 -1.5 6.66
37. 105 0.775 -1.5 6.91; 106 0.775 -1.5 6.66; 107 4 -1.5 3.825; 108 4.4 -1.5 3.825
38. 109 4.4 -1.5 3.5; 110 3.6 -1.5 3.5; 111 3.6 -1.5 3.825; 112 4 -1.5 3.175
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116. 370 2.975 -1.5 4.8; 371 2.725 -1.5 4.8; 372 2.475 -1.5 4.8; 373 2.225 -1.5 4.8
117. 374 2 -1.5 4.8; 375 -0.775 -1.5 5.05; 376 -0.525 -1.5 5.05
118. 377 -0.275 -1.5 5.05; 378 0 -1.5 5.05; 379 0.275 -1.5 5.05
119. 380 0.525 -1.5 5.05; 381 0.775 -1.5 5.05; 382 3.225 -1.5 5.05
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122. 390 1.275 -1.5 5.05; 391 1.525 -1.5 5.05; 392 1.775 -1.5 5.05
123. 393 2.975 -1.5 5.05; 394 2.725 -1.5 5.05; 395 2.475 -1.5 5.05
124. 396 2.225 -1.5 5.05; 397 2 -1.5 5.05; 398 -0.775 -1.5 5.3; 399 -0.525 -1.5 5.3
125. 400 -0.275 -1.5 5.3; 401 0 -1.5 5.3; 402 0.275 -1.5 5.3; 403 0.525 -1.5 5.3
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129. 416 2.975 -1.5 5.3; 417 2.725 -1.5 5.3; 418 2.475 -1.5 5.3; 419 2.225 -1.5 5.3
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132. 472 -0.275 -1.5 6.41; 473 0 -1.5 6.41; 474 0.275 -1.5 6.41
133. 475 0.525 -1.5 6.41; 476 0.775 -1.5 6.41; 477 3.225 -1.5 6.41
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168. 585 5.025 -1.5 4.3; 586 5.025 -1.5 2.95; 587 5.025 -1.5 2.7
169. 588 5.025 -1.5 7.76; 589 5.025 -1.5 7.46; 590 5.025 -1.5 7.16
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171. 594 5.025 -1.5 6.66; 595 5.025 -1.5 4.55; 596 5.025 -1.5 4.8
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173. 600 5.025 -1.5 6.16; 601 5.025 -1.5 5.91; 602 5.025 -1.5 5.66
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175. 606 5.275 -1.5 3.2; 607 5.275 -1.5 4.05; 608 5.275 -1.5 4.3
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177. 612 5.275 -1.5 7.46; 613 5.275 -1.5 7.16; 614 5.275 -1.5 8.01
178. 615 5.275 -1.5 8.26; 616 5.275 -1.5 6.91; 617 5.275 -1.5 6.66
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184. 633 5.625 -1.5 2.7; 634 5.625 -1.5 7.76; 635 5.625 -1.5 7.46
185. 636 5.625 -1.5 7.16; 637 5.625 -1.5 8.01; 638 5.625 -1.5 8.26
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193. 660 -1.025 -1.5 8.01; 661 -1.025 -1.5 8.26; 662 -1.025 -1.5 6.66
194. 663 -1.025 -1.5 6.91; 664 -1.025 -1.5 4.55; 665 -1.025 -1.5 4.8
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201. 684 -1.275 -1.5 8.26; 685 -1.275 -1.5 6.66; 686 -1.275 -1.5 6.91
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216. 730 4.275 -1.5 2.45; 731 4.525 -1.5 2.45; 732 4.775 -1.5 2.45
217. 733 1.025 -1.5 2.45; 734 1.275 -1.5 2.45; 735 1.525 -1.5 2.45
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221. 746 -1.275 -1.5 2.45; 747 -1.625 -1.5 2.45; 748 -0.775 -1.5 2.2
222. 749 -0.525 -1.5 2.2; 750 -0.275 -1.5 2.2; 751 0 -1.5 2.2; 752 0.275 -1.5 2.2
223. 753 0.525 -1.5 2.2; 754 0.775 -1.5 2.2; 755 3.225 -1.5 2.2; 756 3.475 -1.5 2.2
224. 757 3.725 -1.5 2.2; 758 4 -1.5 2.2; 759 4.275 -1.5 2.2; 760 4.525 -1.5 2.2
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227. 769 2.225 -1.5 2.2; 770 2 -1.5 2.2; 771 5.025 -1.5 2.2; 772 5.275 -1.5 2.2
228. 773 5.625 -1.5 2.2; 774 -1.025 -1.5 2.2; 775 -1.275 -1.5 2.2
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235. 795 2.975 -1.5 1.85; 796 2.725 -1.5 1.85; 797 2.475 -1.5 1.85
236. 798 2.225 -1.5 1.85; 799 2 -1.5 1.85; 800 5.025 -1.5 1.85; 801 5.275 -1.5 1.85
237. 802 5.625 -1.5 1.85; 803 -1.025 -1.5 1.85; 804 -1.275 -1.5 1.85
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265. 892 -1.625 -1.5 9.11
266. MEMBER INCIDENCES
267. 1 5 1; 2 6 2; 3 7 3; 4 8 4
268. ELEMENT INCIDENCES SHELL
269. 16 5 13 14 9; 18 11 15 13 5; 19 16 5 9 17; 20 18 11 5 16; 22 9 14 20 19
270. 23 17 9 19 21; 25 19 20 23 22; 26 21 19 22 24; 27 25 26 15 11; 28 27 25 11 18
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516. START GROUP DEFINITION
517. FLOOR
518. ELEMENT
519. _FDN_MOMENT 22 23 25 TO 30 34 TO 40 42 44 46 TO 60 67 68 70 TO 99 106 TO 109 -
520. 111 112 114 TO 117 119 TO 124 126 TO 141 148 TO 151 153 154 156 TO 171 174 -
521. 175 TO 183 185 TO 190 192 TO 197 199 TO 204 206 TO 211 213 TO 218 220 TO 225 -
522. 227 TO 232 234 TO 239 241 TO 252 254 TO 259 261 TO 266 268 TO 273 -
523. 275 TO 280 282 TO 287 289 TO 294 296 TO 301 303 TO 308 310 TO 321 -
524. 323 TO 344 346 TO 367 369 TO 390 392 TO 413 448 TO 543 545 TO 880
525. _FDN_SHEAR 185 TO 190 192 TO 197 199 TO 204 206 TO 211 213 TO 218 220 TO 225 -
526. 227 TO 232 234 TO 239 241 TO 252 254 TO 259 261 TO 266 268 TO 273 -
527. 275 TO 280 282 TO 287 289 TO 294 296 TO 301 303 TO 308 310 TO 321 -
528. 323 TO 344 346 TO 367 369 TO 390 392 TO 413 448 TO 543 545 TO 880
529. END GROUP DEFINITION
530. ELEMENT PROPERTY
531. 16 18 TO 20 22 23 25 TO 30 34 TO 40 42 44 46 TO 60 62 TO 65 67 68 70 TO 99 -
532. 101 103 TO 109 111 112 114 TO 117 119 TO 124 126 TO 142 144 146 TO 151 153 -
533. 154 156 TO 171 174 TO 183 185 TO 190 192 TO 197 199 TO 204 206 TO 211 213 -
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535. 268 TO 273 275 TO 280 282 TO 287 289 TO 294 296 TO 301 303 TO 308 -
536. 310 TO 321 323 TO 344 346 TO 367 369 TO 390 392 TO 413 448 TO 543 -
537. 545 TO 880 THICKNESS 0.6
538. DEFINE MATERIAL START
539. ISOTROPIC CONCRETE
540. E 2.17185E+007
541. POISSON 0.17
542. DENSITY 25
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543. ALPHA 1E-005
544. DAMP 0.05
545. TYPE CONCRETE
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550. 2 PRIS YD 0.8 ZD 0.65
551. CONSTANTS
552. MATERIAL CONCRETE ALL
553. SUPPORTS
554. 16 18 TO 20 22 23 25 TO 30 34 TO 40 42 44 46 TO 60 62 TO 65 67 68 70 TO 99 -
555. 101 103 TO 109 111 112 114 TO 117 119 TO 124 126 TO 142 144 146 TO 151 153 -
556. 154 156 TO 171 174 TO 183 185 TO 190 192 TO 197 199 TO 204 206 TO 211 213 -
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558. 268 TO 273 275 TO 280 282 TO 287 289 TO 294 296 TO 301 303 TO 308 -
559. 310 TO 321 323 TO 344 346 TO 367 369 TO 390 392 TO 413 448 TO 543 -
560. 545 TO 880 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
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18	0.072187
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693	0.075000
694	0.064500
695	0.054000
696	0.052500
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717	0.037625
718	0.031500
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757	0.078750

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774	0.075000
775	0.090000
776	0.052500
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780	0.048125
781	0.045937
782	0.043750
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787	0.048125
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857	0.067500
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859	0.090000
860	0.052500
861	0.075000
862	0.090000
863	0.052500
864	0.043750
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866	0.045937
867	0.048125
868	0.045937
869	0.043750

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       880          0.043750
       881          0.041562
       882          0.043750
       883          0.043750
       884          0.043750
       885          0.041562
       886          0.039375
       887          0.043750
       888          0.052500
       889          0.030625
       890          0.043750
       891          0.052500
       892          0.030625
    TOTAL          52.635086
561. ************************************************************
562. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
563. JOINT LOAD
564. 2 FX 9.41 FY -237.38 FZ -4.97
565. 4 FX 10.18 FY -94.13 FZ 4.75
566. 1 FX -13.40 FY -65.26 FZ -0.52
567. 3 FX -16.00 FY -65.64 FZ 0.40
569. SELFWEIGHT Y -1
570. ELEMENT LOAD
571. 16 18 TO 20 22 23 25 TO 30 34 TO 40 42 44 46 TO 60 62 TO 65 67 68 70 TO 99 -
572. 101 103 TO 109 111 112 114 TO 117 119 TO 124 126 TO 142 144 146 TO 151 153 -
573. 154 156 TO 171 174 TO 183 185 TO 190 192 TO 197 199 TO 204 206 TO 211 213 -
574. 214 TO 218 220 TO 225 227 TO 232 234 TO 239 241 TO 252 254 TO 259 261 TO 266 -
575. 268 TO 273 275 TO 280 282 TO 287 289 TO 294 296 TO 301 303 TO 308 -
576. 310 TO 321 323 TO 344 346 TO 367 369 TO 390 392 TO 413 448 TO 543 -
577. 545 TO 880 PR GY -16.2
578. ***
579. LOAD 2 LOADTYPE NONE  TITLE NORMAL OPERATING FLUID LOAD ( FL1)
580. JOINT LOAD
581. 2 FX 13.00 FY -621.22 FZ -1.20
582. 4 FX 0.79 FY -9.98 FZ 0.67
583. 1 FX -30.22 FY -57.54 FZ -0.74
584. 3 FX 3.26 FY 19.45 FZ -0.75
586. LOAD 3 LOADTYPE NONE  TITLE MAX. FLUID LOAD ( FL2)
587. JOINT LOAD
588. 2 FX 16.09 FY -768.05 FZ -1.48
589. 4 FX 1.01 FY -12.53 FZ 0.83
590. 1 FX -37.30 FY -70.89 FZ -0.91
591. 3 FX 4.07 FY 24.21 FZ -0.92
593. LOAD 4 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
594. JOINT LOAD
595. 2 FX 2.61 FY -74.72 FZ -7.21
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596. 4 FX 1.71 FY -22.93 FZ 7.80
597. 1 FX -4.43 FY -23.34 FZ -0.98
598. 3 FX -3.83 FY -15.78 FZ 0.86
600. LOAD 5 LOADTYPE NONE TITLE SEISMIC LOAD + X (SLX1)
601. JOINT LOAD
602. 2 FX 50.21 FY -166.81 FZ -2.66
603. 4 FX 15.30 FY -70.55 FZ -1.34
604. 1 FX 80.28 FY 162.53 FZ -0.43
605. 3 FX 20.50 FY 85.26 FZ -0.41
607. LOAD 6 LOADTYPE NONE TITLE SEISMIC LOAD - X (SLX2)
608. JOINT LOAD
609. 2 FX -50.21 FY 166.81 FZ 2.66
610. 4 FX -15.30 FY 70.55 FZ 1.34
611. 1 FX -80.28 FY -162.53 FZ 0.43
612. 3 FX -20.50 FY -85.26 FZ 0.41
614. LOAD 7 LOADTYPE NONE TITLE SEISMIC LOAD +Z (SLZ1)
615. JOINT LOAD
616. 2 FX 2.84 FY 132.39 FZ 64.87
617. 4 FX 6.31 FY -137.72 FZ 67.45
618. 1 FX 25.68 FY 86.57 FZ 7.33
619. 3 FX -18.16 FY -81.29 FZ 7.33
621. LOAD 8 LOADTYPE NONE TITLE SEISMIC LOAD -Z (SLZ2)
622. JOINT LOAD
623. 2 FX -2.84 FY -132.39 FZ -64.87
624. 4 FX -6.31 FY 137.72 FZ -67.45
625. 1 FX -25.68 FY -86.57 FZ -7.33
626. 3 FX 18.16 FY 81.29 FZ -7.33
628. LOAD 9 LOADTYPE NONE TITLE WIND LOAD + X (WLX1)
629. JOINT LOAD
630. 2 FX 24.87 FY -142.38 FZ -1.65
631. 4 FX 23.49 FY -104.69 FZ 0.10
632. 1 FX 44.60 FY 143.77 FZ -0.27
633. 3 FX 28.12 FY 117.27 FZ -0.24
635. LOAD 10 LOADTYPE NONE TITLE WIND LOAD - X (WLX2)
636. JOINT LOAD
637. 2 FX -24.92 FY 142.86 FZ 1.66
638. 4 FX -23.55 FY 105.04 FZ -0.10
639. 1 FX -44.71 FY -144.27 FZ 0.27
640. 3 FX -28.19 FY -117.67 FZ 0.24
642. LOAD 11 LOADTYPE NONE TITLE WIND LOAD + Z (WLZ1)
643. JOINT LOAD
644. 2 FX -4.62 FY 128.04 FZ 43.86
645. 4 FX 10.59 FY -123.00 FZ 44.00
646. 1 FX 31.46 FY 175.96 FZ 18.16
647. 3 FX -31.44 FY -181.13 FZ 18.15
649. LOAD 12 LOADTYPE NONE TITLE WIND LOAD - Z (WLZ2)
650. JOINT LOAD
651. 2 FX 4.62 FY -128.04 FZ -43.86
652. 4 FX -10.59 FY 123.00 FZ -44.00
653. 1 FX -31.46 FY -175.96 FZ -18.16
654. 3 FX 31.44 FY 181.13 FZ -18.15
656. LOAD 13 LOADTYPE NONE TITLE TEMPERATURE LOAD (TL)
657. JOINT LOAD
658. 2 FX -3.34 FY 224.99 FZ -10.13
659. 4 FX -38.70 FY 244.03 FZ 8.88
660. 1 FX -92.62 FY -243.87 FZ -1.35

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661. 3 FX -115.60 FY -253.37 FZ -0.06
663. LOAD 14 LOADTYPE NONE  TITLE LATERAL PIPE LOAD ( PL )
664. JOINT LOAD
665. 2 FX -16.93 FY 211.99 FZ 30.12
666. 4 FX -9.78 FY 17.51 FZ 27.43
667. 1 FX -15.99 FY -45.43 FZ 17.43
668. 3 FX -69.53 FY -401.53 FZ 17.35
670. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
671. LOAD COMB 101 1.4DL+1.4FL1
672. 1 1.4 2 1.4
673. LOAD COMB 102 1.4DL+1.4FL2
674. 1 1.4 3 1.4
675. LOAD COMB 103 1.2DL+1.2FL1+1.6LL
676. 1 1.2 2 1.2 4 1.6
677. LOAD COMB 104 1.2DL+1.2FL2+1.6LL
678. 1 1.2 3 1.2 4 1.6
679. LOAD COMB 105 1.2DL+1.2FL1+1.2TL+1.6LL
680. 1 1.2 2 1.2 4 1.6 13 1.2
681. LOAD COMB 106 1.2DL+1.2FL2+1.2TL+1.6LL
682. 1 1.2 3 1.2 4 1.6 13 1.2
683. LOAD COMB 107 1.4DL+1.4FL1+1.4PL
684. 1 1.4 2 1.4 14 1.4
685. LOAD COMB 108 1.4DL+1.4FL2+1.4PL
686. 1 1.4 3 1.4 14 1.4
687. LOAD COMB 109 1.2DL+1.2FL1+1.6LL+1.2PL
688. 1 1.2 2 1.2 4 1.6 14 1.2
689. LOAD COMB 110 1.2DL+1.2FL2+1.6LL+1.2PL
690. 1 1.2 3 1.2 4 1.6 14 1.2
691. LOAD COMB 111 1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL
692. 1 1.2 2 1.2 4 1.6 13 1.2 14 1.2
693. LOAD COMB 112 1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL
694. 1 1.2 3 1.2 4 1.6 13 1.2 14 1.2
695. LOAD COMB 113 1.2DL+0.8WLX1
696. 1 1.2 9 0.8
697. LOAD COMB 114 1.2DL+0.8WLX2
698. 1 1.2 10 0.8
699. LOAD COMB 115 1.2DL+0.8WLZ1
700. 1 1.2 11 0.8
701. LOAD COMB 116 1.2DL+0.8WLZ2
702. 1 1.2 12 0.8
703. LOAD COMB 117 1.2DL+1.6WLX1+0.5LL
704. 1 1.2 9 1.6 4 0.5
705. LOAD COMB 118 1.2DL+1.6WLX2+0.5LL
706. 1 1.2 10 1.6 4 0.5
707. LOAD COMB 119 1.2DL+1.6WLZ1+0.5LL
708. 1 1.2 11 1.6 4 0.5
709. LOAD COMB 120 1.2DL+1.6WLZ2+0.5LL
710. 1 1.2 12 1.6 4 0.5
711. LOAD COMB 121 1.2DL+1.0SLX1+0.5LL+0.5FL1
712. 1 1.2 5 1.0 4 0.5 2 0.5
713. LOAD COMB 122 1.2DL+1.0SLX2+0.5LL+0.5FL1
714. 1 1.2 6 1.0 4 0.5 2 0.5
715. LOAD COMB 123 1.2DL+1.0SLZ1+0.5LL+0.5FL1
716. 1 1.2 7 1.0 4 0.5 2 0.5
717. LOAD COMB 124 1.2DL+1.0SLZ2+0.5LL+0.5FL1
718. 1 1.2 8 1.0 4 0.5 2 0.5
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719. LOAD COMB 125 0.9DL+1.6WLX1
720. 1 0.9 9 1.6
721. LOAD COMB 126 0.9DL+1.6WLX2
722. 1 0.9 10 1.6
723. LOAD COMB 127 0.9DL+1.6WLZ1
724. 1 0.9 11 1.6
725. LOAD COMB 128 0.9DL+1.6WLZ2
726. 1 0.9 12 1.6
727. LOAD COMB 129 0.9DL+1.0SLX1
728. 1 0.9 5 1.0
729. LOAD COMB 130 0.9DL+1.0SLX2
730. 1 0.9 6 1.0
731. LOAD COMB 131 0.9DL+1.0SLZ1
732. 1 0.9 7 1.0
733. LOAD COMB 132 0.9DL+1.0SLZ2
734. 1 0.9 8 1.0
735. ********************************************************************************
736. * UNFACTORED LOAD COMBINATIONS FOR DEFLECTION CHECK
737. LOAD COMB 201 1.0DL+1.0FL1
738. 1 1.0 2 1.0
739. LOAD COMB 202 1.0DL+1.0FL2
740. 1 1.0 3 1.0
741. LOAD COMB 203 1.0DL+1.0FL1+1.0LL+1.0TL
742. 1 1.0 2 1.0 4 1.0 13 1.0
743. LOAD COMB 204 1.0DL+1.0FL2+1.0LL+1.0TL
744. 1 1.0 3 1.0 4 1.0 13 1.0
745. LOAD COMB 205 1.0DL+1.0FL1+0.75LL+0.75TL
746. 1 1.0 2 1.0 4 0.75 13 0.75
747. LOAD COMB 206 1.0DL+1.0FL2+0.75LL+0.75TL
748. 1 1.0 3 1.0 4 0.75 13 0.75
749. LOAD COMB 207 1.0DL+1.0FL1+1.0PL
750. 1 1.0 2 1.0 14 1.0
751. LOAD COMB 208 1.0DL+1.0FL2+1.0PL
752. 1 1.0 3 1.0 14 1.0
753. LOAD COMB 209 1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL
754. 1 1.0 2 1.0 4 1.0 13 1.0 14 1.0
755. LOAD COMB 210 1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL
756. 1 1.0 3 1.0 4 1.0 13 1.0 14 1.0
757. LOAD COMB 211 1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL
758. 1 1.0 2 1.0 4 0.75 13 0.75 14 1.0
759. LOAD COMB 212 1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL
760. 1 1.0 3 1.0 4 0.75 13 0.75 14 1.0
761. LOAD COMB 213 1.0DL+1.0FL1+1.0WLX1
762. 1 1.0 2 1.0 9 1.0
763. LOAD COMB 214 1.0DL+1.0FL1+1.0WLX2
764. 1 1.0 2 1.0 10 1.0
765. LOAD COMB 215 1.0DL+1.0FL1+1.0WLZ1
766. 1 1.0 2 1.0 11 1.0
767. LOAD COMB 216 1.0DL+1.0FL1+1.0WLZ2
768. 1 1.0 2 1.0 12 1.0
769. LOAD COMB 217 1.0DL+1.0FL2+1.0WLX1
770. 1 1.0 3 1.0 9 1.0
771. LOAD COMB 218 1.0DL+1.0FL2+1.0WLX2
772. 1 1.0 3 1.0 10 1.0
773. LOAD COMB 219 1.0DL+1.0FL2+1.0WLZ1
774. 1 1.0 3 1.0 11 1.0
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775. LOAD COMB 220 1.0DL+1.0FL2+1.0WLZ2
776. 1 1.0 3 1.0 12 1.0
777. LOAD COMB 221 1.0DL+1.0FL1+0.7SLX1
778. 1 1.0 2 1.0 5 0.7
779. LOAD COMB 222 1.0DL+1.0FL1+0.7SLX2
780. 1 1.0 2 1.0 6 0.7
781. LOAD COMB 223 1.0DL+1.0FL1+0.7SLZ1
782. 1 1.0 2 1.0 7 0.7
783. LOAD COMB 224 1.0DL+1.0FL1+0.7SLZ2
784. 1 1.0 2 1.0 8 0.7
785. LOAD COMB 225 1.0DL+1.0FL2+0.7SLX1
786. 1 1.0 3 1.0 5 0.7
787. LOAD COMB 226 1.0DL+1.0FL2+0.7SLX2
788. 1 1.0 3 1.0 6 0.7
789. LOAD COMB 227 1.0DL+1.0FL2+0.7SLZ1
790. 1 1.0 3 1.0 7 0.7
791. LOAD COMB 228 1.0DL+1.0FL2+0.7SLZ2
792. 1 1.0 3 1.0 8 0.7
793. LOAD COMB 229 1.0DL+1.0FL1+0.75WLX1+0.75LL
794. 1 1.0 2 1.0 4 0.75 9 0.75
795. LOAD COMB 230 1.0DL+1.0FL1+0.75WLX2+0.75LL
796. 1 1.0 2 1.0 4 0.75 10 0.75
797. LOAD COMB 231 1.0DL+1.0FL1+0.75WLZ1+0.75LL
798. 1 1.0 2 1.0 4 0.75 11 0.75
799. LOAD COMB 232 1.0DL+1.0FL1+0.75WLZ2+0.75LL
800. 1 1.0 2 1.0 4 0.75 12 0.75
801. LOAD COMB 233 1.0DL+1.0FL2+0.75WLX1+0.75LL
802. 1 1.0 3 1.0 4 0.75 9 0.75
803. LOAD COMB 234 1.0DL+1.0FL2+0.75WLX2+0.75LL
804. 1 1.0 3 1.0 4 0.75 10 0.75
805. LOAD COMB 235 1.0DL+1.0FL2+0.75WLZ1+0.75LL
806. 1 1.0 3 1.0 4 0.75 11 0.75
807. LOAD COMB 236 1.0DL+1.0FL2+0.75WLZ2+0.75LL
808. 1 1.0 3 1.0 4 0.75 12 0.75
809. LOAD COMB 237 1.0DL+1.0FL1+0.525SLX1+0.75LL
810. 1 1.0 2 1.0 4 0.75 5 0.525
811. LOAD COMB 238 1.0DL+1.0FL1+0.525SLX2+0.75LL
812. 1 1.0 2 1.0 4 0.75 6 0.525
813. LOAD COMB 239 1.0DL+1.0FL1+0.525SLZ1+0.75LL
814. 1 1.0 2 1.0 4 0.75 7 0.525
815. LOAD COMB 240 1.0DL+1.0FL1+0.525SLZ2+0.75LL
816. 1 1.0 2 1.0 4 0.75 8 0.525
817. LOAD COMB 241 1.0DL+1.0FL2+0.525SLX1+0.75LL
818. 1 1.0 3 1.0 4 0.75 5 0.525
819. LOAD COMB 242 1.0DL+1.0FL2+0.525SLX2+0.75LL
820. 1 1.0 3 1.0 4 0.75 6 0.525
821. LOAD COMB 243 1.0DL+1.0FL2+0.525SLZ1+0.75LL
822. 1 1.0 3 1.0 4 0.75 7 0.525
823. LOAD COMB 244 1.0DL+1.0FL2+0.525SLZ2+0.75LL
824. 1 1.0 3 1.0 4 0.75 8 0.525
825. LOAD COMB 245 0.6DL+1.0WLX1
826. 1 0.6 9 1.0
827. LOAD COMB 246 0.6DL+1.0WLX2
828. 1 0.6 10 1.0
829. LOAD COMB 247 0.6DL+1.0WLZ1
830. 1 0.6 11 1.0
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831. LOAD COMB 248 0.6DL+1.0WLZ2
832. 1 0.6 12 1.0
833. LOAD COMB 249 0.6DL+0.6FL1+0.7SLX1
834. 1 0.6 2 0.6 5 0.7
835. LOAD COMB 250 0.6DL+0.6FL1+0.7SLX2
836. 1 0.6 2 0.6 6 0.7
837. LOAD COMB 251 0.6DL+0.6FL1+0.7SLZ1
838. 1 0.6 2 0.6 7 0.7
839. LOAD COMB 252 0.6DL+0.6FL1+0.7SLZ2
840. 1 0.6 2 0.6 8 0.7
841. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**845**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**784**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**841**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 701/ 48/ 147 DOF
TOTAL PRIMARY LOAD CASES = 14, TOTAL DEGREES OF FREEDOM = 2547
TOTAL LOAD COMBINATION CASES = 84 SO FAR.
SIZE OF STIFFNESS MATRIX = 375 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 20.6/ 8159.5 MB

```
842. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 2,2020 TIME= 10:20:45 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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ATTACHMENT 4 COMMENT RESOLUTION LOG SHEET C01

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT**
PROJECT NO.:	**216**
STAGE:	**4 - PROJECT EXECUTION.**
REVIEW LOG DOC. NO. & REV.:	
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-002224, 27 April 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 27 April 2020	Comment by	EPC Response DATE: 04 May 2020	Status	Remarks
			General Comments (cc)				
MD-216-3310-EG-CV-CAL-1213-A01	3310	1	Structural concrete is 35 Mpa	Mr. Doron Golan - Bechtel	Incorporated	Closed	